AGRO NEXUS | FarmWise

PROJECT DATA SCIENCE REPORT

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INTRODUCTION

1.1 BUSINESS CONTEXT AND PROBLEM DEFINITION

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INTRODUCTION

1.1 Business Context and Problem Definition



Agriculture is a fundamental pillar of the global economy, providing food security and employment for millions of people. However, the sector is facing unprecedented challenges due to climate change, inefficient resource management, and the increasing demand for higher yields. Farmers often struggle with limited access to real-time data, making it difficult to make informed decisions about irrigation, fertilization, and disease control.

In Tunisia, agriculture plays a crucial role in economic development, but farmers face rising production costs, water scarcity, and unpredictable weather conditions. Traditional farming methods do not always optimize resource usage, leading to waste and environmental impact. Moreover, crop diseases and pest infestations can severely impact yields if not detected and treated early.

To overcome these challenges, FarmWise proposes an AI-powered solution that enables farmers to optimize their agricultural practices using data-driven insights. By leveraging machine learning, satellite imagery, and historical agricultural data, FarmWise aims to increase efficiency, reduce waste, and improve productivity. The project focuses on resource management, disease detection, and land segmentation to ensure that farmers can make the best possible decisions to maximize their yields and profitability.

INTRODUCTION

1.2 Project Objectives

The main objective of the FarmWise project is to revolutionize agriculture through AI-driven solutions that provide personalized recommendations to farmers. These recommendations will help optimize resources, detect plant diseases early, and improve agricultural planning. The specific objectives of the project are:

1.Develop an Intelligent Agricultural Recommendation Platform

- Provide personalized advice on irrigation, fertilization, and crop selection based on soil data, climate conditions, and yield history.
- Help farmers make data-driven decisions to increase productivity and sustainability.

2.Build an Automatic Plant Disease Detection Model

- Use AI and computer vision to analyze plant images and detect diseases with high accuracy.
- Reduce excessive pesticide use by targeting only affected crops, improving crop health and sustainability.

3.Segment Farmers and Agricultural Land for Better Resource Allocation

- Use data-driven clustering techniques to categorize farms based on size, crop type, soil condition, and climate factors.
- Optimize agricultural offerings and support strategies to enhance productivity.

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BUSINESS UNDERSTANDING

2.1 TDSP METHODOLOGY OVERVIEW

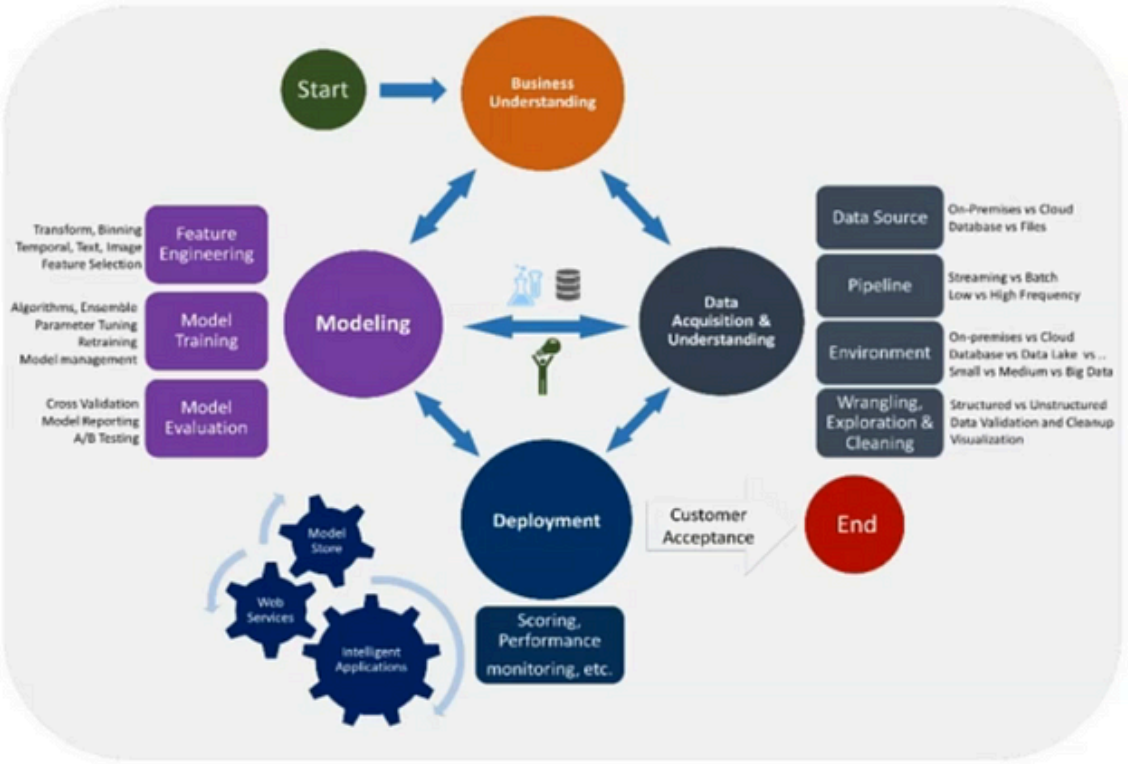
2.2 PROBLEM UNDERSTANDING

BUSINESS UNDERSTANDING

2.1 TDSP Methodology Overview

TDSP methodology

The Team Data Science Process (TDSP) is a structured methodology designed to guide data science projects from conception to deployment. Unlike traditional approaches, TDSP emphasizes collaboration, iterative development, and business alignment, ensuring that data-driven solutions meet real-world needs.



TDSP consists of five main phases:

1. **Business Understanding** – Defining the project's business objectives and expected impact.
2. **Data Acquisition and Understanding** – Collecting, cleaning, and exploring relevant data.
3. **Modeling** – Developing, testing, and refining machine learning models.
4. **Deployment** – Integrating models into real-world applications for end users.
5. **Customer Acceptance** – Validating the solution with stakeholders and ensuring usability.

BUSINESS UNDERSTANDING

2.2 Problem Understanding

The agricultural sector is increasingly turning to data-driven solutions to enhance productivity, sustainability, and resource efficiency. Farmers face growing challenges, including climate change, inefficient resource allocation, unpredictable weather patterns, and crop diseases that threaten food security. These issues impact yields, increase costs, and reduce profitability, making it essential to integrate advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) to improve decision-making.

One of the most pressing concerns in agriculture is the inefficient management of resources such as water, fertilizers, and pesticides. Misuse of these essential inputs not only increases production costs but also contributes to soil degradation and environmental pollution. Furthermore, early detection of plant diseases is critical to preventing significant yield losses, yet traditional disease monitoring methods are often manual, slow, and inaccurate.

Additionally, farmers frequently struggle with real-time decision-making, as they lack access to accurate and timely insights regarding irrigation, fertilization, and crop health. Similarly, understanding land value and agricultural potential remains a challenge, making it difficult for farmers and investors to optimize land use and make informed investment decisions.

Machine learning provides an opportunity to overcome these challenges by analyzing soil, climate, and historical yield data to generate predictive insights. By leveraging AI-powered decision-making tools, FarmWise aims to help farmers optimize resource allocation, detect crop diseases early, and forecast land prices, ultimately leading to higher efficiency, increased productivity, and long-term agricultural sustainability.

BUSINESS UNDERSTANDING

Business Objectives 

The objective of this project is to enhance agricultural productivity and sustainability by leveraging AI and data-driven decision-making tools. The FarmWise platform will provide farmers with real-time recommendations for resource management, disease detection, and land optimization.

More specifically, FarmWise will:

- **Provide accurate fertilizer dosage recommendations** to optimize crop yields and reduce fertilizer waste across different regions of Tunisia.
- **Identify crop diseases early** using **AI-powered image analysis** to prevent losses and improve treatment strategies.
- **Predict yield** and **maturity time** based on localisation and crop type.
- **Recommend optimal crop rotation strategies** to enhance soil fertility and long-term productivity.
- **Predict agricultural land prices in Tunisia** to support **informed investment and land-use planning.**
- **Classify spontaneous plants** (volunteer plants) to determine their medicinal value or potential harm to crops.

BUSINESS UNDERSTANDING

Data Science Objectives 

The primary aim of the data science aspect of this project is to utilize machine learning, computer vision, and predictive analytics to assist farmers in making data-driven decisions.

The specific objectives include:

- **Build a model capable of predicting** the optimal fertilizer dosage based on available data: soil properties, crop characteristics, fertilizer types, application methods, and regional information.
- **Develop a computer vision model** to detect early signs of crop diseases and pests by analyzing plant images.
- **Create a machine learning model** to develop a predictive system for estimating agricultural yield and crop maturity time.
- **Build a recommendation system** to suggest the best crop rotation strategies, balancing profitability and soil health.
- **Predict land prices in Tunisia** by analyzing factors like location, market trends, and soil quality, aiding investment decisions.
- **Automate the classification of volunteer plants** to identify harmful weeds and useful medicinal plants.

BUSINESS UNDERSTANDING

Sustainable Development Goals (SDGs)

This project contributes to several United Nations SDGs by promoting sustainable farming practices, improving resource efficiency, and addressing key global challenges like climate change and hunger. Through data science and AI, the project supports the creation of a more sustainable and equitable agricultural future.

Here's a brief description for each SDG:



SDG 13: Climate Action 🌍
This project supports climate action by optimizing resource use (water, fertilizers, pesticides) to reduce environmental impact and help mitigate climate change in agriculture.



SDG 15: Life on Land 🌱
By enhancing crop health, disease detection, and biodiversity through crop rotation recommendations, this project promotes sustainable farming that protects land ecosystems.



SDG 12: Responsible Consumption and Production 🌱
This project encourages responsible consumption and production by recommending optimal resource use (fertilizers, pesticides) and sustainable crop rotation practices.



SDG 8: Decent Work and Economic Growth 📈
The project fosters economic growth by improving productivity and profitability in agriculture, providing farmers with data-driven insights to make better financial decisions.



SDG 2: Zero Hunger 🌾
By improving crop yields and disease management, the project helps enhance food security and combat hunger by boosting agricultural productivity.



SDG 6: Clean Water and Sanitation 💧
The project promotes efficient water use in agriculture by predicting water needs for crops, supporting sustainable water management and conservation efforts.

03

DATA ACQUISITION AND UNDERSTANDING

BO 1

PROVIDE ACCURATE FERTILIZER DOSAGE RECOMMENDATIONS TO OPTIMIZE CROP YIELDS AND REDUCE FERTILIZER WASTE ACROSS DIFFERENT REGIONS OF TUNISIA.

I -DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS (EDA)

1-DATA OVERVIEW

THE DATASET CONSISTS OF 10,000 ROWS AND 20 FEATURES.
FEATURES INCLUDE:

- FERTILIZER-RELATED ATTRIBUTES (FERTILIZER NAME , FERTILIZER TYPE, COMPOSITION: N%, P%, K%,APPLICATION METHOD ,PRICE PER KG (TND))
- CROP-RELATED ATTRIBUTES (SPECIFIC CROP , NUTRIENT REQUIREMENTS: CROP P , CROP N , CROP K , ROOT DEPTH)
- SOIL-RELATED ATTRIBUTES (SOIL NUTRIENT LEVELS:SOIL P , SOIL K , SOIL N , PH)
- REGIONAL INFORMATION (STATE, REGION OF USE)
- TARGET VARIABLE: USAGE (DOSAGE) (CONTINUOUS)

- **CATEGORICAL FEATURES** REPRESENT APPROXIMATELY 30% OF THE DATA, WHILE THE REMAINING 70% ARE NUMERICAL.

2-DATA OVERVIEW

THE DATASET HAS MINOR MISSING VALUES:

- **SOIL_P**: 212 MISSING VALUES
- **SOIL_K**: 207 MISSING VALUES
- **SOIL_N**: 202 MISSING VALUES

-THE MISSING RATE IS BELOW 2% FOR EACH, MAKING IMPUTATION MANAGEABLE WITHOUT INTRODUCING MAJOR BIASES.

3-TARGET VARIABLE ANALYSIS: USAGE (DOSAGE)

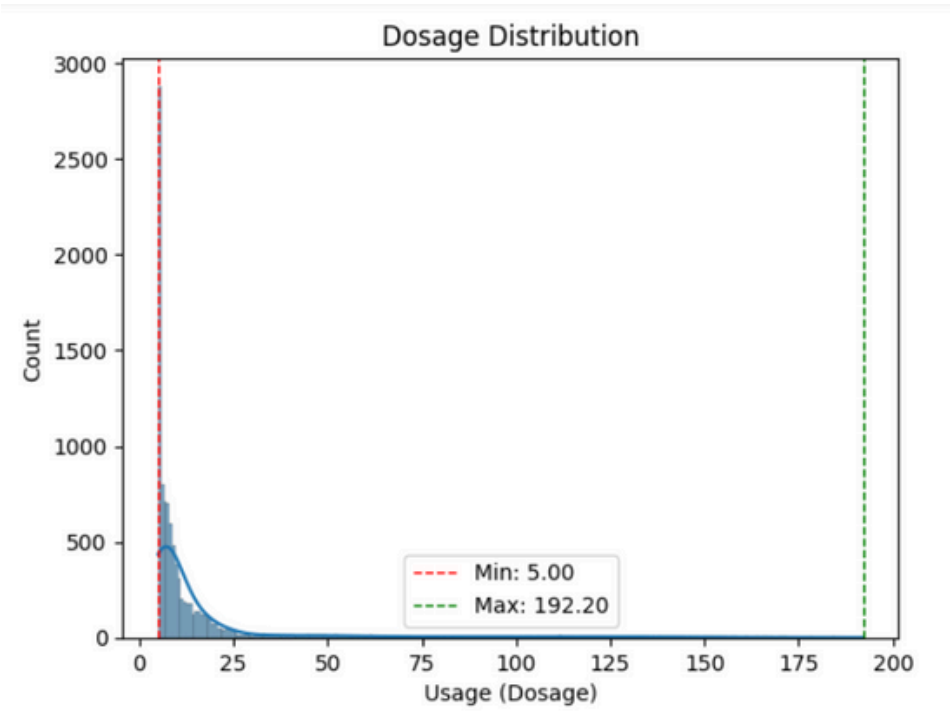
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4-FERTILIZER-SPECIFIC DOSAGE VARIATION

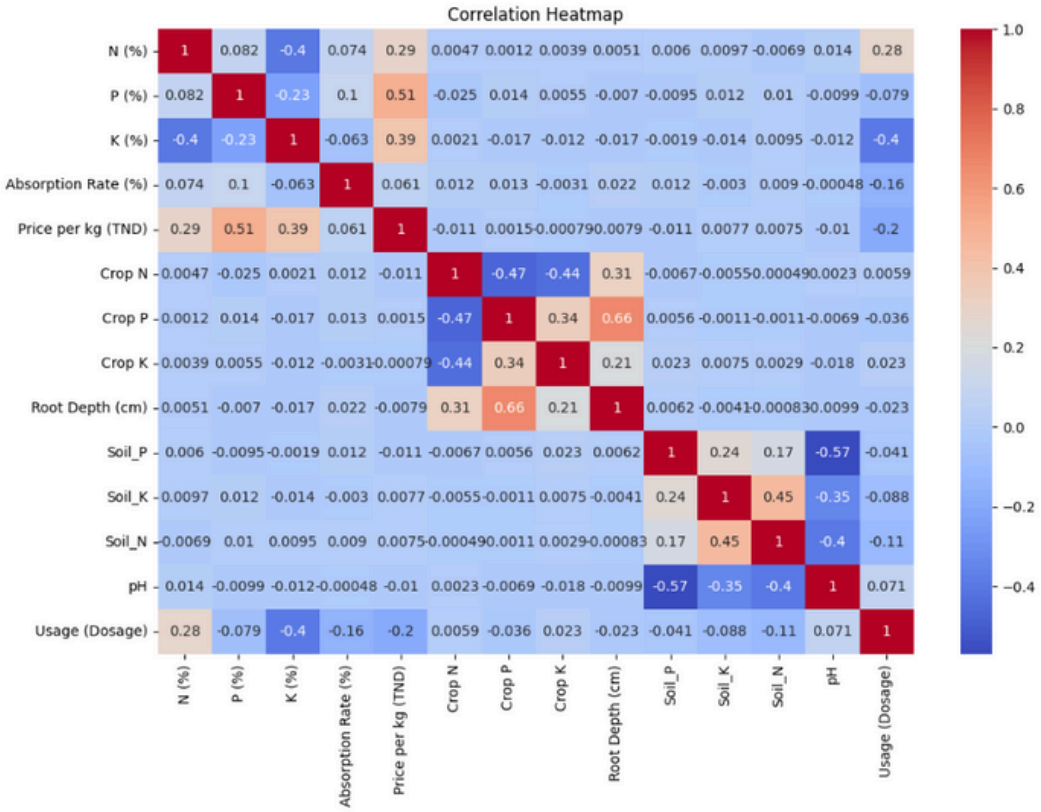
A DETAILED INSPECTION OF THE **USAGE (DOSAGE)** VARIABLE WAS CONDUCTED. THE HISTOGRAM OF DOSAGE VALUES SHOWS A RIGHT-SKEWED DISTRIBUTION, WITH THE MAJORITY OF FERTILIZER DOSAGES CONCENTRATED BETWEEN 5 AND 40 UNITS. HOWEVER, THE MAXIMUM DOSAGE RECORDED IS 192.2 UNITS, COMPARED TO A MINIMUM OF 5.0 UNITS, REPRESENTING A NEARLY 38-FOLD RANGE.



THIS WIDE SPREAD AND SKEWNESS HIGHLIGHT THE NEED TO POTENTIALLY TREAT OUTLIERS AND CONSIDER APPLYING TRANSFORMATIONS DURING THE MODELING PHASE, DEPENDING ON HOW THEY AFFECT REGRESSION MODEL PERFORMANCE.

5. CORRELATION ANALYSIS)

TO UNDERSTAND THE RELATIONSHIPS BETWEEN NUMERIC VARIABLES, A CORRELATION HEATMAP WAS GENERATED.



THE CORRELATION HEATMAP DISPLAYS THE RELATIONSHIPS BETWEEN VARIOUS AGRICULTURAL VARIABLES, WITH CORRELATION COEFFICIENTS RANGING FROM -1 TO 1. HERE'S A BREAKDOWN OF THE KEY INSIGHTS:

STRONG POSITIVE CORRELATIONS (CLOSE TO 1):

- PRICE PER KG (TND) AND P (%): MODERATE POSITIVE CORRELATION (0.51), SUGGESTING THAT HIGHER PHOSPHORUS CONTENT MAY BE ASSOCIATED WITH HIGHER PRICES.
- USAGE (DOSAGE) AND N (%): MODERATE POSITIVE CORRELATION (0.28), INDICATING THAT HIGHER NITROGEN CONTENT MIGHT LEAD TO INCREASED FERTILIZER USAGE.

STRONG NEGATIVE CORRELATIONS (CLOSE TO -1):

- N (%) AND K (%): STRONG NEGATIVE CORRELATION (-0.4), MEANING HIGHER NITROGEN LEVELS ARE ASSOCIATED WITH LOWER POTASSIUM LEVELS, AND VICE VERSA.
- USAGE (DOSAGE) AND K (%): STRONG NEGATIVE CORRELATION (-0.4), IMPLYING THAT HIGHER POTASSIUM LEVELS MAY REDUCE THE NEED FOR FERTILIZER USAGE.

WEAK CORRELATIONS (CLOSE TO 0):

- MOST VARIABLES RELATED TO CROP NUTRIENTS (CROP N, CROP P, CROP K), SOIL PROPERTIES (SOIL N, SOIL P, SOIL K), AND PH SHOW VERY WEAK CORRELATIONS (VALUES BETWEEN -0.03 AND 0.03), INDICATING MINIMAL LINEAR RELATIONSHIPS WITH OTHER VARIABLES.

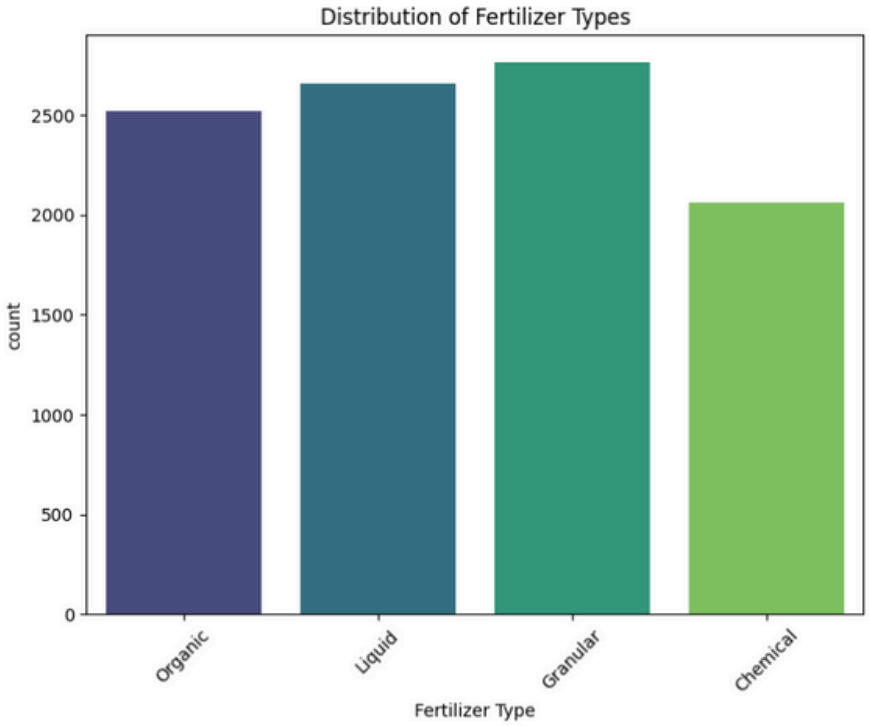
NOTABLE OBSERVATIONS:

- P (%) HAS A MODERATE POSITIVE CORRELATION WITH PRICE PER KG (TND) (0.51) BUT A WEAK NEGATIVE CORRELATION WITH USAGE (DOSAGE) (-0.079).
- K (%) SHOWS STRONG NEGATIVE CORRELATIONS WITH N (%) (-0.4) AND USAGE (DOSAGE) (-0.4), SUGGESTING THAT POTASSIUM MAY PLAY A ROLE IN REDUCING NITROGEN LEVELS AND FERTILIZER USAGE.
- ABSORPTION RATE (%) HAS WEAK CORRELATIONS WITH MOST VARIABLES, INDICATING IT MAY NOT BE STRONGLY INFLUENCED BY THE OTHER FACTORS LISTED.

SUMMARY:

- THE HEATMAP HIGHLIGHTS SIGNIFICANT RELATIONSHIPS BETWEEN NUTRIENT CONTENT (N, P, K), PRICING, AND FERTILIZER USAGE, WHILE MOST OTHER VARIABLES SHOW MINIMAL CORRELATIONS. THIS SUGGESTS THAT NITROGEN, PHOSPHORUS, AND POTASSIUM ARE KEY FACTORS INFLUENCING AGRICULTURAL OUTCOMES IN THIS DATASET.

6. FERTILIZER ANALYSIS

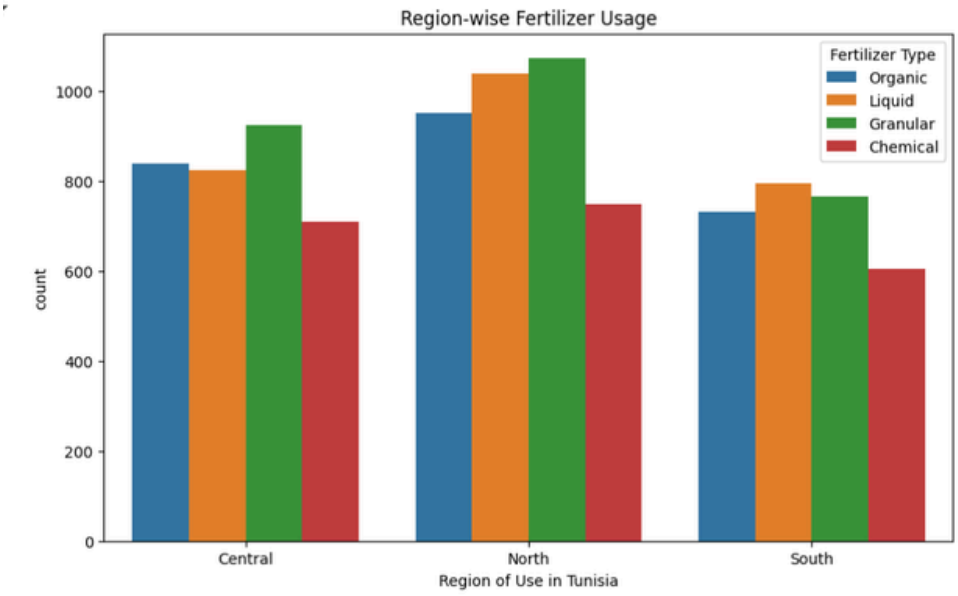


THE DISTRIBUTION OF FERTILIZER TYPE SHOWS THAT GRANULAR FERTILIZERS DOMINATE THE DATASET, FOLLOWED BY LIQUID, ORGANIC, AND CHEMICAL TYPES.

7. REGIONAL AND STATE DISTRIBUTION

7. 1. REGION-WISE FERTILIZER USAGE

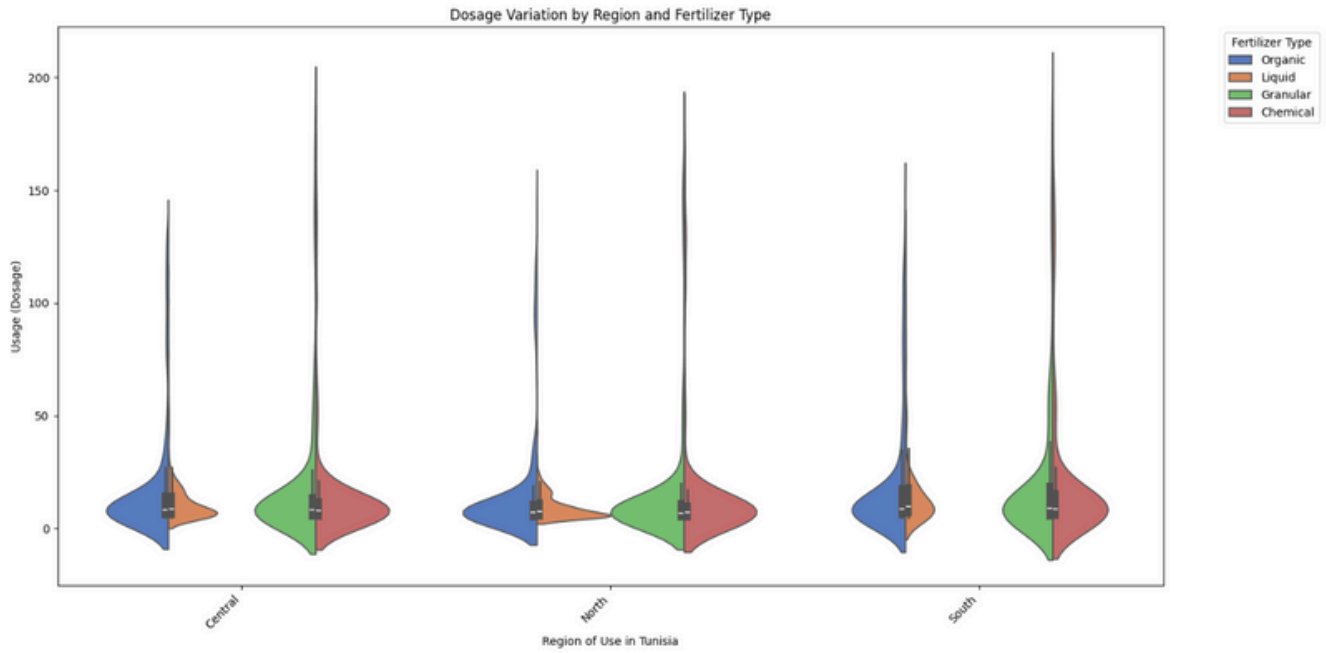
THE BAR PLOT VISUALIZES THE DISTRIBUTION OF FERTILIZER TYPES USED ACROSS THE THREE MAIN REGIONS OF TUNISIA: CENTRAL, NORTH, AND SOUTH.



THE BAR PLOT VISUALIZES THE DISTRIBUTION OF FERTILIZER TYPES USED ACROSS THE THREE MAIN REGIONS OF TUNISIA: **CENTRAL, NORTH, AND SOUTH.**
SEVERAL CLEAR PATTERNS EMERGE:

- **NORTH REGION:**
  - THE NORTH SHOWS THE HIGHEST OVERALL FERTILIZER USAGE ACROSS ALL TYPES.
  - GRANULAR FERTILIZERS ARE THE MOST DOMINANT IN THE NORTH, FOLLOWED CLOSELY BY LIQUID FERTILIZERS.
  - ORGANIC FERTILIZERS ALSO HAVE A STRONG PRESENCE, WHILE CHEMICAL FERTILIZERS ARE THE LEAST USED AMONG THE FOUR TYPES.
- **CENTRAL REGION:**
  - FERTILIZER USAGE IS MORE BALANCED IN THE CENTRAL REGION.
  - GRANULAR FERTILIZERS SLIGHTLY LEAD USAGE, FOLLOWED BY ORGANIC AND LIQUID FERTILIZERS WITH VERY CLOSE COUNTS.
  - CHEMICAL FERTILIZERS AGAIN ARE THE LEAST PREFERRED, THOUGH THEIR USE IS SLIGHTLY HIGHER THAN IN THE NORTH.
- **SOUTH REGION:**
  - FERTILIZER USAGE IS LOWER OVERALL IN THE SOUTH COMPARED TO THE NORTH AND CENTRAL REGIONS.
  - LIQUID FERTILIZERS ARE THE MOST USED TYPE IN THE SOUTH, FOLLOWED CLOSELY BY GRANULAR AND ORGANIC FERTILIZERS.
  - CHEMICAL FERTILIZERS ARE THE LEAST POPULAR HERE AS WELL, WITH THE LOWEST COUNTS AMONG ALL REGIONS.

7. 2. DOSAGE VARIATION BY REGION AND FERTILIZER



WHAT THE PLOT TELLS US:
EACH "VIOLIN" REPRESENTS THE DISTRIBUTION (DENSITY) OF FERTILIZER USAGE FOR EACH FERTILIZER TYPE IN EACH REGION.
THE WIDTH OF EACH VIOLIN AT A GIVEN DOSAGE LEVEL SHOWS HOW COMMON THAT DOSAGE IS. (WIDER = MORE OBSERVATIONS AT THAT DOSAGE.)
INSIDE EACH VIOLIN, YOU HAVE A BOXPLOT: SHOWING **THE MEDIAN (WHITE DOT), INTERQUARTILE RANGE (THICK BAR), AND POTENTIAL OUTLIERS (THIN LINES).**

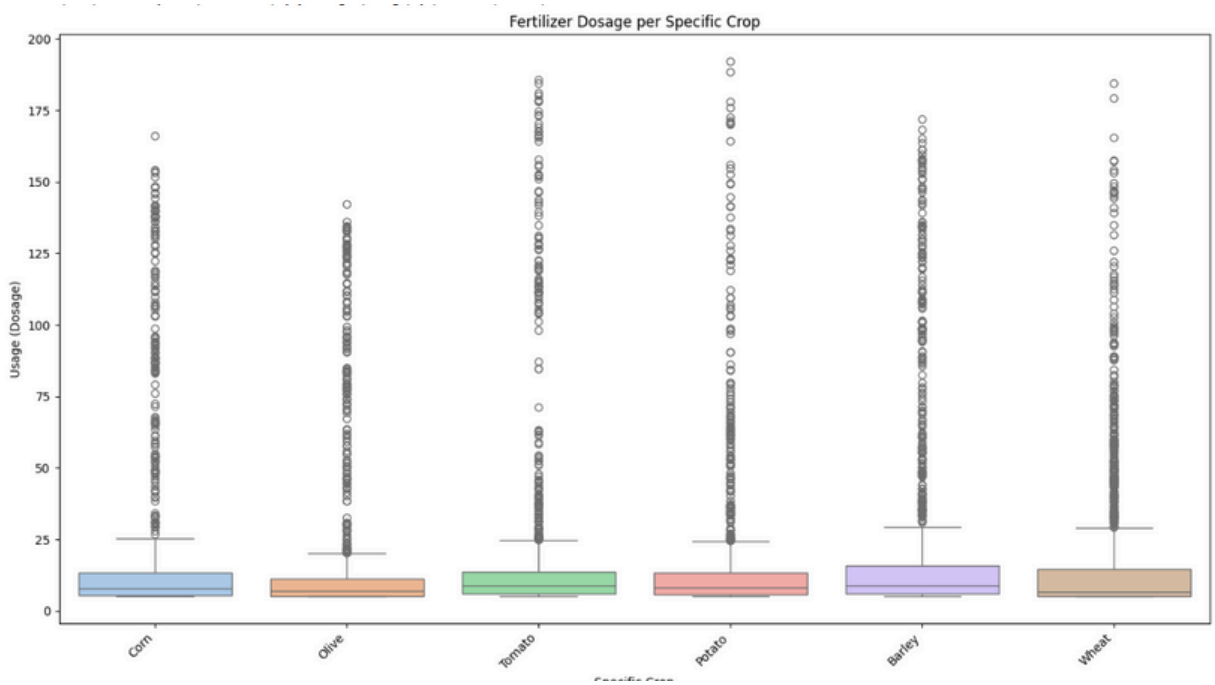
OBSERVATIONS:

- ORGANIC FERTILIZERS (BLUE) SHOW A WIDE DOSAGE RANGE ACROSS ALL REGIONS, WITH MOST USAGE BETWEEN 5–15 UNITS, BUT SOME OUTLIERS REACHING NEAR 200 UNITS.
- LIQUID FERTILIZERS (ORANGE) ARE APPLIED IN LOWER AND MORE CONSISTENT DOSES, WITH NARROWER DISTRIBUTIONS AND FEWER EXTREME VALUES.
- GRANULAR (GREEN) AND CHEMICAL FERTILIZERS (RED) DISPLAY MODERATE MEDIAN USAGE (~10 UNITS) BUT STILL HAVE SOME HIGH OUTLIERS ABOVE 100.
- REGIONAL PATTERNS: CENTRAL AND NORTH REGIONS HAVE SIMILAR FERTILIZER USAGE DISTRIBUTIONS, WHILE THE SOUTH SHOWS SLIGHTLY MORE SPREAD, PARTICULARLY FOR GRANULAR FERTILIZERS.

SUMMARY:

- ORGANIC AND CHEMICAL FERTILIZERS HAVE THE HIGHEST VARIABILITY IN DOSAGE.
- LIQUID FERTILIZERS ARE USED MORE UNIFORMLY AND CONSERVATIVELY ACROSS TUNISIA.
- OUTLIERS ARE PRESENT FOR ALL FERTILIZER TYPES, ESPECIALLY ORGANIC AND CHEMICAL.

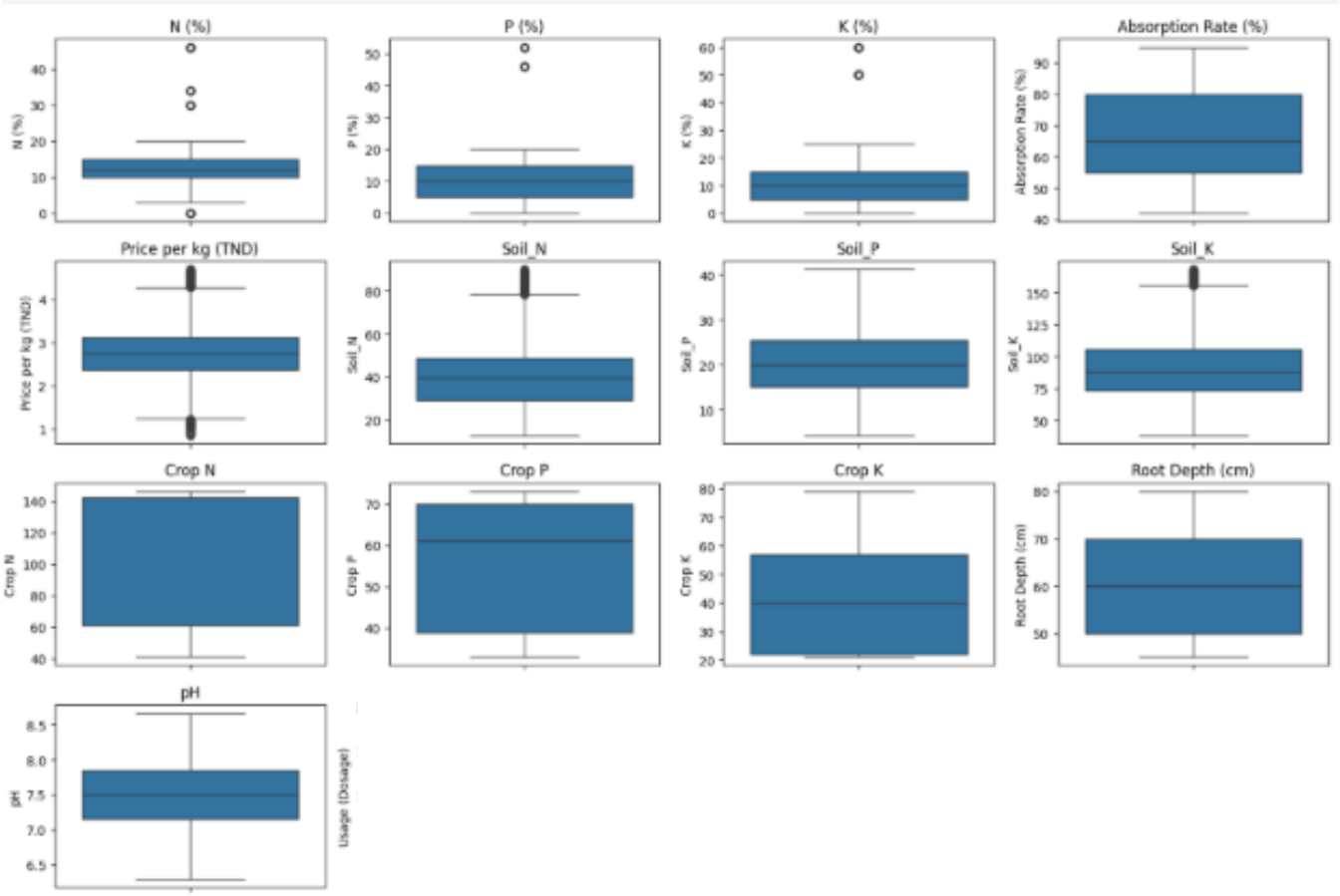
8. CROP-SPECIFIC FERTILIZER DOSAGE



OBSERVATIONS:

- FOR ALL CROPS, MOST DOSAGE VALUES FALL BETWEEN **5 AND 25 UNITS**, WITH MEDIAN VALUES AROUND 10–15 UNITS**.**
- **TOMATO AND POTATO CROPS** HAVE HIGHER MEDIANS AND WIDER DOSAGE SPREADS, INDICATING THEY TYPICALLY REQUIRE MORE FERTILIZER COMPARED TO OLIVE AND WHEAT, WHICH HAVE THE LOWEST MEDIANS.
- **CORN** SHOWS A MEDIAN SLIGHTLY HIGHER THAN WHEAT AND OLIVE BUT SHARES A SIMILAR PATTERN OF MANY OUTLIERS.
- **OUTLIERS** ARE FREQUENT FOR ALL CROPS, WITH SOME FIELDS APPLYING DOSAGES EXCEEDING 150 UNITS, INDICATING OCCASIONAL VERY HEAVY FERTILIZER PRACTICES.
- CROP-TO-CROP COMPARISON: DESPITE SMALL DIFFERENCES, THE GENERAL DISTRIBUTION SHAPE IS SIMILAR ACROSS CROPS:
- MOST FERTILIZER USE STAYS MODERATE, BUT EVERY CROP HAS A FEW CASES OF EXTREMELY HIGH FERTILIZER APPLICATION.

8. OUTLIERS



FERTILIZER COMPOSITION:

N (%) (NITROGEN):

- MEDIAN AROUND 20%, MOST VALUES BETWEEN 15–25%. SOME OUTLIERS BELOW 5% AND ABOVE 35%.

P (%) (PHOSPHORUS):

- MEDIAN NEAR 15%, SPREAD FROM 10–20%, WITH OCCASIONAL OUTLIERS ABOVE 30%.

K (%) (POTASSIUM):

- MEDIAN 15–20%, BROADER SPREAD (10–30%) WITH OUTLIERS REACHING 50%.

ABSORPTION RATE (%):

- MEDIAN ABOUT 60%, GENERALLY BETWEEN 40–80% WITH NO EXTREME OUTLIERS.

PRICE PER KG (TND):

- MEDIAN AROUND 4–5 TND, MOSTLY 3–6 TND, WITH A FEW LOW-PRICE OUTLIERS (~1 TND).

SOIL NUTRIENT CONTENT:

SOIL_N:

- MEDIAN NEAR 45 UNITS, SPREAD FROM 30–60, WITH HIGH OUTLIERS (>80).

SOIL_P:

- MEDIAN AROUND 20 UNITS, RANGING 15–30 UNITS.

SOIL_K:

- HIGHER MEDIAN (~100 UNITS), SPREAD FROM 80–120, WITH OUTLIERS ABOVE 150.

CROP NUTRIENT DEMAND:

CROP N:

- HIGH MEDIAN (~120 UNITS), SPREAD 100–140 UNITS.

CROP P:

- MEDIAN AROUND 60 UNITS.

CROP K:

- MEDIAN ABOUT 50 UNITS.

OTHER FEATURES:

ROOT DEPTH:

- MEDIAN AROUND 50 CM, WITH MOST CROPS BETWEEN 40–60 CM.

PH:

- MEDIAN ~7.5, RANGING FROM 7.0 TO 8.0 — NEUTRAL TO SLIGHTLY BASIC SOILS.

KEY INSIGHTS:

- CROPS DEMAND HIGH NITROGEN; SOIL POTASSIUM IS NATURALLY ABUNDANT.
- FERTILIZER PRICES ARE STABLE, WITH SOME VERY CHEAP PRODUCTS.
- SOIL CONDITIONS (NUTRIENTS, PH) ARE FAVORABLE FOR AGRICULTURE.
- OUTLIERS EXIST MAINLY IN FERTILIZER COMPOSITION AND SOIL POTASSIUM.

II -DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS (EDA)

THE DATA PREPARATION PHASE INVOLVED SEVERAL CRITICAL STEPS TO ENSURE THAT THE DATASET WAS CLEAN, CONSISTENT, AND SUITABLE FOR MODELING FERTILIZER DOSAGE PREDICTIONS. EACH PREPROCESSING STEP WAS CAREFULLY DESIGNED TO IMPROVE THE QUALITY AND INTERPRETABILITY OF THE DATA, WHILE MINIMIZING BIASES AND INFORMATION LOSS.

1. INITIAL EXPLORATION

WE BEGAN BY ANALYZING THE DISTRIBUTION AND SKEWNESS OF THE TARGET VARIABLE, USAGE (DOSAGE), USING VISUAL INSPECTION TECHNIQUES. A HISTOGRAM COMBINED WITH A KERNEL DENSITY ESTIMATE (KDE) PLOT WAS USED TO ASSESS THE OVERALL SHAPE OF THE DISTRIBUTION, AND THE NUMERICAL SKEWNESS COEFFICIENT WAS CALCULATED TO DETECT POTENTIAL ASYMMETRY.

2. HANDLING MISSING VALUES

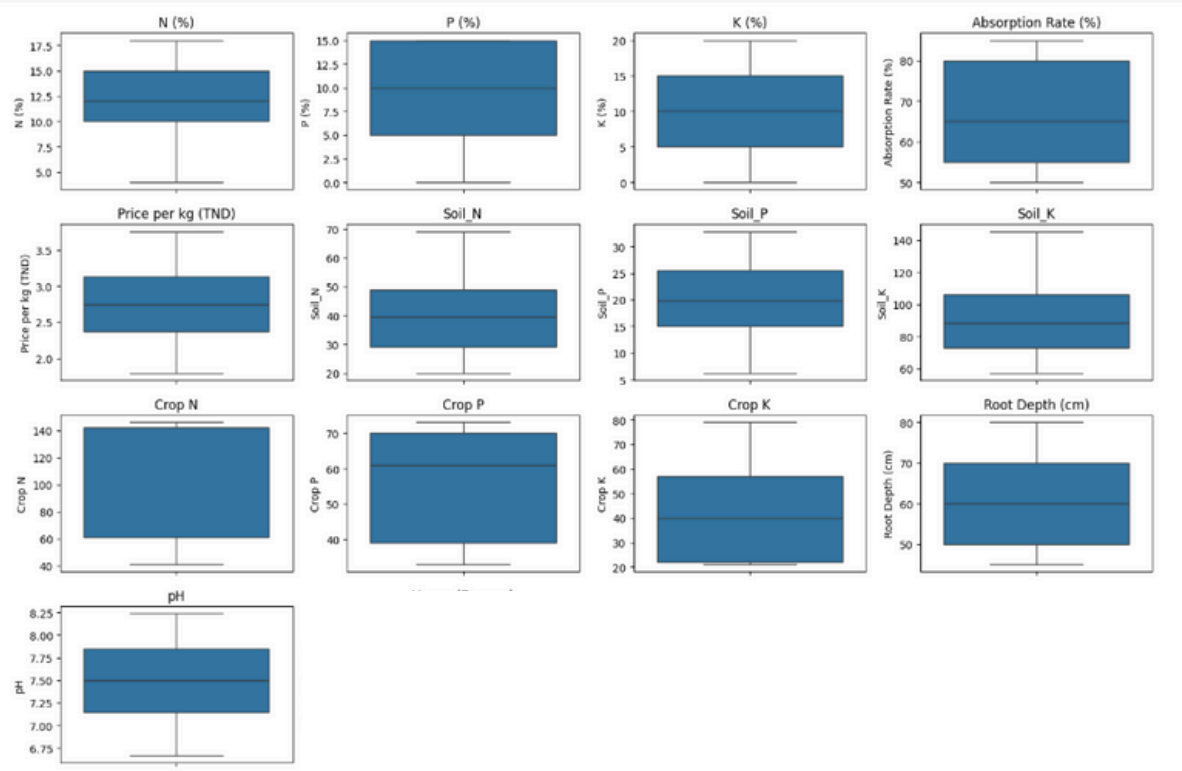
FOR SOIL NUTRIENT VARIABLES (SOIL_N, SOIL_P, SOIL_K), MISSING VALUES WERE IMPUTED BASED ON THE MEDIAN VALUE WITHIN EACH CORRESPONDING REGION. IF NO REGIONAL DATA WAS AVAILABLE FOR A GIVEN NUTRIENT, THE OVERALL DATASET MEDIAN WAS USED AS A FALLBACK. ROWS MISSING THE TARGET VARIABLE (USAGE (DOSAGE)) WERE REMOVED FROM THE DATASET TO AVOID INCONSISTENCIES DURING MODEL TRAINING.
THE REMAINING MISSING VALUES ACROSS THE DATASET WERE THEN REVIEWED TO ENSURE MINIMAL DATA LOSS AND HIGH INTEGRITY.

3. SKEWNESS REDUCTION

FEATURES EXHIBITING SIGNIFICANT SKEWNESS (ABSOLUTE SKEWNESS GREATER THAN 1.5) WERE IDENTIFIED AND SUBJECTED TO LOGARITHMIC TRANSFORMATION USING A LOG1P FUNCTION. THIS STEP AIMED TO NORMALIZE THE DISTRIBUTIONS AND IMPROVE MODEL PERFORMANCE, ESPECIALLY FOR LINEAR MODELS SENSITIVE TO FEATURE DISTRIBUTIONS.

4. OUTLIER DETECTION AND TREATMENT

BOXPLOTS WERE GENERATED FOR ALL NUMERIC FEATURES TO VISUALIZE OUTLIERS. EXTREME VALUES WERE THEN TREATED BY CAPPING EACH FEATURE AT THE 5TH AND 95TH PERCENTILES, REDUCING THE INFLUENCE OF EXTREME OBSERVATIONS WITHOUT REMOVING VALUABLE DATA POINTS.



5. FEATURE ENGINEERING

SEVERAL ADDITIONAL FEATURES WERE ENGINEERED TO ENRICH THE DATASET AND BETTER CAPTURE AGRONOMIC RELATIONSHIPS:

- **NPK RATIO**: MEASURES HOW NITROGEN-HEAVY A FERTILIZER FORMULATION IS RELATIVE TO PHOSPHORUS AND POTASSIUM.
- **NUTRIENT GAPS (N_GAP, P_GAP, K_GAP):** REPRESENTS THE DIFFERENCE BETWEEN CROP NUTRIENT NEEDS AND THE CORRESPONDING FERTILIZER CONTENTS.
- **SOIL NUTRIENT SUFFICIENCY (SOIL_N_SUFF, SOIL_P_SUFF, SOIL_K_SUFF):** ESTIMATES HOW MUCH THE SOIL ALREADY CONTRIBUTES TOWARD MEETING THE CROP'S NUTRIENT REQUIREMENTS.
- **EFFECTIVE NUTRIENT CONTENT (EFFECTIVE_N, EFFECTIVE_P, EFFECTIVE_K):** CALCULATES THE AMOUNT OF FERTILIZER NUTRIENTS THAT ARE ACTUALLY AVAILABLE TO THE PLANT AFTER CONSIDERING THE FERTILIZER'S ABSORPTION RATE.

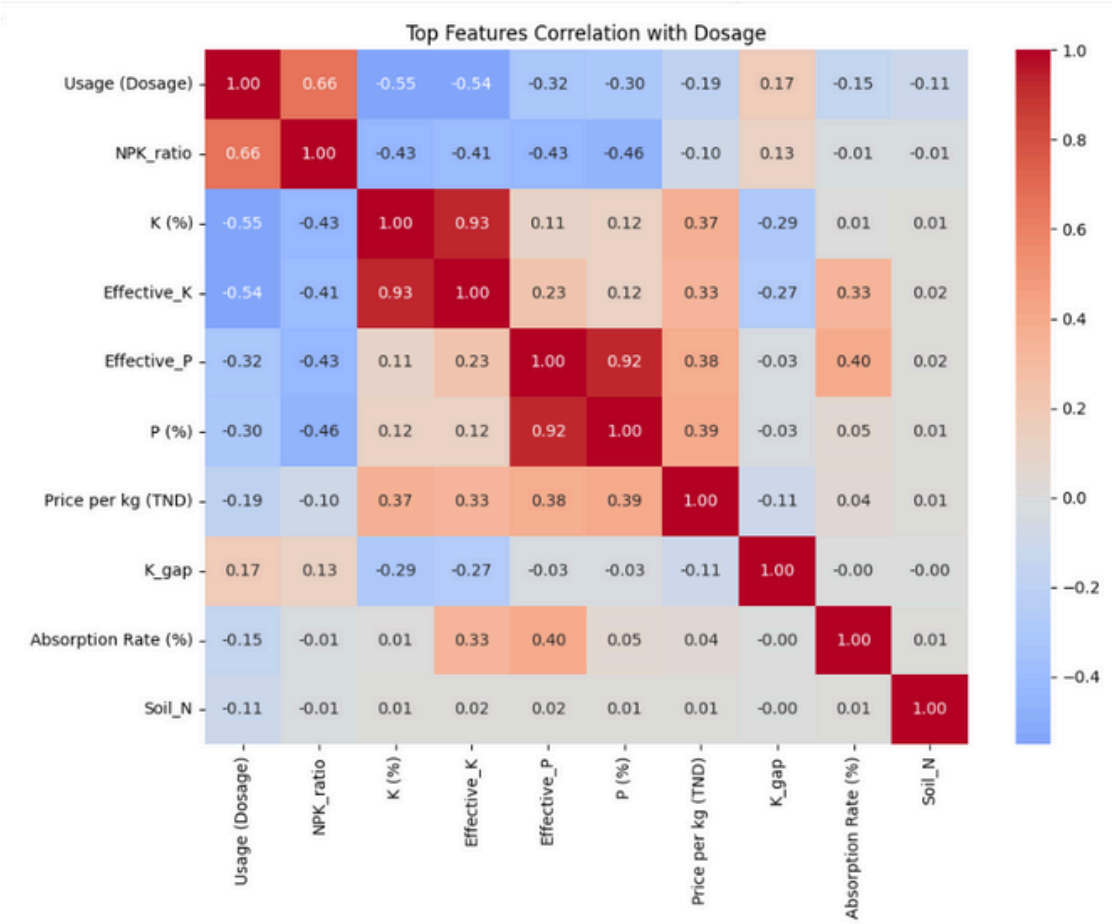
EACH ENGINEERED FEATURE WAS CAREFULLY MOTIVATED BY DOMAIN KNOWLEDGE IN AGRONOMY AND FERTILIZER MANAGEMENT.

6. CORRELATION ANALYSIS

TO UNDERSTAND RELATIONSHIPS WITHIN THE DATA:

- A FULL CORRELATION HEATMAP WAS GENERATED FOR ALL NUMERICAL VARIABLES.
- TOP FEATURES MOST CORRELATED WITH THE TARGET (USAGE (DOSAGE)) WERE IDENTIFIED.
- KEY FINDINGS INCLUDED:
  - FERTILIZER NPK RATIO HAD A **STRONG POSITIVE CORRELATION** WITH DOSAGE.
  - HIGHER POTASSIUM CONTENT AND EFFECTIVE POTASSIUM AVAILABILITY WERE **STRONGLY ASSOCIATED** WITH LOWER DOSAGE REQUIREMENTS.
  - SOIL NUTRIENT RICHNESS AND FERTILIZER EFFICIENCY METRICS SHOWED **MODERATE TO WEAK NEGATIVE** CORRELATIONS WITH DOSAGE.

ADDITIONALLY, POTENTIAL ISSUES OF MULTICOLLINEARITY WERE DETECTED, PARTICULARLY BETWEEN ORIGINAL NUTRIENT PERCENTAGES AND THEIR CORRESPONDING "EFFECTIVE" VALUES (E.G., K (%) AND EFFECTIVE_K), PROMPTING CONSIDERATIONS FOR DIMENSIONALITY REDUCTION LATER.



7. CATEGORICAL ENCODING

CATEGORICAL FEATURES SUCH AS FERTILIZER NAMES, TYPES, REGIONS, STATES, CROP TYPES, AND APPLICATION METHODS WERE ENCODED NUMERICALLY USING LABEL ENCODING TECHNIQUES. ENCODERS WERE STORED FOR FUTURE DECODING OR INTERPRETABILITY NEEDS.

8. FINAL DATA CLEANING AND EXPORT

A FINAL CHECK FOR MISSING VALUES WAS PERFORMED, CONFIRMING THE READINESS OF THE DATASET. THE FULLY PREPROCESSED DATASET WAS SAVED TO A CSV FILE FOR REUSE AND ARCHIVAL PURPOSES.
NUMERIC AND CATEGORICAL COLUMNS WERE CLEARLY SEPARATED TO FACILITATE FURTHER PROCESSING DURING MODEL DEVELOPMENT.

9. TRAIN-TEST SPLIT

THE DATASET WAS THEN DIVIDED INTO TRAINING AND TESTING SETS, USING AN 80/20 SPLIT.
THE FEATURE SET (X) EXCLUDED THE TARGET VARIABLE (USAGE (DOSAGE)) AND, OPTIONALLY, THE FERTILIZER NAME TO PREVENT DATA LEAKAGE.
THE CORRESPONDING TARGET (Y) CONTAINED ONLY THE DOSAGE VALUES TO BE PREDICTED.
FINALLY, THE ORDER OF FEATURES USED IN TRAINING WAS RECORDED, ENSURING CONSISTENCY FOR FUTURE MODELING STEPS AND DEPLOYMENT.

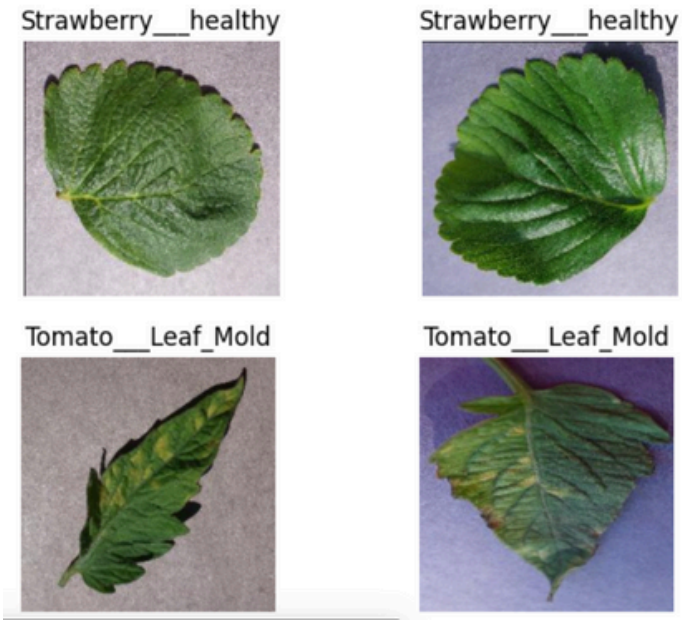
BO2

BUSINESS OBJECTIVE (BO2) :
PLANT DISEASE DETECTION

1-Data Overview

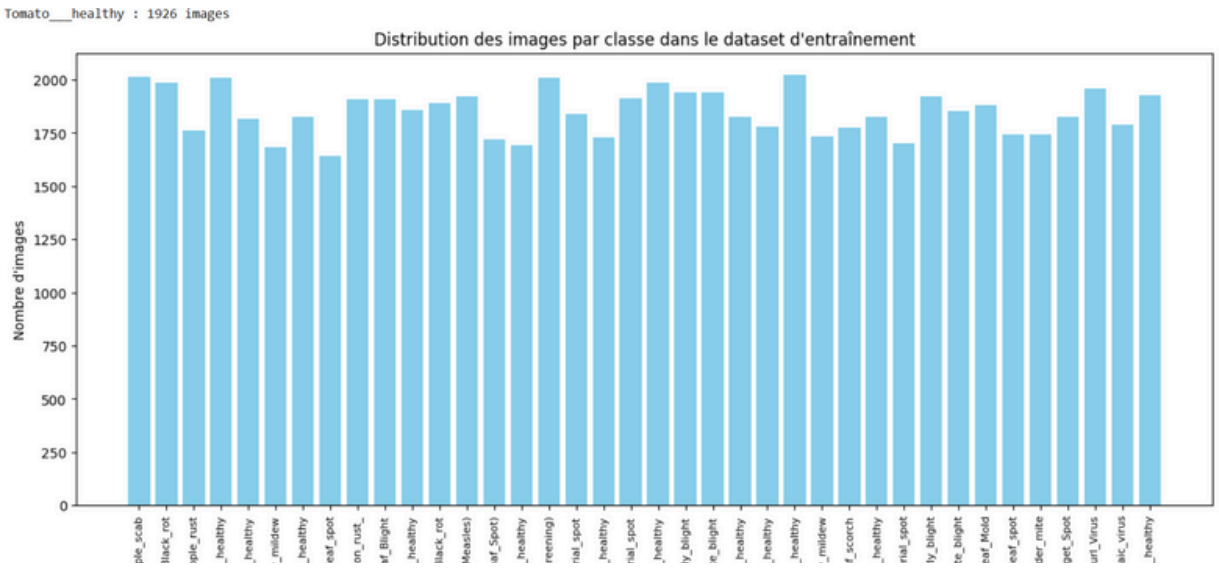


Data upload and visualization of a few images



The dataset was uploaded into the working environment and organized into directories based on plant species and disease categories. To better understand the data, a few sample images from different classes were visualized. This initial exploration helped confirm the quality of the images and observe the variations in plant appearance between healthy and diseased conditions.

2-Checking the Distribution of the data



The dataset is well-balanced, with an approximately equal number of images for each class. This balance is important to prevent the model from being biased toward any particular class, ensuring fair and accurate predictions across all plant disease categories.

3-Data Preparation

Image Size and Batch Size:

For this project, all images were resized to 150x150 pixels to ensure consistency and optimize training speed. The batch size was set to 32, balancing memory usage and gradient updates.

Training Data Augmentation:

To improve the model's generalization and prevent overfitting, a variety of data augmentation techniques were applied using the ImageDataGenerator class. These included:

- **Rescaling:** Pixel values normalized to the range [0, 1] by dividing by 255.
- **Rotation:** Random rotations up to 20 degrees.
- **Shifting:** Horizontal and vertical shifts up to 20% of the image dimensions.
- **Shearing:** Shear transformations with a range of 0.2.
- **Zooming:** Random zoom within a range of 20%.
- Horizontal Flipping: Random flips along the horizontal axis.
- **Fill Mode:** New pixels generated after transformations were filled using the nearest strategy.

Validation and Test Data Processing:

For the validation and test sets, only rescaling was applied to maintain the integrity of the evaluation without introducing augmented variations.

Data Generators:

Three separate data generators were created:

- **Training Generator:** Applied augmentation and loaded images from the training directory.
- **Validation Generator:** Loaded validation images without augmentation for evaluation during training.
- **Test Generator:** Loaded test images without augmentation, and shuffle was disabled to preserve the order for accurate performance analysis.

Class Verification:
After setting up the generators, the class indices were printed to verify correct label mapping.

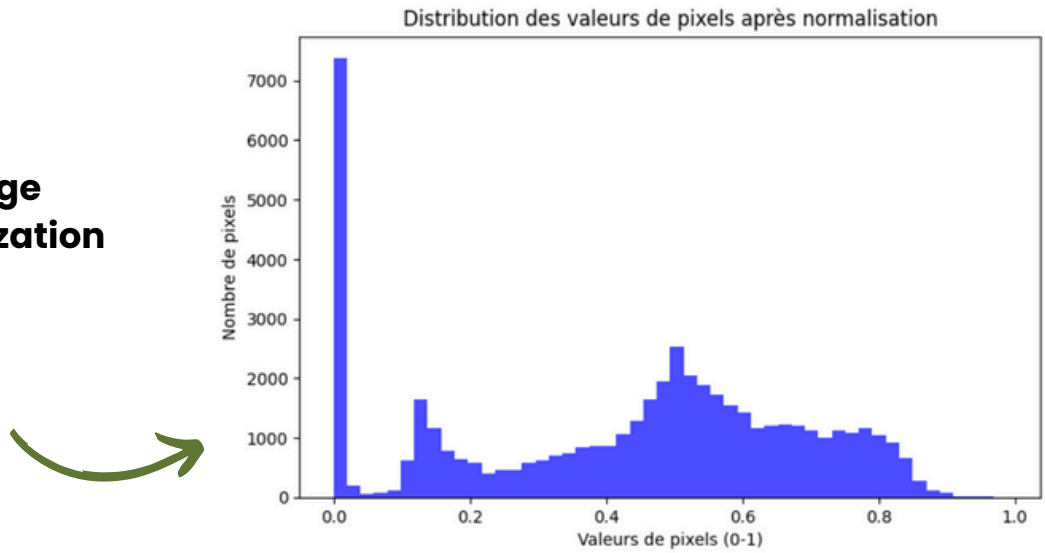
In this project, we designed an image classification model based on a transfer learning approach using MobileNetV2 as the backbone. First, we activated mixed precision training to accelerate computations on compatible GPUs.

We initialized a MobileNetV2 model pre-trained on ImageNet, set its layers to non-trainable to preserve learned features, and built a custom Sequential model by adding a GlobalAveragePooling2D layer, followed by a Dense layer with 256 neurons (ReLU activation) and a Dropout layer to prevent overfitting. Finally, a Dense output layer with 38 neurons and softmax activation was added to predict the 38 plant disease classes.

The model was compiled with the Adam optimizer (learning rate 0.0001) and the categorical cross-entropy loss function, while tracking accuracy, precision, and recall as metrics. To improve model generalization, we used an ImageDataGenerator to apply various data augmentation techniques, such as random rotations, shifts, shearing, zooms, brightness changes, and flips. Images were resized to 128x128 pixels to balance training speed and performance.

For training, we split the data into training and validation subsets (80/20 split) and generated batches with a size of 32. We also incorporated important callbacks such as EarlyStopping (to halt training when no improvement is observed) and ReduceLROnPlateau (to reduce the learning rate on performance plateaus), helping to optimize convergence. Overall, this pipeline enables efficient and robust model training for multi-class plant disease detection.

4-Verification of image quality after normalization



Interpretation of the Pixel Value Histogram

Significant peak at 0.0 (black)

A large portion of the image is very dark (many black or near-black pixels). Possible causes: black background, strong shadows, or transformation artifacts.

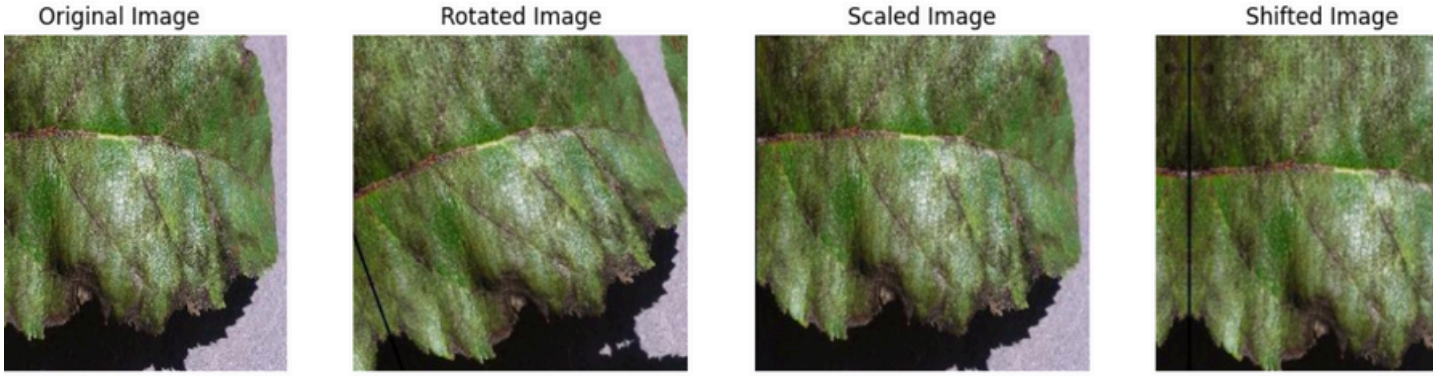
Pixel distribution between 0.1 and 0.8

The pixels are widely distributed, indicating visible details and good contrast. Several secondary peaks suggest important variations in brightness (dark, medium, and bright areas).

Low presence of pixels near 1.0 (pure white)

The image does not have many very bright areas (no overexposure). This can be beneficial if your dataset has a well-balanced brightness distribution.

5-Validation of Image Quality After Augmentation



The transformations have been effectively applied, and the quality of the images has been preserved. This demonstrates that the augmentation techniques used are suitable for enhancing the dataset without degrading the visual information.

BO3

DEVELOP A PREDICTIVE MODEL TO ESTIMATE AGRICULTURAL YIELD AND CROP MATURITY TIME.

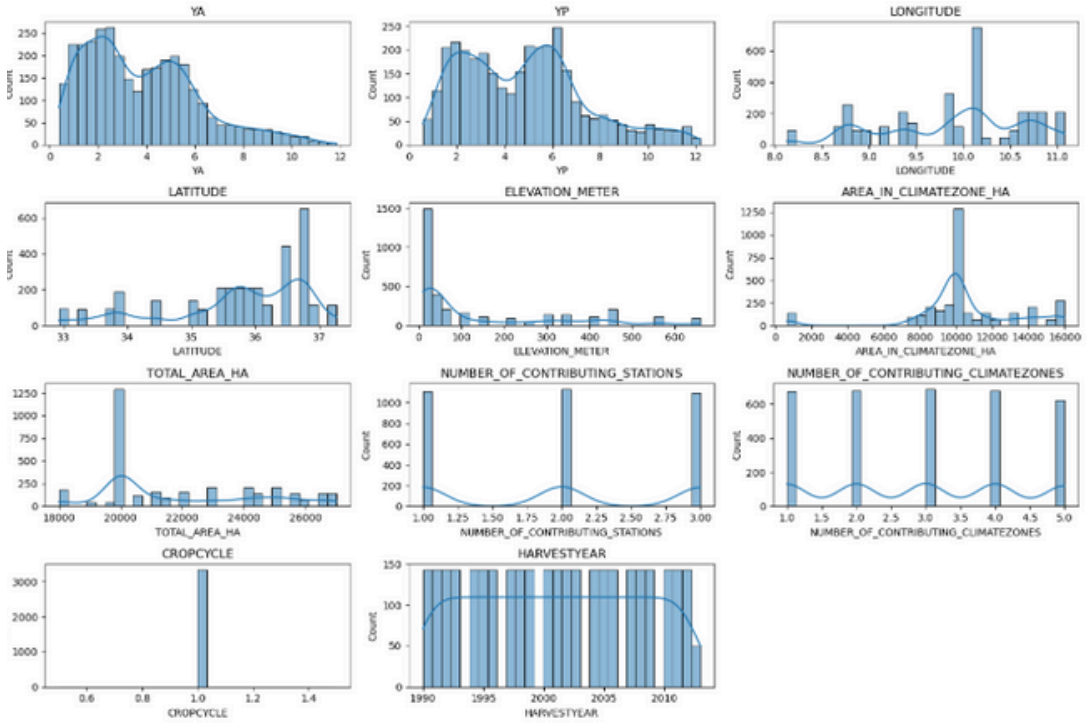
I -DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS (EDA)

1-DATA OVERVIEW

THE DATASET CONTAINS 3,339 ROWS AND 22 FEATURES.
FEATURES INCLUDE:

- STATION-RELATED ATTRIBUTES: STATIONNAME, CLIMATEZONE, COUNTRY, SOURCE_TYPE, MANAGEMENT, AND IRRIGATION_TYPE.
- CROP-RELATED ATTRIBUTES: CROP, YA (YIELD), YP (PRODUCTION), CROPCYCLE, SOWINGDATE, CULTIVARMATURITY, AND HARVESTYEAR.
- GEOGRAPHICAL ATTRIBUTES: LONGITUDE, LATITUDE, ELEVATION_METER, AND AREA_IN_CLIMATEZONE_HA.
- LAND AND CLIMATE DATA: TOTAL_AREA_HA, NUMBER_OF_CONTRIBUTING_STATIONS, NUMBER_OF_CONTRIBUTING_CLIMATEZONES, AND DATA_TYPE.

Categorical features account for about 45% of the data, such as crop type, climate zone, and irrigation type, while the remaining 55% are **numerical features** like yield, area, and elevation.



----Distribution of Numerical Features----

2-DATA INFORMATIONS

In this section, we aim to better understand the structure and content of our dataset before applying any transformation or modeling.

The dataset used consists of agricultural observations regarding wheat cultivation in Tunisia, specifically in the Beja region.

Each record describes an agricultural campaign (harvest year) with various characteristics such as:

- **STATIONNAME**: the name of the weather station (e.g., Beja),
- **CLIMATEZONE**: the associated climate zone (e.g., Humid),
- **CROP**: the type of crop (e.g., Wheat),
- **COUNTRY**: the country (Tunisia),
- **YA**: observed agricultural yield (in tons per hectare),
- **YP**: potential yield simulated (in tons per hectare),
- **Source_Type**: the data source (e.g., Station),
- **LONGITUDE, LATITUDE, ELEVATION_METER**: geographic coordinates and altitude of the station,
- **NUMBER_OF_CONTRIBUTING_STATIONS**and **NUMBER_OF_CONTRIBUTING_CLIMATEZONES**: number of stations and climate zones contributing to the observation,
- **Data_Type, MANAGEMENT, CROPCYCLE, HARVESTYEAR, SOWINGDATE, CULTIVARMATURITY, USEDMODEL, Irrigation_Type**: additional information about crop management, sowing date, variety maturity, and irrigation.

The dataset is sourced from a compilation of simulated and measured data, based on years of research and analysis conducted by Tunisian researchers. It is often used to analyze yield gaps between actual agricultural practices and theoretical potential.

We would like to sincerely thank the researchers for their valuable contributions to agricultural science and for making this data available for further analysis and innovation.

```
0   STATIONNAME
1   CLIMATEZONE
2   CROP
3   COUNTRY
4   YA
5   YP
6   Source_Type
7   LONGITUDE
8   LATITUDE
9   ELEVATION_METER
10  AREA_IN_CLIMATEZONE_HA
11  TOTAL_AREA_HA
12  NUMBER_OF_CONTRIBUTING_STATIONS
13  NUMBER_OF_CONTRIBUTING_CLIMATEZONES
14  Data_Type
15  MANAGEMENT
16  CROPCYCLE
17  HARVESTYEAR
18  SOWINGDATE
19  CULTIVARMATURITY
20  USEDMODEL
21  Irrigation_Type
```

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- **LONGITUDE, LATITUDE, ELEVATION_METER**: geographic coordinates and altitude of the station,
- **NUMBER_OF_CONTRIBUTING_STATIONS**and **NUMBER_OF_CONTRIBUTING_CLIMATEZONES**: number of stations and climate zones contributing to the observation,
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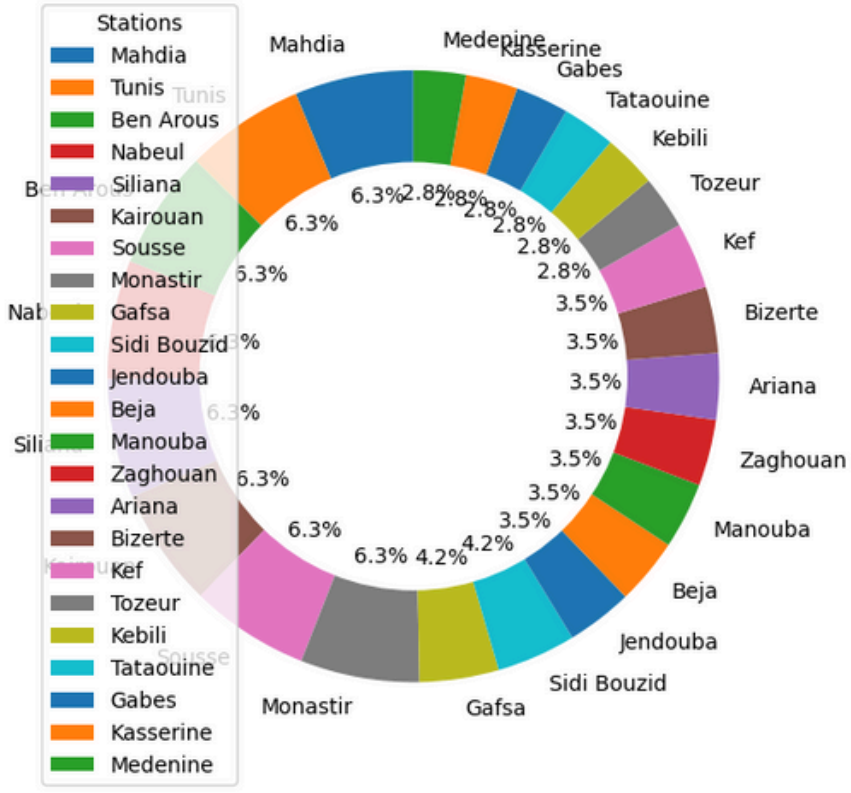
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```
0   STATIONNAME
1   CLIMATEZONE
2   CROP
3   COUNTRY
4   YA
5   YP
6   Source_Type
7   LONGITUDE
8   LATITUDE
9   ELEVATION_METER
10  AREA_IN_CLIMATEZONE_HA
11  TOTAL_AREA_HA
12  NUMBER_OF_CONTRIBUTING_STATIONS
13  NUMBER_OF_CONTRIBUTING_CLIMATEZONES
14  Data_Type
15  MANAGEMENT
16  CROPCYCLE
17  HARVESTYEAR
18  SOWINGDATE
19  CULTIVARMATURITY
20  USEDMODEL
21  Irrigation_Type
```

3-DETAILED DESCRIPTION OF KEY VARIABLES

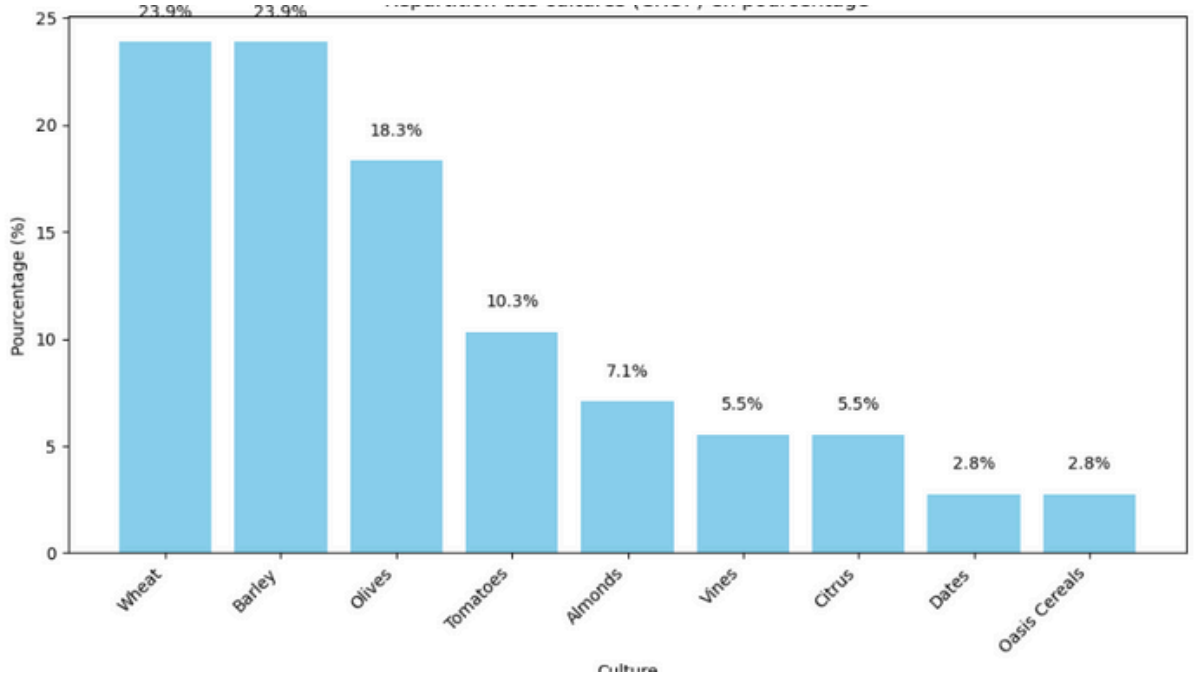
1. STATIONNAME

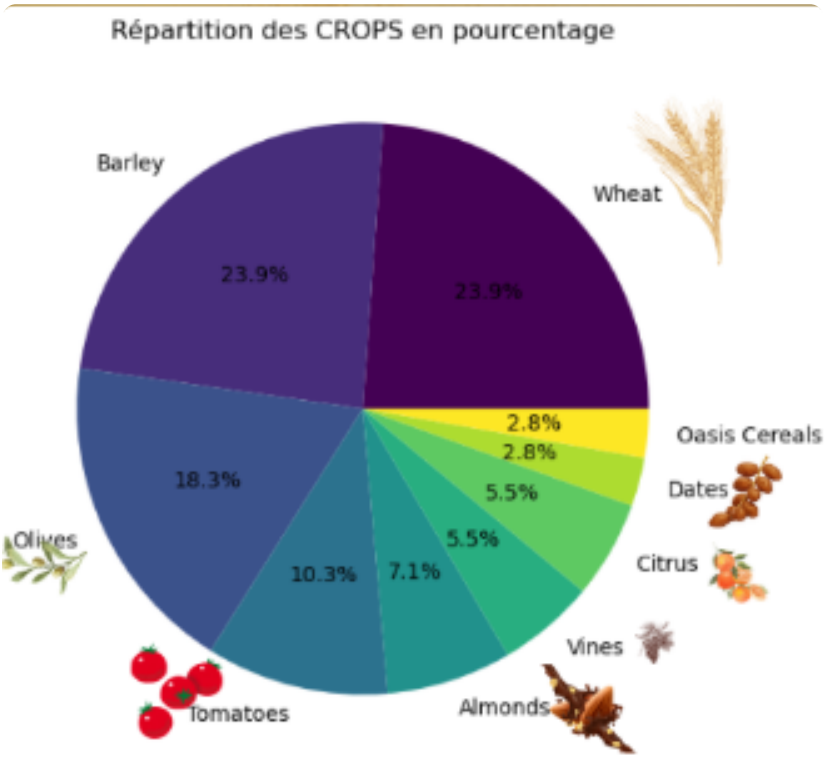
STATIONNAME column in the dataset refers to the different meteorological and agricultural data collection stations across Tunisia. These stations are distributed throughout the country, representing various regions with distinct climatic conditions, such as coastal, inland, and desert zones. Each station is positioned in a specific geographical location, providing valuable insights into regional variations in climate, soil conditions, and agricultural practices. By analyzing the distribution of stations, we can observe how different areas of Tunisia contribute to the dataset, with each station offering data relevant to the crops grown and the environmental factors influencing agricultural productivity. For instance, stations located in coastal regions might experience more temperate climates, while those in inland or desert areas could be characterized by harsher, arid conditions. Understanding the geographical spread of these stations is crucial for assessing regional agricultural yield, maturity times, and other related factors, as the local climate plays a significant role in crop growth. This distribution also allows for targeted recommendations and predictions based on specific regional needs and challenges, thereby aiding in more precise agricultural planning and management.



1. CROP

The **CROP** column in the dataset includes various types of crops cultivated across Tunisia, each of which has specific agricultural requirements, growth cycles, and climatic preferences. These crops include staple grains such as wheat and barley, which are widely grown in the more temperate regions of Tunisia, as well as olives, which thrive in the Mediterranean climate and are a major agricultural product of the country. Tomatoes and citrus fruits are commonly grown in the coastal areas, benefiting from Tunisia's mild winters and warm summers. Almonds, which require a dry, mild climate, are typically cultivated in areas with less rainfall. Oasis cereals are grown in Tunisia's desert regions, where the irrigation systems are essential for crop survival. Vineyards produce grapes in regions that have a balance of warmth and moisture, contributing to Tunisia's wine industry. Dates are primarily cultivated in the southern oasis regions, where the hot and dry climate is ideal for date palms. Each of these crops plays a vital role in Tunisia's agricultural economy and is sensitive to local climatic conditions, such as temperature, rainfall, and soil type. By analyzing the distribution and performance of these crops, we can gain a deeper understanding of their growth patterns, maturation times, and yield potential in different regions of Tunisia. This can help predict crop productivity and suggest the most suitable agricultural practices for each type of crop based on local environmental conditions.

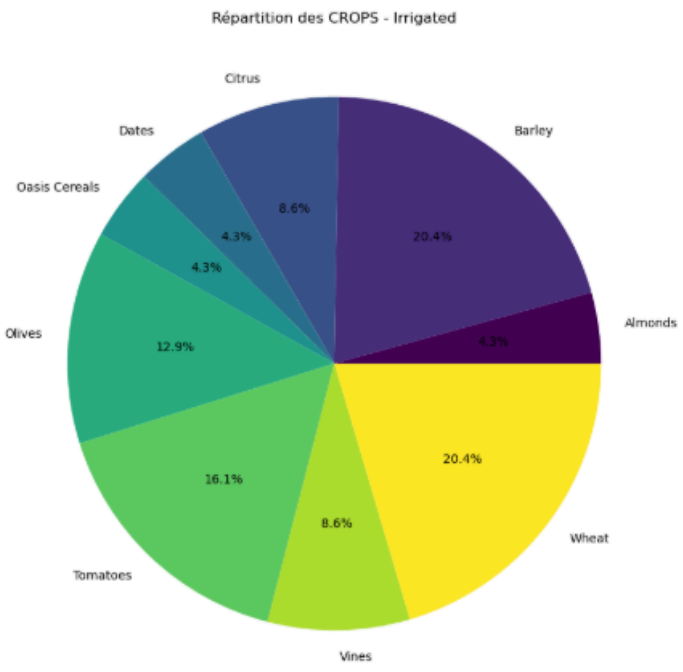


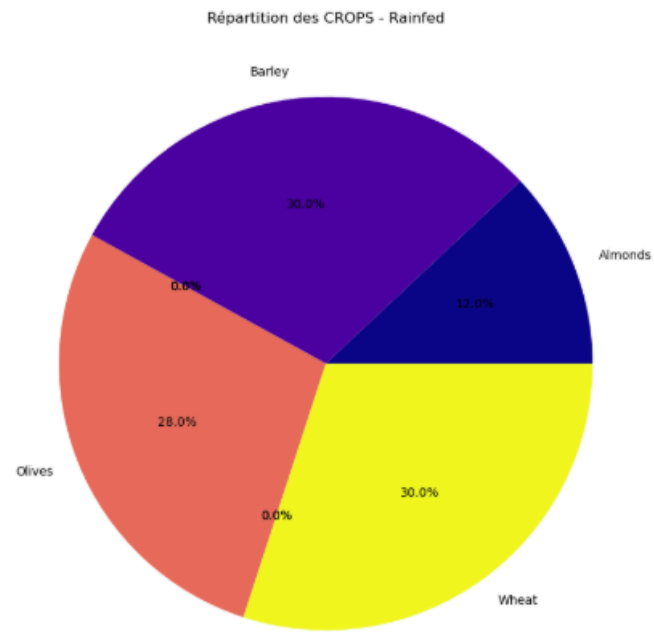


Irrigation Types

The dataset includes two types of irrigation systems: Irrigated and Rainfed.

- Irrigated crops are those that receive water through artificial means, typically using irrigation systems such as drip or sprinkler irrigation. This method is used in regions where natural rainfall is insufficient to meet the water needs of the crops.
- Rainfed crops, on the other hand, rely solely on natural rainfall to provide water. These crops are grown in regions where precipitation is adequate to support their growth without the need for additional irrigation.





4. CORRELATION ANALYSIS)

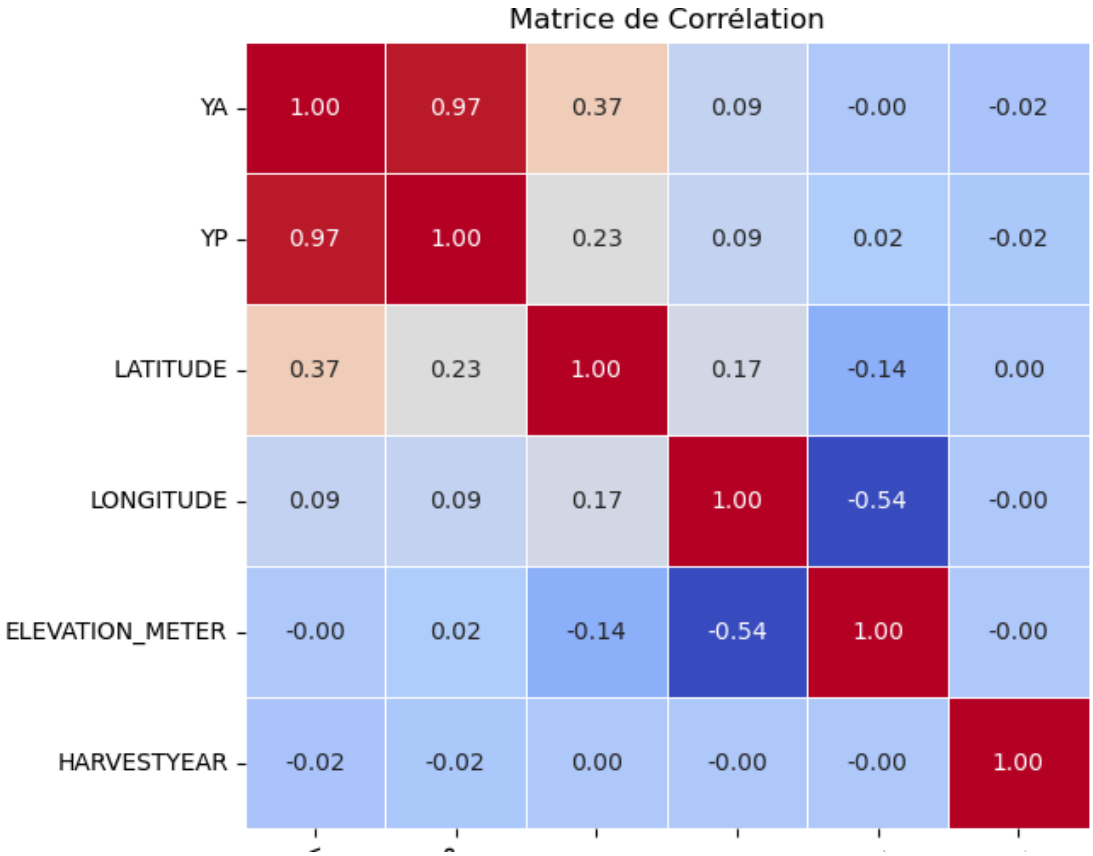
The correlation matrix displayed above summarizes the relationships between key numerical variables in the dataset: YA (Actual Yield), YP (Potential Yield), ELEVATION_METER (elevation of the station), and LATITUDE.

The color gradient from blue to red indicates the strength and direction of the correlation, with red representing strong positive correlations and blue indicating negative or weak correlations.

Key observations include:

- There is a very strong positive correlation (0.97) between YA and YP. This indicates that the actual yields and the potential yields are highly related: when the potential yield increases, the actual yield tends to increase as well.
- LATITUDE shows a moderate positive correlation with YA (0.37) and YP (0.23). This suggests that the geographic location (north-south positioning) may have a slight influence on yield outcomes.
- ELEVATION_METER has very weak correlations with both yield variables (YA and YP), and even shows a slightly negative correlation with LATITUDE (-0.14), indicating that elevation does not play a significant role in explaining yield variations in this dataset.

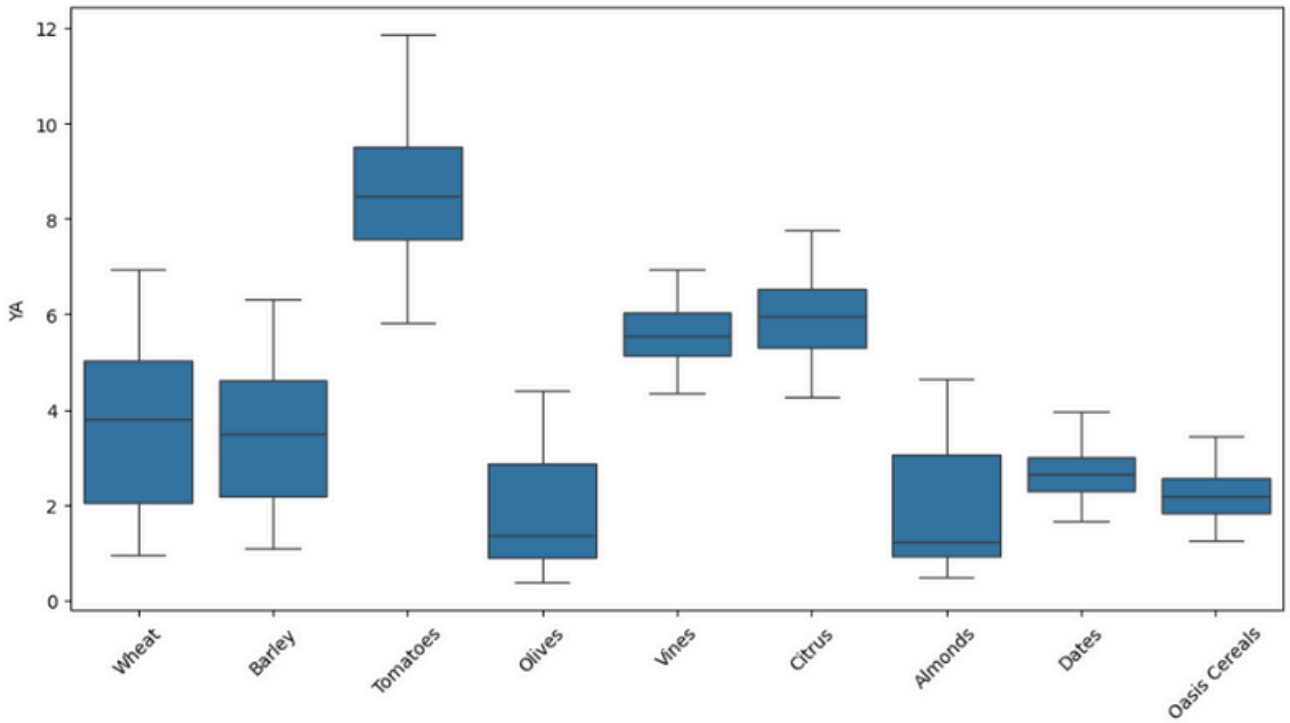
Overall, the correlation analysis suggests that yield variability is much more strongly associated with the potential yield and slightly with latitude, while elevation seems to have minimal influence.



5.SUMMARY OF CROP DATA

The table below presents the mean and standard deviation values for each crop type, providing an overview of the central tendency and the variability of the data across the different crops.

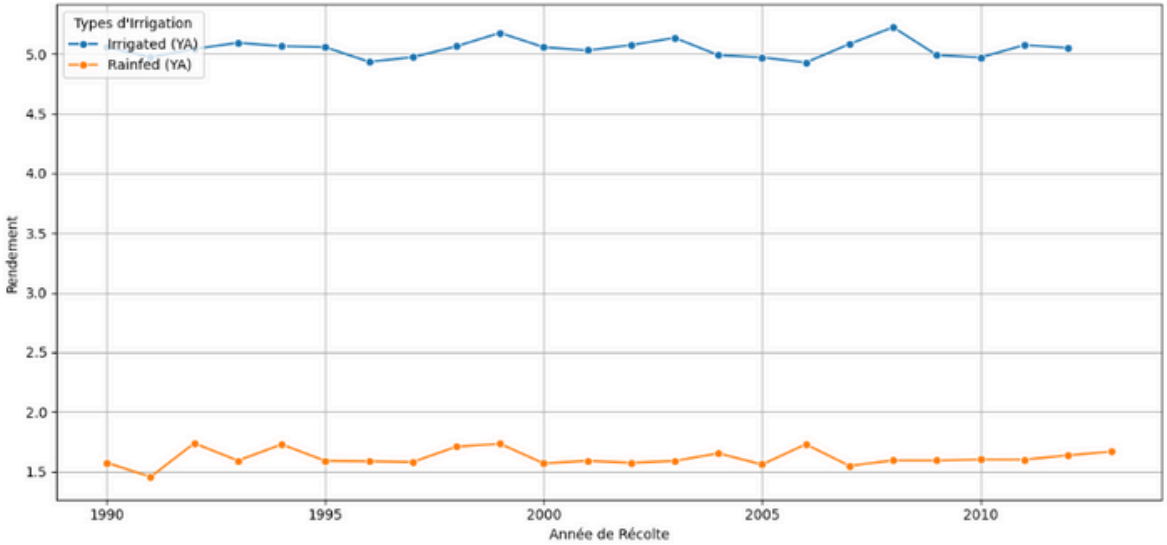
- Almonds have a mean value of 1.89 with a standard deviation of 1.21, indicating a relatively low average and moderate variability in the data.
- Barley shows a mean of 3.45 and a standard deviation of 1.36, suggesting a higher average with noticeable variability in the dataset.
- Citrus crops have a mean of 5.95 and a lower standard deviation of 0.83, indicating more consistency around the average for this crop type.
- Dates have a mean of 2.67 with a standard deviation of 0.53, reflecting a moderate average and relatively low variability.
- Oasis Cereals show a mean of 2.20 with a standard deviation of 0.48, implying a lower average and minimal fluctuation in the values.
- Olives also have a mean of 1.88 with a standard deviation of 1.13, which is similar to almonds in terms of both central tendency and variability.
- Tomatoes have the highest mean of 8.57 with a standard deviation of 1.29, indicating a high average with significant variation.
- Vines show a mean of 5.57 and a standard deviation of 0.61, representing a higher average with relatively low variability compared to other crops.
- Wheat has a mean of 3.59 with a standard deviation of 1.62, showing a moderate average and relatively high variability in the data.



6.IRRIGATION TYPE VALUES FOR CROPS

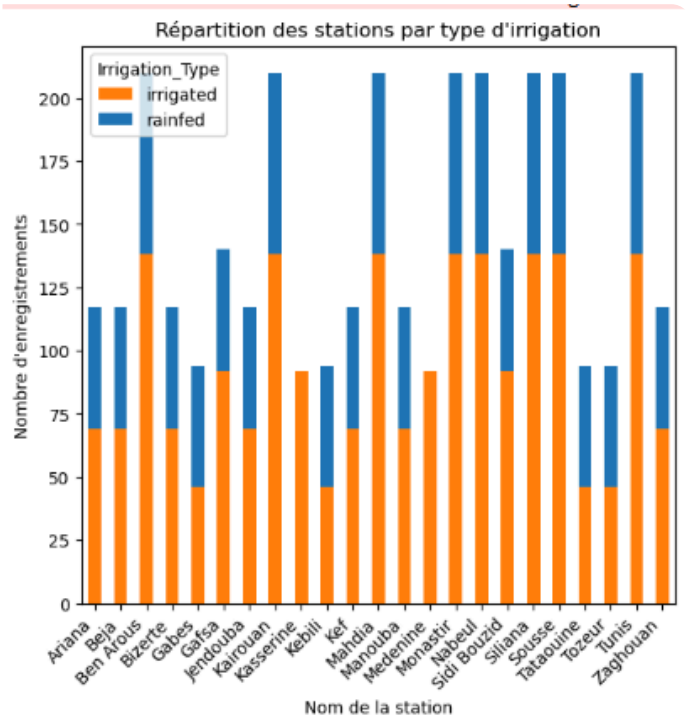
The table below summarizes the distribution of crops under two irrigation types: Irrigated and Rainfed, showing the values for each crop's cultivation under these two irrigation systems.

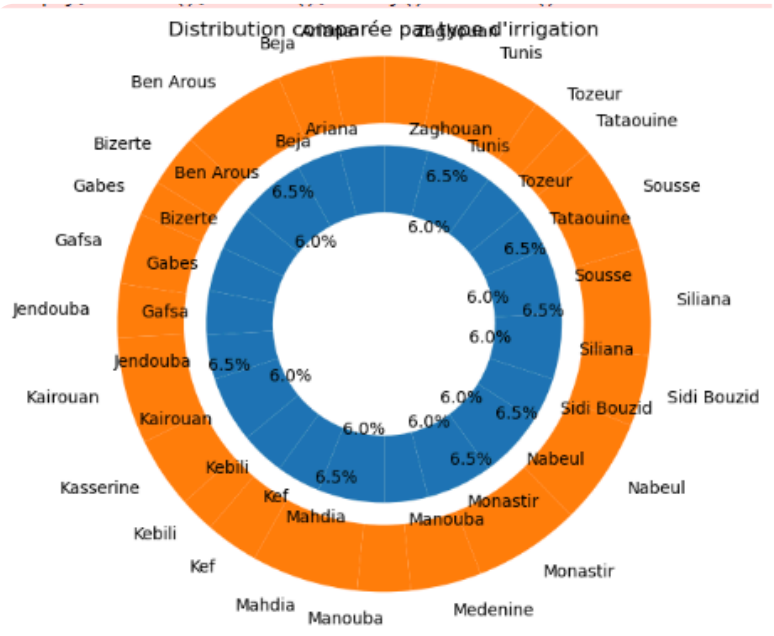
- Almonds under irrigation have a value of 4.30, while the rainfed value is significantly higher at 12.00, suggesting that almonds are more commonly cultivated with natural rainfall.
- Barley shows a similar trend, with 20.43 under irrigation and 30.00 under rainfed conditions, indicating that barley can grow in both irrigated and rainfed environments, but is more dependent on natural rainfall.
- Citrus is almost exclusively cultivated through irrigation, with a value of 8.60 under irrigation and 0 under rainfed conditions, suggesting that citrus crops require irrigation for successful cultivation.
- Dates also show the same pattern as citrus, being cultivated entirely through irrigation with a value of 4.30 and 0 under rainfed conditions.
- Oasis Cereals are cultivated under irrigation with a value of 4.30, but have no rainfed values, reflecting their dependence on irrigation in Tunisia's dry regions.
- Olives are cultivated in both irrigation and rainfed systems, with 12.90 for irrigation and 28.00 under rainfed, indicating that olives are more commonly grown with natural rainfall but can also be irrigated in drier regions.
- Tomatoes show a balance, with 16.13 under irrigation but 0 under rainfed, meaning that they depend primarily on irrigation for optimal growth.
- Vines are cultivated almost exclusively through irrigation, with a value of 8.60, and have no cultivation under rainfed conditions, showing a high reliance on irrigation systems.
- Wheat exhibits a similar pattern, with 20.43 under irrigation and 30.00 under rainfed, suggesting that wheat can grow in both systems, but is also frequently rainfed in Tunisia.



7.IRRIGATION TYPE VALUES FOR CROPS

- The provided visualizations illustrate the distribution of agricultural stations in Tunisia by irrigation type. The bar chart on the left shows the number of stations for each governorate, separated into irrigated (orange) and rainfed (blue) categories. It is evident that in most regions, irrigated stations outnumber rainfed ones, with particularly high counts observed in governorates such as Sousse, Sidi Bouzid, and Kairouan. The donut chart on the right complements this by presenting the proportional distribution of stations by governorate and irrigation type. Each segment represents a governorate, with the inner ring reflecting rainfed stations and the outer ring showing irrigated stations. The percentages indicate that the distribution is relatively balanced across regions, with most governorates contributing between 6% and 6.5% of the total stations, regardless of irrigation type. This detailed breakdown highlights both the regional concentration and the overall parity in the deployment of irrigation methods across Tunisia.





II -DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS (EDA)

The data preparation phase is a crucial step in developing a predictive model aimed at estimating agricultural yield and crop maturity time. This phase involves several critical steps to ensure that the dataset is clean, consistent, and suitable for modeling. Each preprocessing step is designed to enhance the quality and interpretability of the data while minimizing biases and information loss.

1. INITIAL EXPLORATION

We commenced our analysis by examining the distribution and skewness of the target variable, Yield (YA), employing various visual inspection techniques. To evaluate the overall shape of the distribution, we utilized a histogram alongside a kernel density estimate (KDE) plot. This combination allowed us to visualize the frequency of yield values and identify any underlying patterns. Additionally, we calculated the numerical skewness coefficient to quantify the degree of asymmetry in the distribution. This coefficient is crucial for understanding whether the yield data is symmetrically distributed or if it exhibits skewness, which could influence the performance of our predictive models.

2. HANDLING MISSING VALUES

A thorough examination of the dataset reveals that there are no missing values across all columns. This completeness is advantageous as it ensures that the analysis can proceed without the need for imputation or removal of incomplete records. The absence of missing values enhances the reliability of the dataset, allowing for more accurate insights and conclusions to be drawn from the data. Furthermore, the presence of complete data at this level assures us that we have collected high-quality data, which is essential for building robust predictive models. This commitment to data quality not only strengthens our analysis but also increases the confidence in the results derived from the dataset.

```
# Identifier les colonnes avec des valeurs manquantes
print(df.isnull().sum())
```

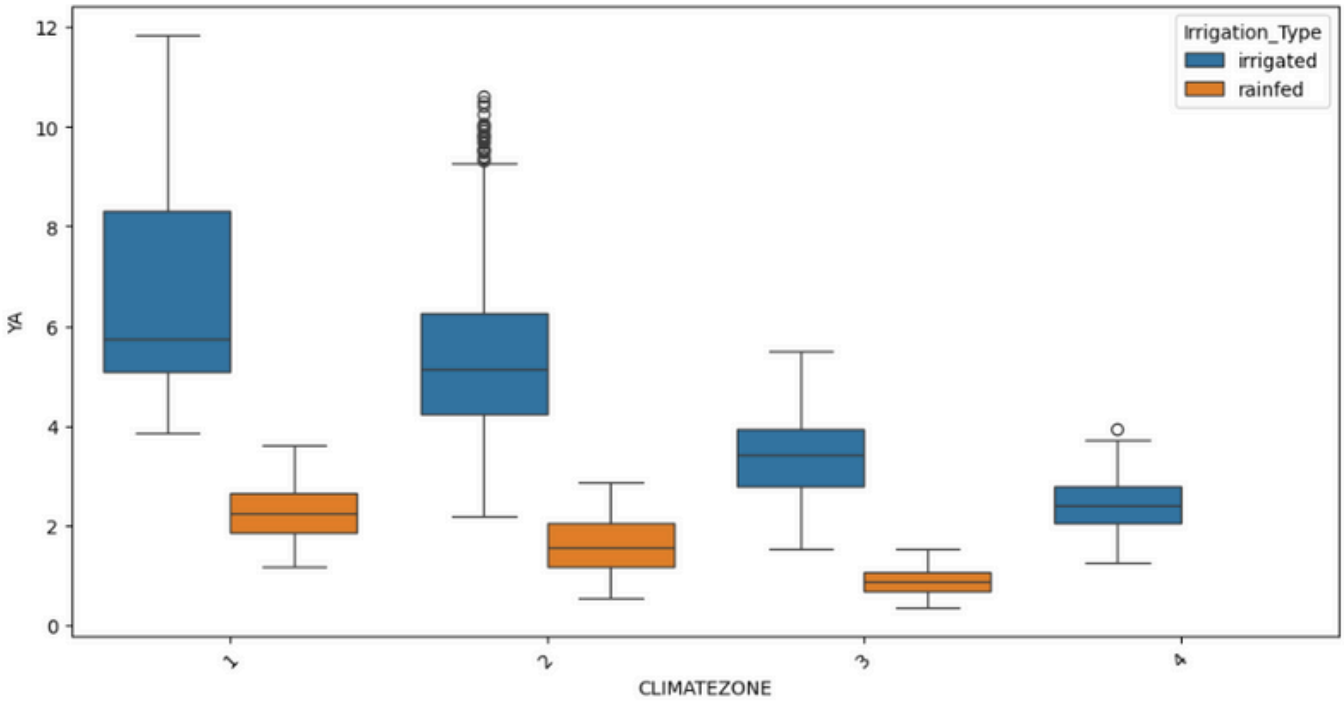
```
STATIONNAME                          0
CLIMATEZONE                          0
CROP                                 0
COUNTRY                              0
YA                                   0
YP                                   0
Source_Type                          0
LONGITUDE                            0
LATITUDE                             0
ELEVATION_METER                      0
AREA_IN_CLIMATEZONE_HA               0
TOTAL_AREA_HA                        0
NUMBER_OF_CONTRIBUTING_STATIONS      0
NUMBER_OF_CONTRIBUTING_CLIMATEZONES  0
Data_Type                            0
MANAGEMENT                           0
CROPCYCLE                            0
HARVESTYEAR                          0
SOWINGDATE                           0
CULTIVARMATURITY                     0
USEDMODEL                            0
Irrigation_Type                      0
```

3. SKEWNESS REDUCTION

Upon initial exploration, it was observed that certain columns exhibit skewness, which may impact the validity of statistical analyses and model performance. Skewness refers to the asymmetry in the distribution of data values. In this dataset, the columns YA and YP may show positive skewness, indicating that there are a few high yield values that could disproportionately influence the mean.
To address this, transformations such as logarithmic or square root transformations could be considered in future analyses to normalize the distributions. However, no transformations were applied at this stage, as the focus is on understanding the data's current state.

4. OUTLIER DETECTION AND TREATMENT

Outlier detection is a critical step in data analysis, as outliers can significantly skew results and lead to misleading interpretations. In this dataset, a preliminary analysis using box plots and scatter plots was conducted to identify potential outliers in the YA and YP columns.
The analysis revealed that there were no significant outliers present in the dataset. This finding is beneficial as it simplifies the analysis process, eliminating the need for outlier treatment methods such as removal or capping. However, it is essential to remain vigilant in future analyses, as outliers can emerge with new data or different analytical methods.

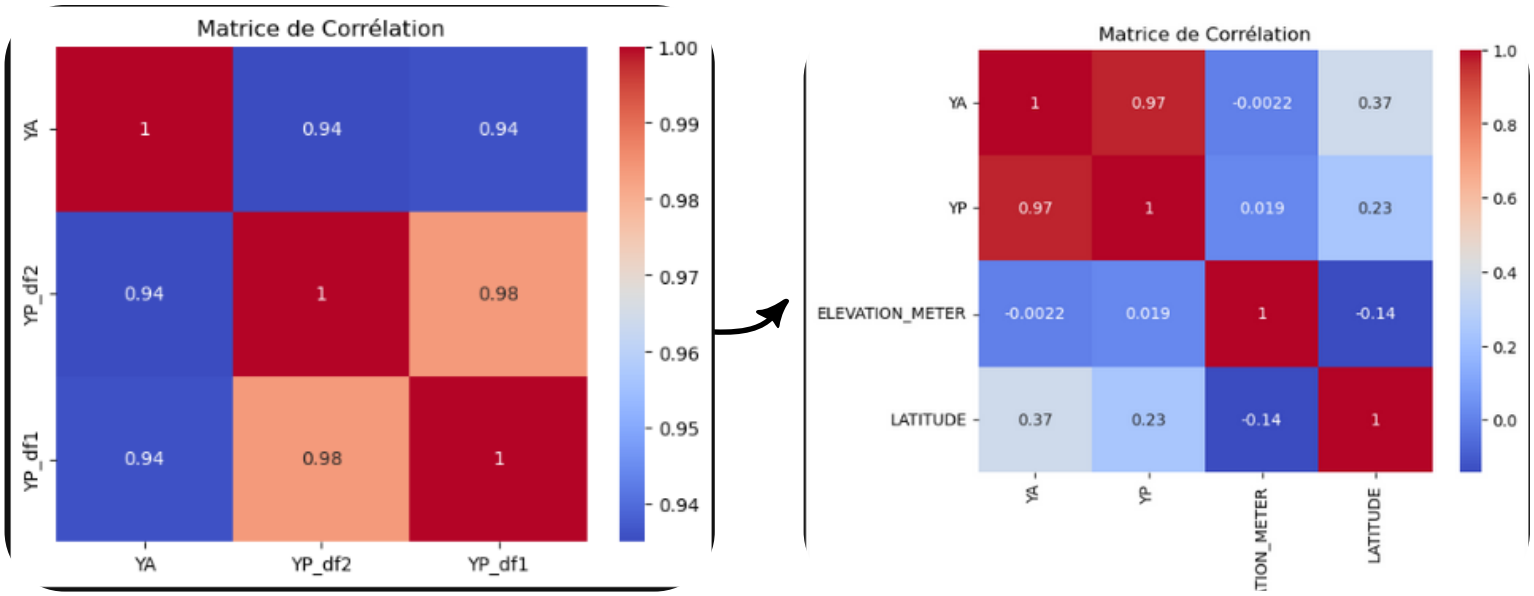


5. CORRELATION ANALYSIS)

Correlation analysis is essential for understanding the relationships between different variables in the dataset. A correlation matrix was generated to assess the strength and direction of relationships between key variables, particularly focusing on YA, YP, and other numerical features.

- Key Findings:
  - The correlation between YA and YP is 0.94, indicating a very strong positive relationship. This suggests that as potential yield increases, actual yield tends to increase as well.
  - LATITUDE and LONGITUDE show a correlation of 0.99, which is expected due to their geographical nature.
  - Moderate correlations were observed between ELEVATION_METER and YA (0.56) and YP (0.55), indicating that elevation may influence crop yields.
  - Other variables such as HARVESTYEAR and SOWING_MONTH showed weak correlations with YA and YP, suggesting that these factors may have less impact on yield outcomes.

The correlation heatmap visually represents these relationships, highlighting areas of interest for further analysis. Understanding these correlations is crucial for developing predictive models and making informed agricultural decisions.



6.FEATURE ENCODING:

I encoded categorical variables into numerical formats where necessary, making them suitable for machine learning algorithms. Specifically, I utilized the LabelEncoder from scikit-learn to convert categorical variables into numerical representations. This encoding process allowed me to transform categorical data into a format that can be processed by machine learning models.

6. NORMALIZATION:

I applied normalization techniques to certain numerical features to ensure that they were on a similar scale, which is important for many modeling techniques. I employed the StandardScaler from scikit-learn to normalize the numerical features, which scales the data to have a mean of 0 and a standard deviation of 1. This normalization step is crucial for preventing features with large ranges from dominating the model, and it ensures that all features contribute equally to the model's predictions.

=> These encoding and normalization steps were essential in preparing the dataset for modeling, as they enabled me to transform categorical data into a numerical format and scale numerical features to a common range. By doing so, I ensured that the dataset was in a suitable format for machine learning algorithms, which is necessary for developing accurate predictive models.

7. FINAL DATA CLEANING AND EXPORT

A final check for missing values was conducted, confirming that the dataset was fully prepared for analysis. This verification process ensured that all columns contained complete data, which is essential for reliable modeling. The fully preprocessed dataset was then saved to a CSV file for future use and archival purposes, allowing for easy access and sharing of the cleaned data. To enhance usability, numeric and categorical columns were distinctly separated, facilitating more efficient processing during subsequent model development.

8. TRAIN-TEST SPLIT

THE DATASET WAS SUBSEQUENTLY DIVIDED INTO TRAINING AND TESTING SETS, EMPLOYING AN 80/20 SPLIT. THIS DIVISION IS CRITICAL FOR EVALUATING THE MODEL'S PERFORMANCE ON UNSEEN DATA, ENSURING THAT THE MODEL CAN GENERALIZE EFFECTIVELY. THE FEATURE SET (X) WAS CONSTRUCTED BY EXCLUDING THE TARGET VARIABLE, YIELD (YA), AND ANY OTHER IDENTIFIERS TO PREVENT DATA LEAKAGE. THE CORRESPONDING TARGET VARIABLE (Y) CONTAINED ONLY THE YIELD VALUES THAT WERE TO BE PREDICTED. ADDITIONALLY, THE ORDER OF FEATURES UTILIZED IN TRAINING WAS METICULOUSLY RECORDED, ENSURING CONSISTENCY FOR FUTURE MODELING STEPS AND DEPLOYMENT. THIS STRUCTURED APPROACH TO DATA PREPARATION IS VITAL FOR MAINTAINING THE INTEGRITY AND REPRODUCIBILITY OF THE MODELING PROCESS.

04

Create crop rotation plans that maintain soil fertility and ensure sustainable farming .

1. Data Collection

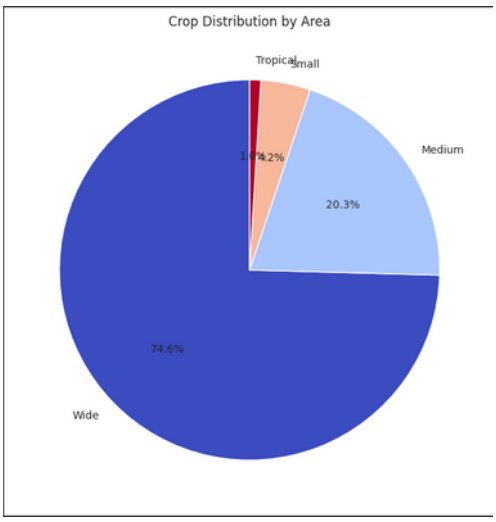
The dataset used for this project contains historical crop data, including crop yields, environmental factors , and soil health indicators. The data spans several years, allowing the LSTM network to learn from temporal patterns and trends. Additionally, a graph structure is built to represent the relationships between crops, soil health, and environmental factors, allowing the GCN to learn spatial dependencies that influence crop performance.

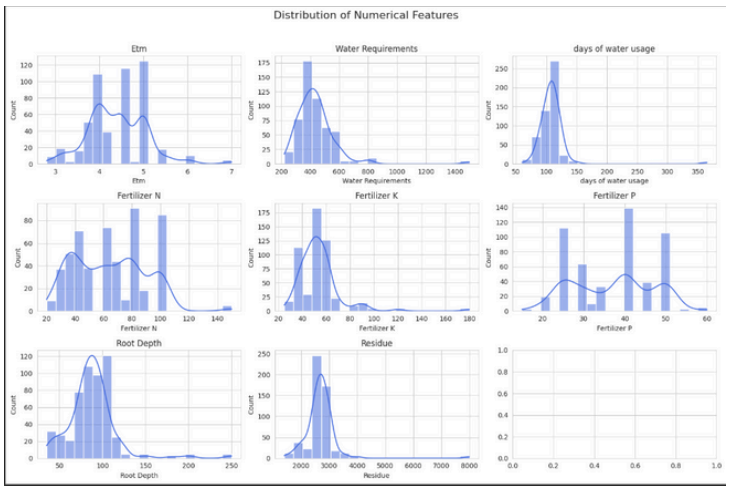
2. 🗎 Dataset Description

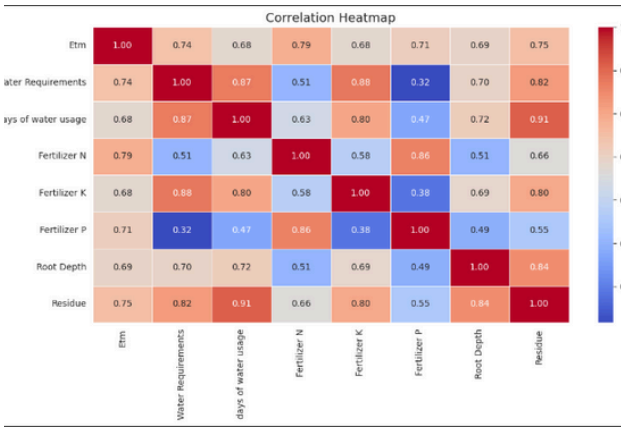
The dataset used in this study is a merged agricultural dataset containing both static and sequential features.

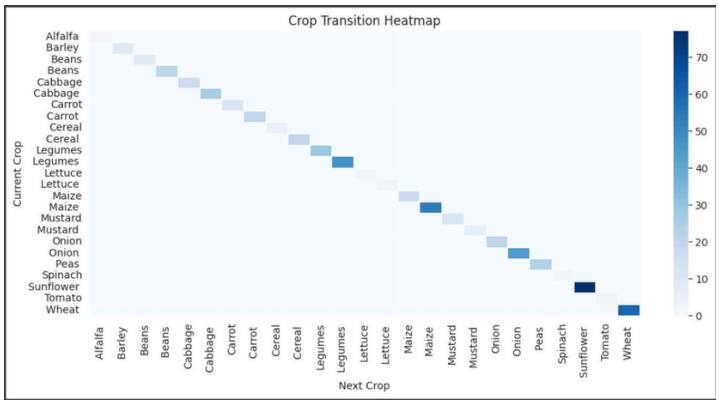
It is designed for a multi-class classification task, where each sample is associated with a target class representing different crop types or agricultural conditions.

- Static Features:
- These are non-temporal attributes such as soil properties, geographic information, or environmental constants.
- They are used as input for graph-based models (e.g., Graph Convolutional Networks).
- Sequential Features:
- These represent time-dependent data sequences, such as rainfall, temperature variation, or crop growth stages over time.
- They are processed through sequential models like Long Short-Term Memory (LSTM) networks.
- Target Variable:
- The target column indicates the crop type (or agricultural condition) and contains 16 distinct classes initially, ranging from class 0 to class 15.
- However, the dataset exhibited severe class imbalance, with some classes represented by only 1–2 samples.









Data Preparation

The preparation of the dataset was a crucial step to ensure the quality and reliability of the predictive modeling for crop rotation. The main steps performed were:

1. Data Loading and Cleaning

- The dataset (merged_dataset.csv) was loaded using Pandas with proper encoding and separator configuration.
- Columns related to numerical measurements (e.g., "Etm", "Water Requirements", "Fertilizer N", etc.) were explicitly converted to numeric types, with invalid entries coerced into NaN.
- Missing values were handled systematically:
  - Categorical variables were filled with their mode (most frequent value).
  - Numerical variables were filled with their median value to minimize the impact of outliers.

2. Feature Engineering

- A new feature, Total Fertilizer Index, was created by summing the quantities of nitrogen (N), phosphorus (P), and potassium (K) fertilizers, reflecting overall fertilizer usage.
- A Water per Root Depth feature was computed to analyze the relationship between irrigation requirements and root system development.

3. Handling Outliers and Rare Classes

- Outliers were not explicitly removed but handled via robust filling strategies (median/mode).
- Rare classes in the target variable (Next Crop) were filtered out by keeping only crops that appeared at least twice in the dataset, ensuring enough samples for model learning.

4. Feature Encoding

- Static Features:
  - Categorical static features (e.g., Soil type, Area, Pest, Disease) were one-hot encoded using pandas.get_dummies().
  - Numerical static features (e.g., Etm, Water Requirements) were standardized (mean=0, std=1) using StandardScaler to ensure equal weighting during model training.
- Sequential Features:
  - Sequential historical crops (Crop (t-2), Crop (t-1), Crop (t)) were label encoded separately into integer codes while maintaining a mapping for later decoding.

5. Data Splitting

- The data was split into training (70%), validation (15%), and testing (15%) sets using stratified sampling to maintain the distribution of the target classes across all splits.
- Separate arrays were prepared for:
  - Static features (X_static_train, X_static_val, X_static_test)
  - Sequential crop history (X_seq_train, X_seq_val, X_seq_test)
  - Target labels (y_train, y_val, y_test)

6. Sanity Checks

- After preparation, dimensions and data types were checked to ensure consistency.
- Correlation analyses and feature distribution plots were generated to visualize the relationships and distributions in the data.

BO 5

PREDICT LAND PRICES IN TUNISIA BY ANALYZING FACTORS LIKE LOCATION, MARKET TRENDS, AND SOIL QUALITY, AIDING INVESTMENT DECISIONS.

I -DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS (EDA)

In this initial phase of the project, the main objective was to understand the structure and content of the available dataset in order to properly prepare the following steps of preprocessing and modeling.

The analyzed file was an Excel document named "data.xlsx", successfully loaded using the pandas library. Upon loading, a quick inspection of the first few rows of the dataset was performed to ensure that the data was read correctly. This initial check revealed that the dataset contains information related to various land properties, characterized by multiple variables.

The dataset consists of **428 observations** and **28 variables.** Among the main features, we identified standard characteristics related to real estate land, including:

- **State**: The governorate where the land is located (e.g., Ariana, Tunis, Zaghouan).
- **Category**: The type of land, distinguishing between residential properties and agricultural land.
- **Location**: The specific site within the governorate.
- **Price**: The selling price of the land.
- **Surface**: The land area in square meters.
- **LNDF**, **CLAF**, **Drain**: Attributes related to the nature or quality of the land.

Several technical columns related to soil and water parameters were also present, labeled with prefixes such as **D1_PHAQ, D2_TAWC, D1_ELCO, etc.**

An initial analysis of the distribution of the **Category** variable showed that:

- **315 properties** were categorized as residential ("habitation"),
- **113 properties** were categorized as agricultural ("terrain agricole").

This distribution indicates a clear dominance of residential land in the studied sample, which could influence the overall modeling process.

The initial dataset exploration also revealed the presence of **missing values**, particularly within the **Drain** variable. These gaps will need to be addressed in the subsequent cleaning and imputation stages to ensure robust analysis.

Overall, the dataset offers a rich variety of **quantitative** and **qualitative** variables, providing a solid foundation for predictive modeling. However, special attention will be necessary to:

- Handle missing data appropriately,
- Standardize categorical variables,
- Consider dimensionality reduction if some features prove redundant or less relevant for prediction.

This understanding phase has confirmed that the data is exploitable and has helped define the necessary improvement areas to ensure the quality and reliability of future analyses and models.

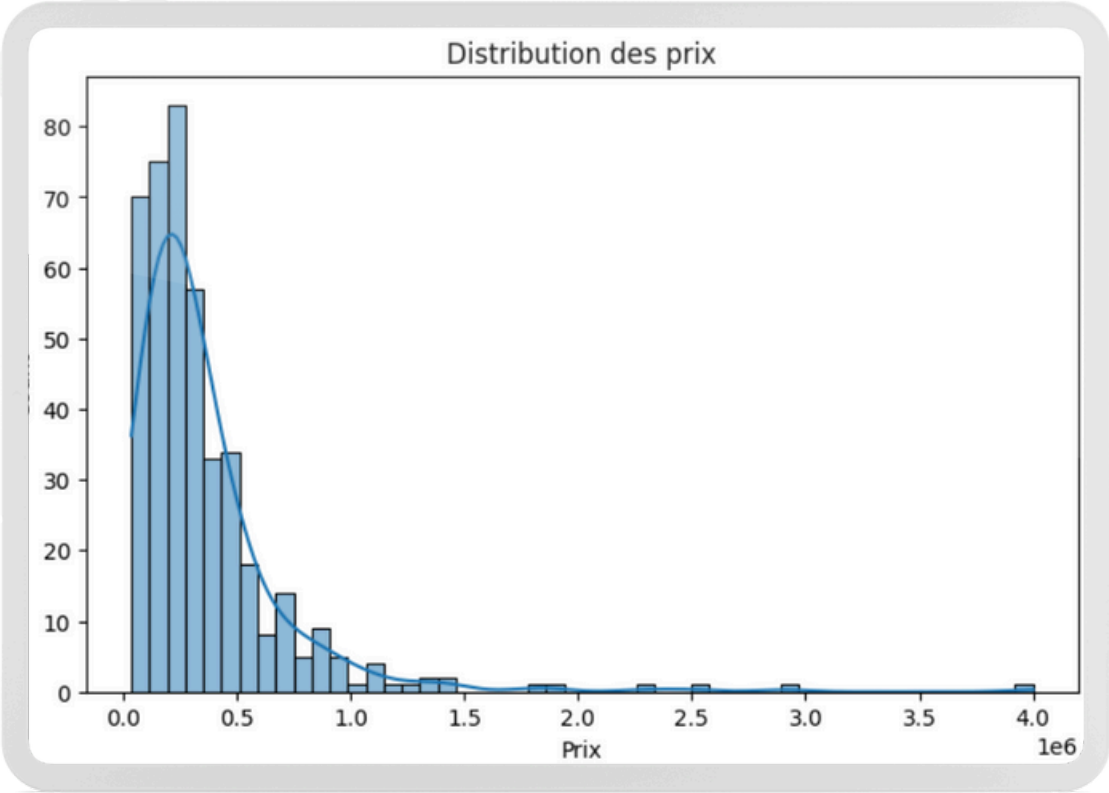
After completing the data understanding and preparation phases, we carried out a visual exploration, a crucial step to:

- Understand the distribution of variables,
- Detect anomalies or underlying trends,
- Identify key correlations that could guide future modeling choices.

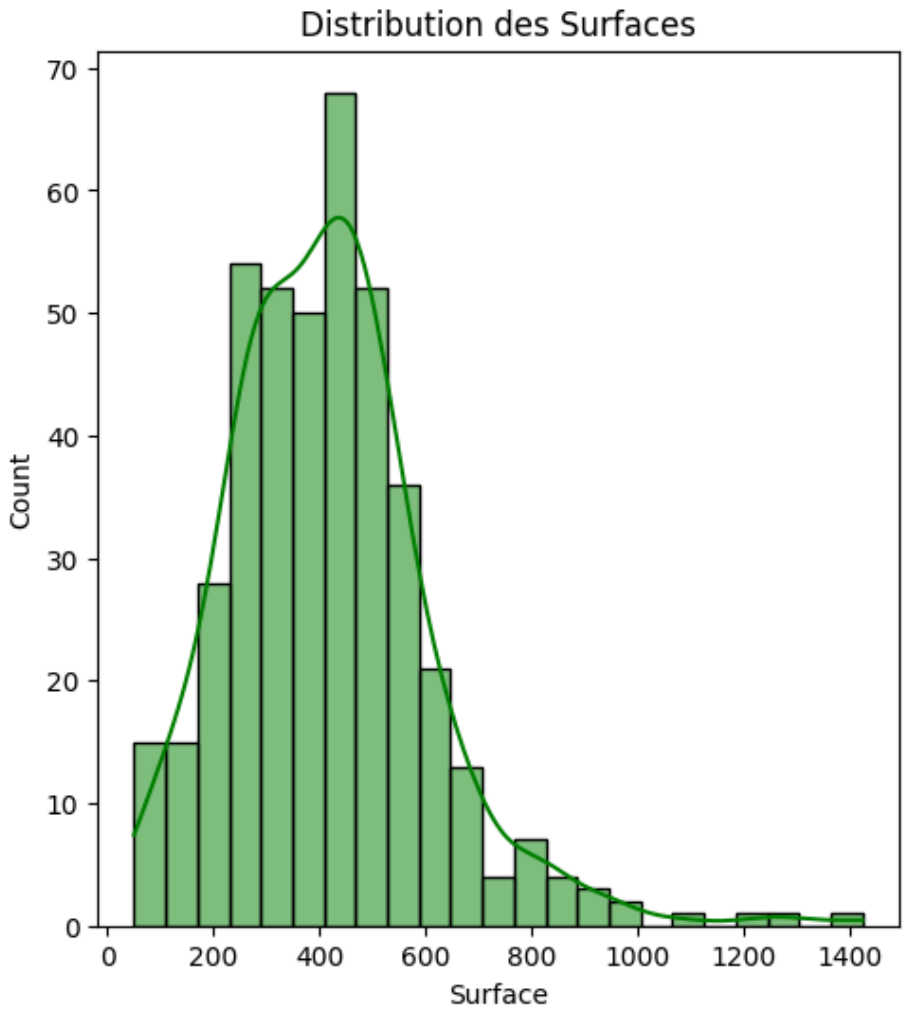
1. Distribution Analysis of Key Variables

We started by examining the distribution of the main variables:

- Price: The "Price" variable exhibits a highly skewed distribution, with a concentration of properties at lower price points and a few extremely high-value outliers.



- Surface Area: Similarly to price, the surface area shows a non-uniform distribution, with the majority of properties being small to medium in size, and only a few large properties.



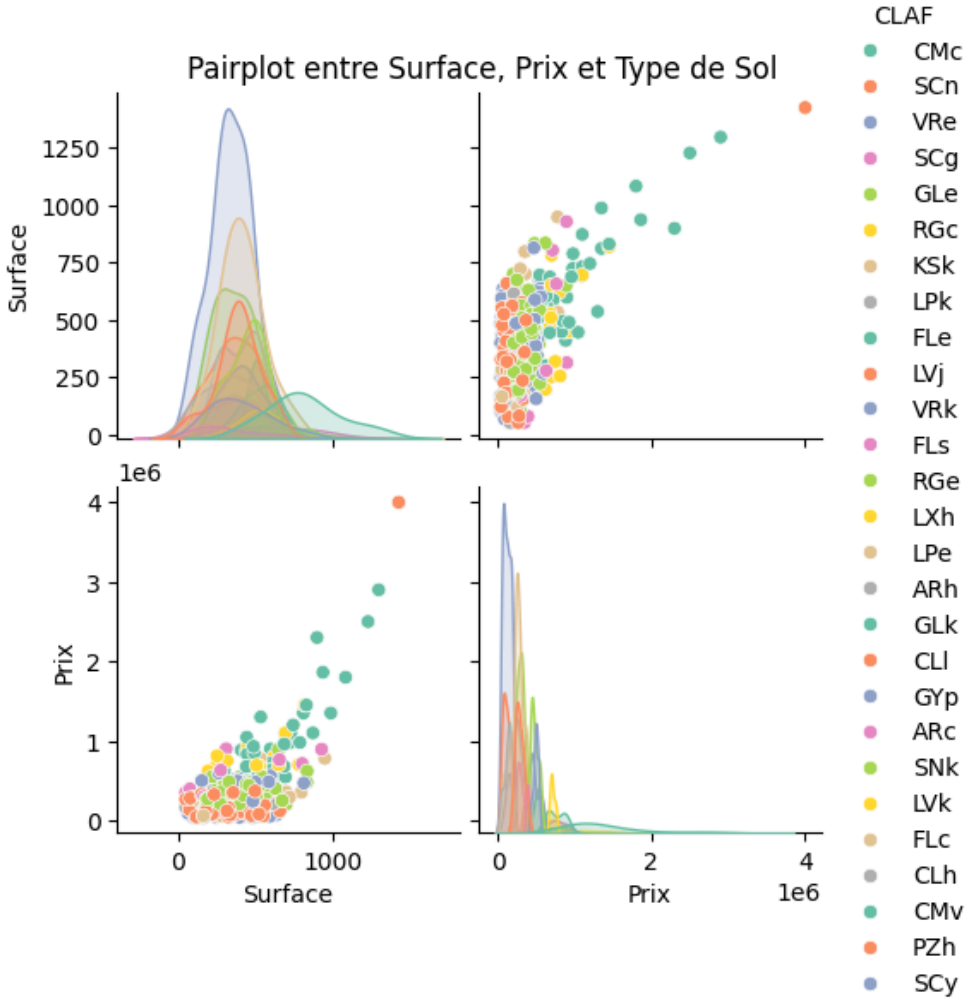
The distribution of surface areas is right-skewed, with most values concentrated between 200 and 600 units. A few larger surface areas extend beyond 800 units, indicating the presence of some outliers or rare large observations.

3. Analysis by Categorical Features

We further visualized how different categories influence the land price:

- Boxplot of Price by State: This plot shows that regions such as Tunis and Ariana exhibit significantly higher average prices compared to others.

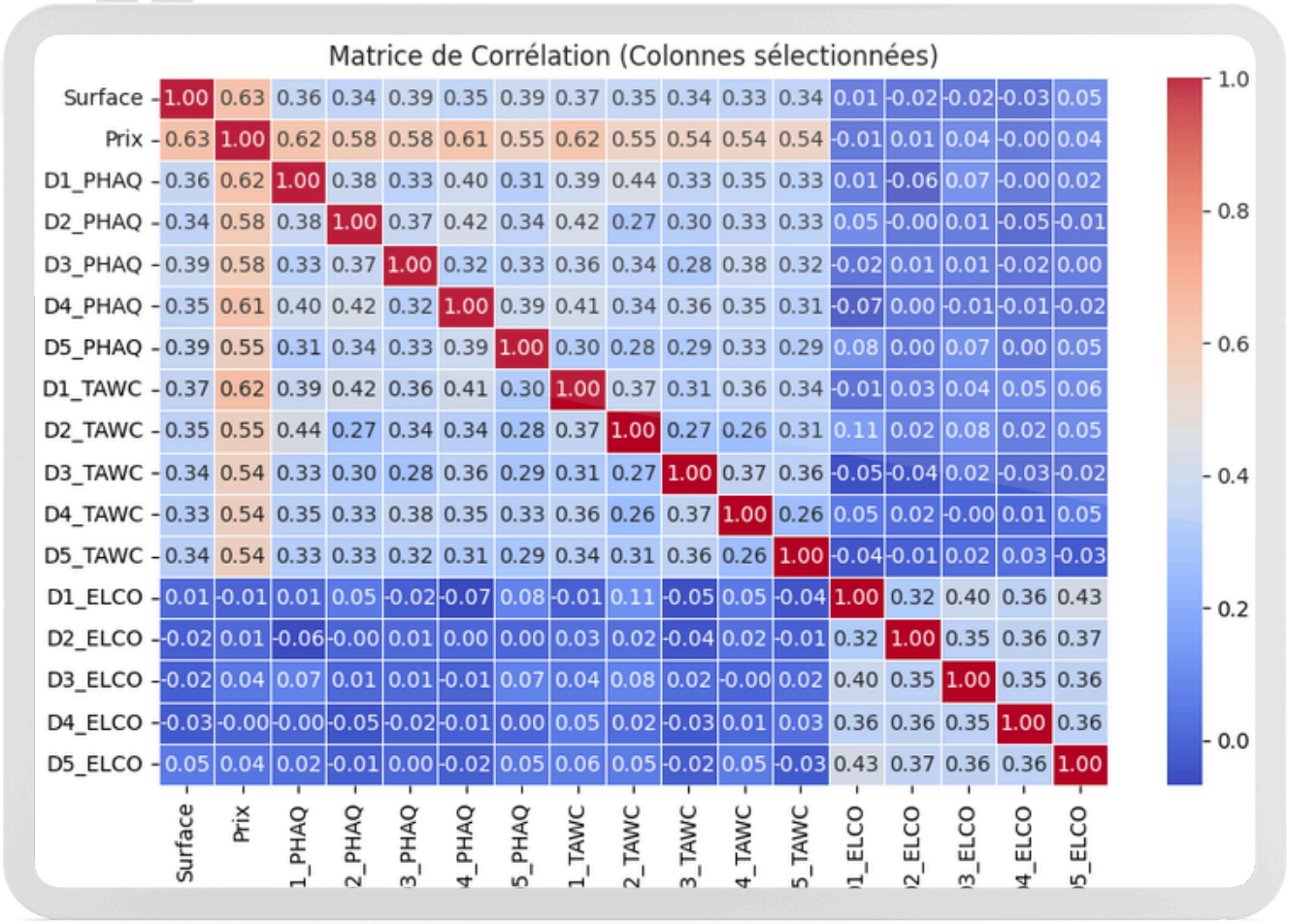
- Boxplot of Price by CLAF (Soil Type): The analysis revealed that certain soil types (e.g., GYp, KSk) are associated with higher average prices.



4. Correlations Among Numerical Variables

A correlation heatmap was generated to assess relationships between the main numerical variables (Price, Surface, TAWC, PHAQ, etc.).
Key observations include:

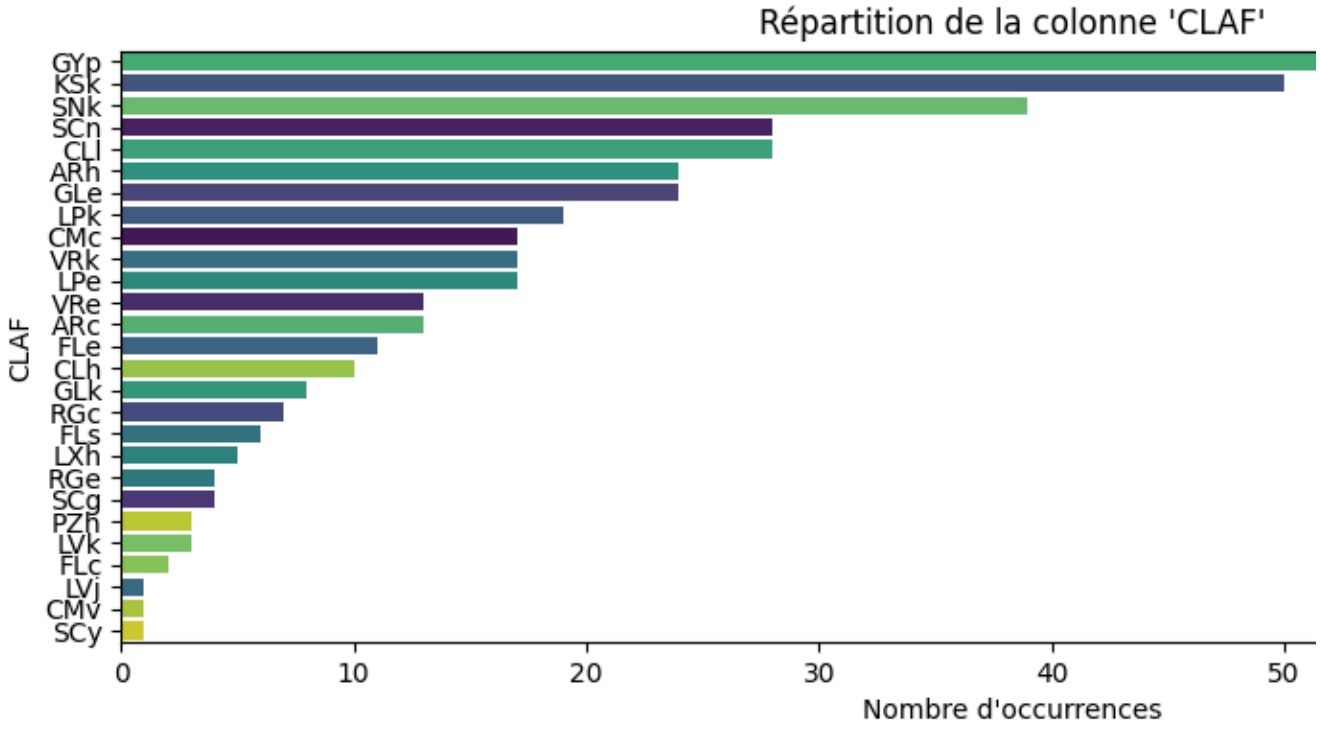
- Price and Surface show a strong positive correlation (r ≈ 0.63).
- Several soil-related parameters (such as TAWC and PHAQ) also exhibit interesting correlations with the Price variable.



5. Analysis of Categorical Variable Distributions

To better understand the distribution of key categorical features:

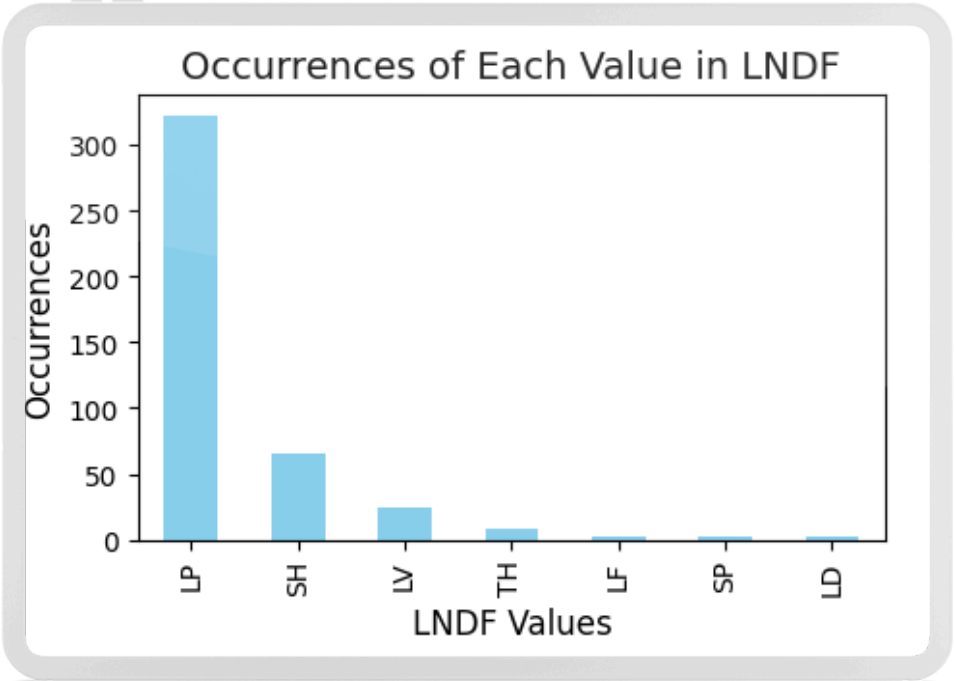
- Distribution of CLAF (dominant soil types):



- Distribution of LNDF (primary soil texture):

This bar chart shows the frequency of LNDF values.

LP is the most common, appearing nearly 300 times, while other categories occur far less frequently.

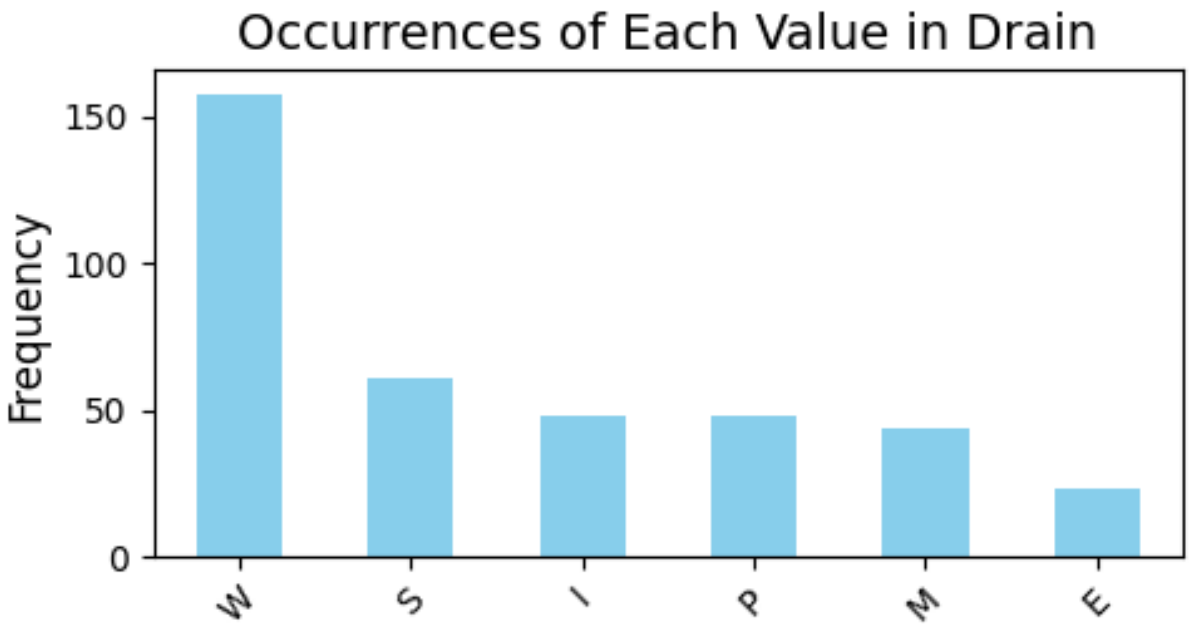


- Distribution of Drain (soil drainage types):

The soil drainage classes are categorized as follows:

E : Excessively well drained
I : Imperfectly drained
M : Moderately well drained
P : Poorly drained
S : Somewhat excessively well drained
V : Very poorly drained
W : Well drained

This bar chart shows the frequency of Drain values.
W is the most common, appearing nearly 150 times, while other categories occur far less frequently.



- Distribution of Soil Properties: TAWC, ELCO, and PHAQ

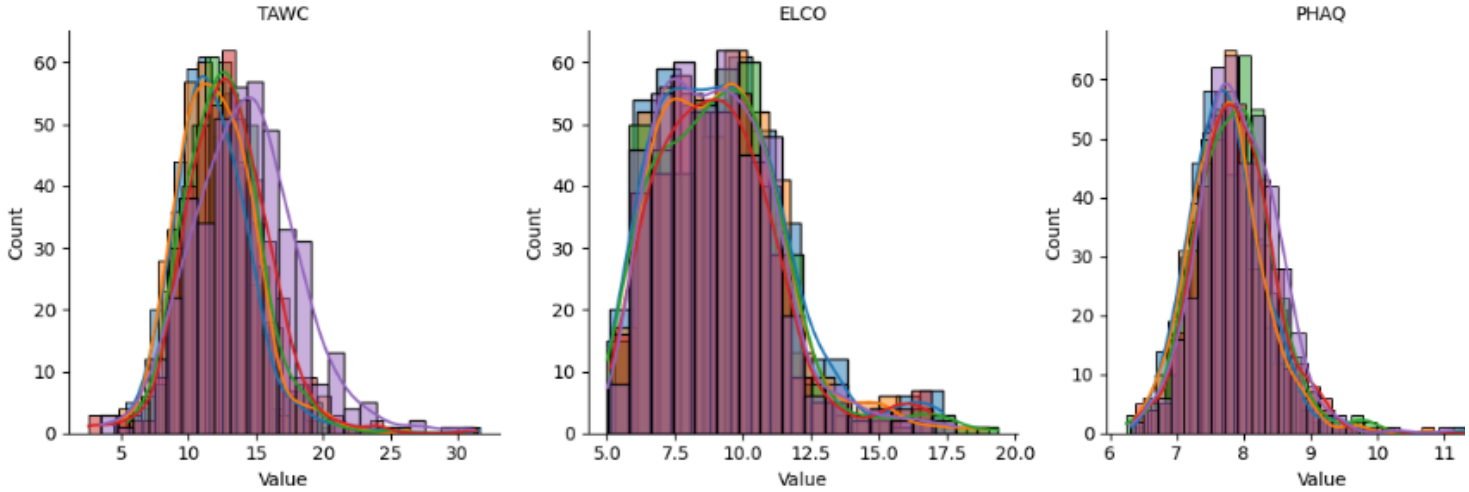
The figure presents the distributions of three key soil properties:

- TAWC (Total Available Water Capacity): The distribution is slightly right-skewed, with most values concentrated between 10 and 15, indicating that a majority of the soil samples have a moderate capacity to retain water.

- ELCO (Electrical Conductivity): The distribution also shows a right skew, suggesting that most soils have low to moderate salinity levels, with a few samples exhibiting higher conductivity.

- PHAQ (Soil pH in water): The pH distribution is relatively narrow and slightly right-skewed, centering around neutral values (close to 7–8), which is typical for many agricultural soils and favorable for plant growth.

- Overall, all three variables display a positively skewed (right-skewed) distribution, meaning that extreme high values are less frequent but still present in the datasets.



- Distribution of PSCL (FAO texture class):

The figure shows the distribution of PSCL values, representing FAO soil texture classes, across different soil depths (D1 to D5):

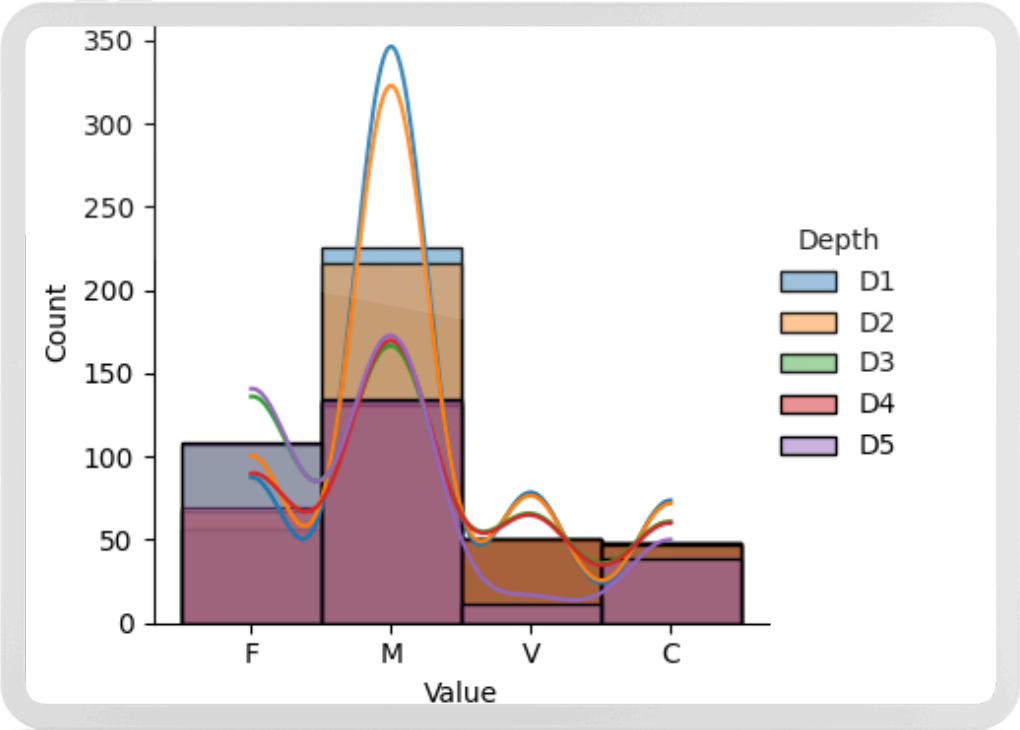
- The majority of soil samples fall into class M (Medium texture), indicating that medium-textured soils are the most common across all depths.

- Class F (Fine texture) is the second most frequent, suggesting the presence of finer soils, particularly in the upper layers.

- V (Very fine texture) and C (Coarse texture) classes are much less represented, implying that very fine or coarse soils are rare in the studied area.

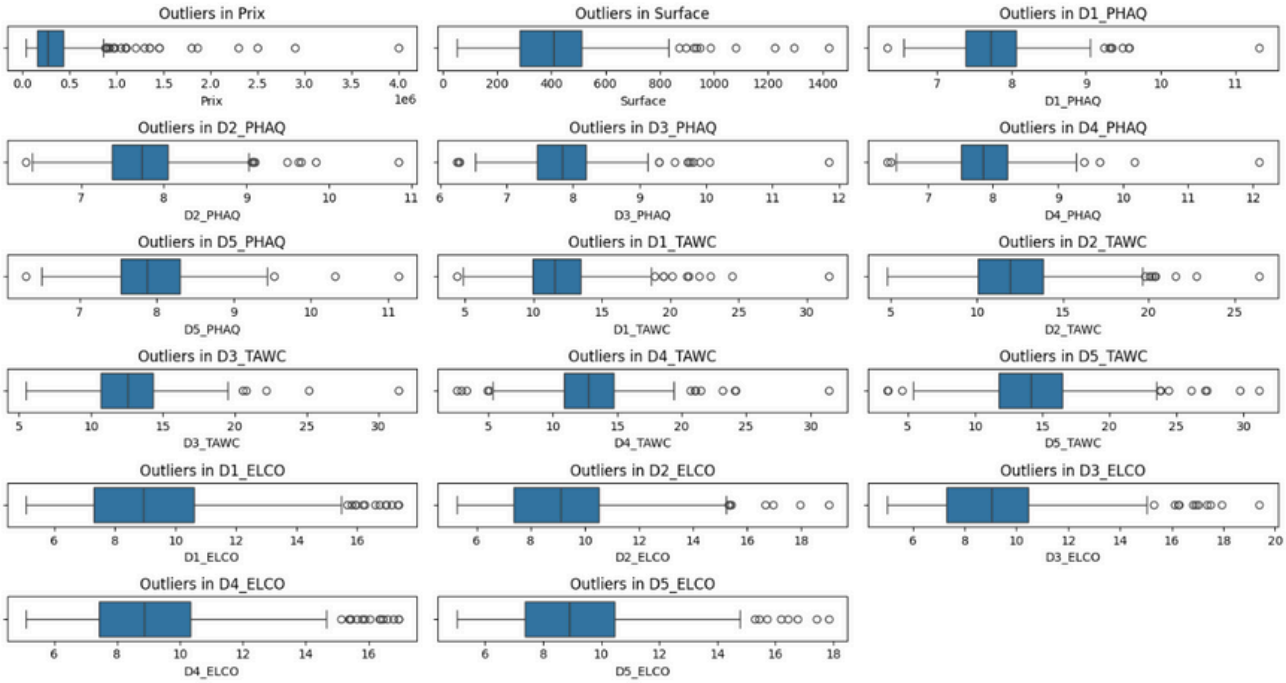
- The general pattern is consistent across all depths, although slight variations can be observed in the proportion of each class.

Overall, medium-textured soils dominate across all sampled depths, which could be favorable for balanced water retention and drainage.



6.Outlier Analysis Interpretation

The figure above presents a series of boxplots illustrating the distribution and detection of outliers across multiple variables related to land price prediction in Tunisia. Each boxplot highlights the presence of potential outliers — values that fall significantly outside the typical range of the data.



- Prix (Price): There are several extreme outliers, with some land prices significantly higher than the majority of the dataset. This suggests a few exceptionally high-value properties that could influence predictive models if not handled carefully.

- Surface (Area): The surface area also shows many outliers on the higher end, indicating that while most properties have moderate sizes, some exceptionally large plots exist.

- PH, TAWC, and ELCO Measurements: Variables such as D1_PHAQ, D2_PHAQ, D3_TAWC, D4_ELCO, and others exhibit moderate outlier behavior. Most of the data is tightly grouped, but a few points fall far from the median values. This pattern is consistent across different depths (D1 to D5), suggesting natural variability in soil characteristics at different layers.

- General Observation: While most variables display a reasonable distribution with a few notable outliers, Prix and Surface stand out for having particularly extreme values. This finding underlines the importance of considering robust scaling or outlier treatment techniques during model development to avoid skewed predictions.

II -DATA UNDERSTANDING AND EXPLORATORY DATA ANALYSIS (EDA)

Data preparation is a critical step in any Machine Learning project. The objective is to ensure that the dataset is clean, consistent, and properly structured to meet the requirements of advanced modeling techniques.

1.1 Exploratory Data Analysis (EDA):

The first part of this phase consisted of performing an initial Exploratory Data Analysis (EDA) to:

- Understand the overall structure of the dataset.
- Identify missing values.
- Detect potential outliers.
- Analyze the distribution of both numerical and categorical variables.

This early insight helped establish a clear diagnosis of the dataset's quality and highlighted specific areas that required intervention during preprocessing.

1.2 Combining Multi-Layer Columns

The dataset contained information collected from different layers (D1 to D5) for certain soil characteristics. To simplify and enhance the dataset, several columns were intelligently combined into a single representative feature:

- For PHAQ and TAWC, the mean across the five layers was calculated.
- For PSCL, the most frequent value (mode) among the five layers was selected.

This approach allowed us to:

- Reduce the dimensionality of the dataset,
- Preserve a synthetic and realistic representation of each property,
- Improve the robustness of models by limiting redundancy.

1.3 Handling Missing Values

Missing values were managed carefully to maintain the integrity of the dataset:

- For continuous variables, missing values were imputed using the mean value.
- For categorical variables, missing values were filled with the most frequent category.
- In cases of substantial missingness across multiple fields, certain rows were excluded to ensure data quality.

This process guaranteed that the data remained usable for all machine learning algorithms without introducing bias.

1.4 Encoding of Categorical Variables:

Since machine learning algorithms require numerical inputs, categorical features were encoded following two main strategies:

- Automatic Mapping: Categorical features such as State, Emplacement, Category (Catégorie), and CLAF were transformed into unique numerical identifiers.
- Hierarchical Mapping: Domain-specific expertise was used to assign an ordinal encoding to features like LNDF (soil type), Drain (soil drainage quality), and PSCL (soil texture).

This encoding process was crucial to preserve the relationships and importance among categories rather than treating them as arbitrary numbers.
Examples:

- For the State and Emplacement columns, a unique ID was generated for each unique value.
- For the LNDF variable, a hierarchy based on agronomic significance was applied.
- Drain and PSCL were similarly encoded based on their respective impact on soil quality.

This ensured that the categorical features contributed meaningfully to the predictive modeling phase.

1.5 Final Data Processing:

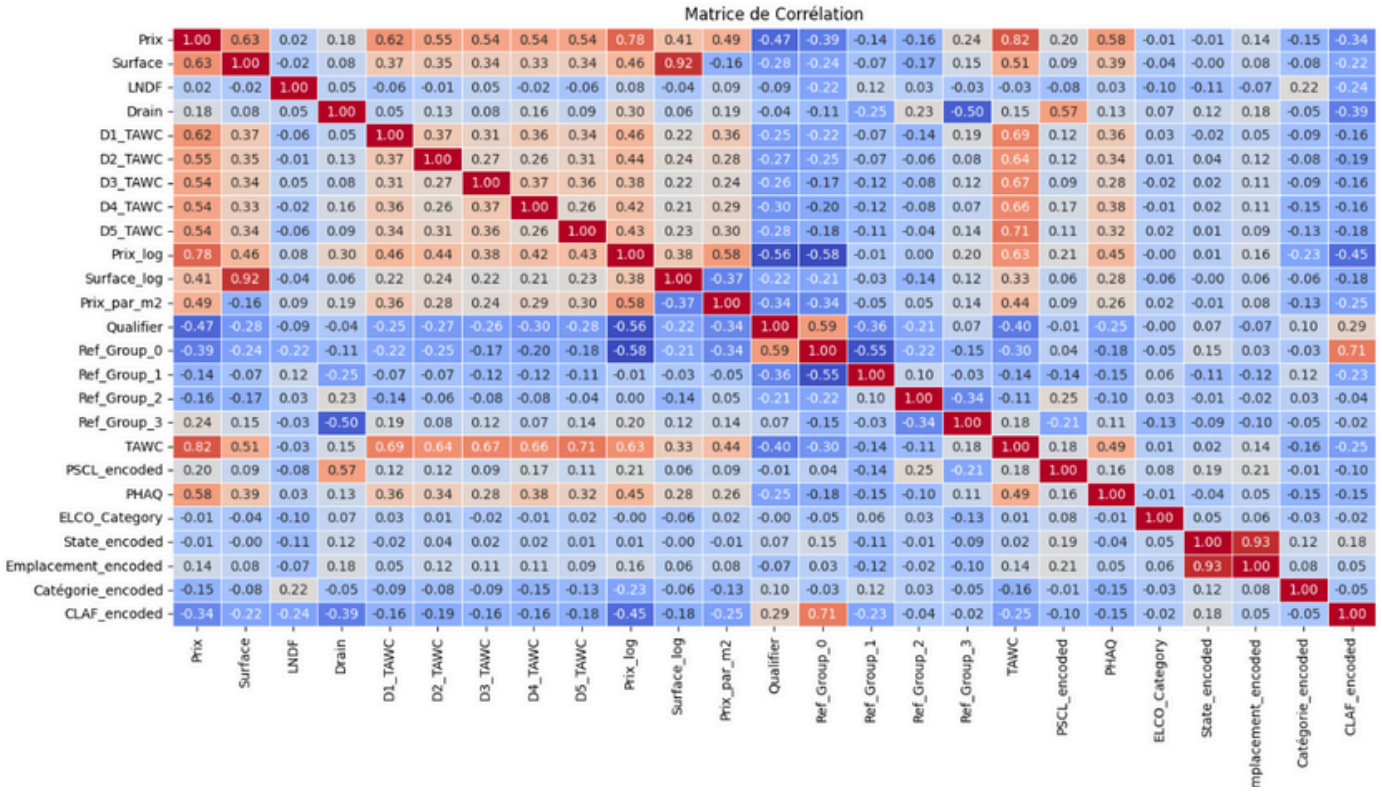
After combining features and encoding categorical variables:

- All obsolete columns corresponding to individual soil layers (D1_, D2_, D3_, D4_, D5_) were removed.
- Only the features deemed relevant for the prediction task were retained.
- Continuous variables were standardized when necessary, particularly for linear-based models.
- Finally, the dataset was split into two distinct subsets: 80% for training and 20% for testing.

1.6 Correlation Analysis after Data Preparation:

Once the data cleaning, transformation, and encoding steps were completed, a correlation analysis was performed to better understand the relationships between the numerical variables.
A correlation matrix was generated, focusing on key variables such as Price, Surface, TAWC, PHAQ, and the encoded categorical variables.
Key observations from the correlation matrix include:

- Price and Surface showed a strong positive correlation, confirming that larger properties generally have higher prices.
- TAWC (Total Available Water Capacity) and PHAQ (Physical and Hydraulic Attributes of Soil) also exhibited interesting correlations with Price, suggesting that soil characteristics can significantly influence land valuation.
- Some engineered features (such as the hierarchical encoding of Drain, LNDF, and PSCL) also showed mild correlations with Price, validating the relevance of the domain-driven encoding approach.



This correlation study helped to:

- Confirm the importance of key predictors for modeling,
- Detect possible multicollinearity issues,
- Guide feature selection for the final models.

Overall, the correlation matrix provided crucial insights that strengthened the strategic choices for the subsequent modeling phase.

BO 6

IDENTIFY VOLUNTEER PLANTS AND DETERMINE IF THEY ARE HELPFUL WEEDS, MEDICINAL PLANTS OR HARMFUL WEEDS.

I -DATA ACQUISITION

THE GOAL OF THIS PHASE IS TO GATHER A RELIABLE DATASET TO SUPPORT THE CLASSIFICATION OF VOLUNTEER PLANTS INTO THREE CATEGORIES: MEDICINAL PLANTS, HARMFUL WEEDS, AND USEFUL WEEDS. THE DATASET NEEDED TO INCLUDE BOTH IMAGE DATA FOR MODEL TRAINING AND DESCRIPTIVE METADATA TO ENRICH THE CLASSIFICATION CONTEXT

IMAGE DATA COLLECTION

TO BUILD THE VISUAL DATASET:

- WEB SCRAPING: WE CONDUCTED TARGETED WEB SCRAPING TO COLLECT PLANT IMAGES.
- ORGANIZATION: EACH PLANT SPECIES WAS STORED IN A DISTINCT FOLDER NAMED AFTER THE SPECIES.
  INSIDE EACH FOLDER, MULTIPLE IMAGES REPRESENTING THE SPECIES WERE SAVED. THIS ORGANIZATION ALLOWED EASY LABELING AND RETRIEVAL DURING MODEL TRAINING.

METADATA CREATION

IN PARALLEL WITH THE IMAGE COLLECTION:

- RESEARCH PHASE: WE MANUALLY RESEARCHED EACH PLANT SPECIES USING TRUSTED ONLINE BOTANICAL RESOURCES AND HERBAL MEDICINE BOOK REFERENCES.
- CSV FILE GENERATION:

WE CREATED A STRUCTURED CSV FILE LISTING EACH SPECIES.

  - FOR EACH SPECIES, WE ADDED:
    - THE TYPE OF PLANT (HARMFUL WEED, USEFUL WEED, OR MEDICINAL PLANT).
    - A DESCRIPTION EXPLAINING EITHER ITS HARMFULNESS OR ITS BENEFITS.
    - PRACTICAL GUIDELINES ON HOW TO USE IT (FOR BENEFICIAL PLANTS) OR MANAGE IT (FOR HARMFUL WEEDS).

A COMPLETE AND WELL-ORGANIZED DATASET WAS BUILT BY COMBINING WEB SCRAPING FOR IMAGES AND MANUAL RESEARCH FOR METADATA. THIS METHOD ENSURES THAT BOTH THE VISUAL AND DESCRIPTIVE INFORMATION OF EACH VOLUNTEER PLANT ARE CAPTURED, SUPPORTING THE ACHIEVEMENT OF OUR DATA SCIENCE OBJECTIVE.

II -DATA UNDERSTANDING

AFTER COMPLETING THE DATA ACQUISITION PHASE, THE NEXT STEP WAS TO EXPLORE AND UNDERSTAND THE COLLECTED DATA. THIS PHASE FOCUSED ON VALIDATING THE QUALITY, STRUCTURE, AND DISTRIBUTION OF THE DATASET TO ENSURE IT IS SUITABLE FOR MACHINE LEARNING TASKS

DATA INSPECTION AND VALIDATION

IIMAGE FORMAT VERIFICATION

WE BEGAN BY EXAMINING THE TYPES OF IMAGE FILES COLLECTED.
IT WAS VERIFIED THAT ALL IMAGES WERE IN .JPG FORMAT, ENSURING CONSISTENCY ACROSS THE DATASET AND ELIMINATING THE NEED FOR FORMAT CONVERSION.

ISAMPLE EXTRACTION

A FEW RANDOM SAMPLES FROM THE DATASET WERE EXTRACTED AND MANUALLY INSPECTED TO CONFIRM THAT:

- THE IMAGES WERE RELEVANT TO THEIR CORRESPONDING SPECIES.
- THE FILE NAMING AND ORGANIZATION WERE CORRECT.





CLASS EXTRACTION

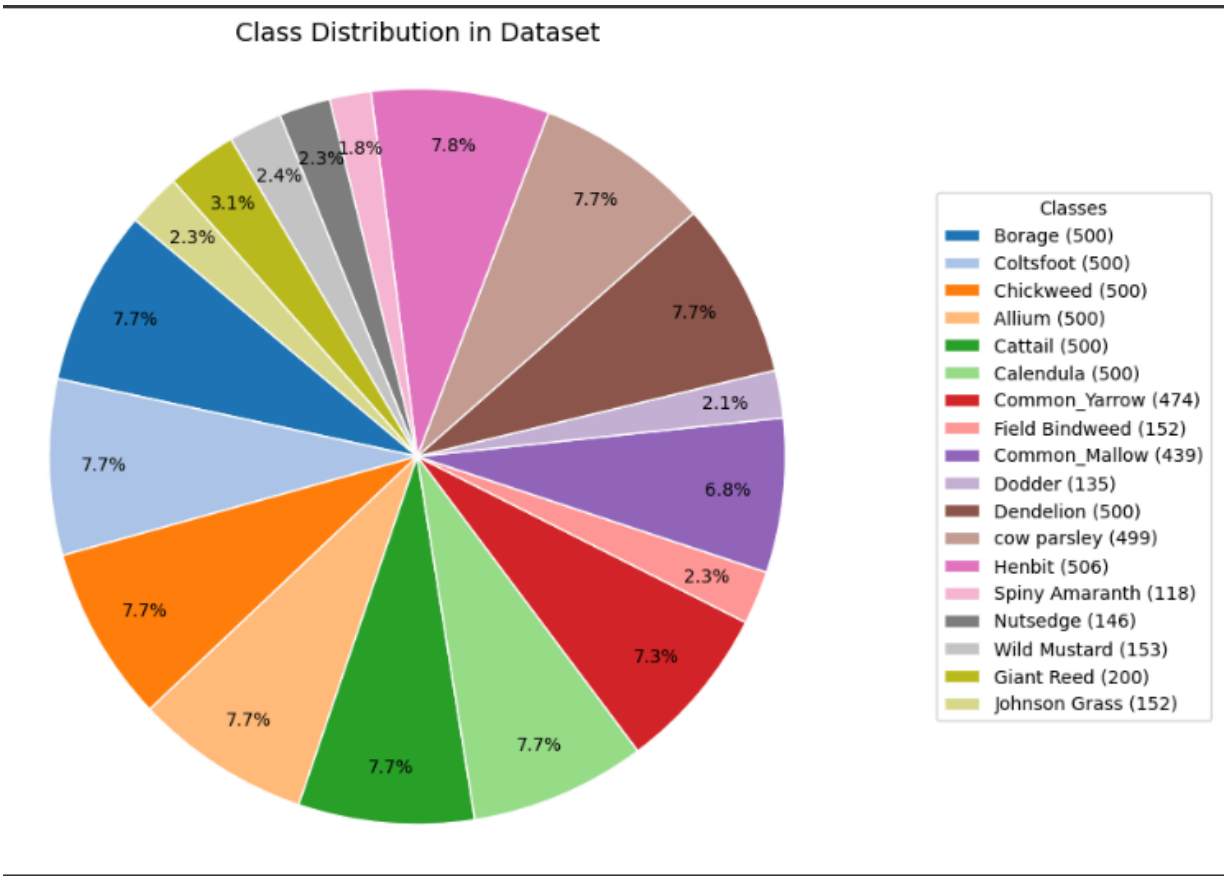
- THE NAMES OF THE PLANT SPECIES (CLASSES) WERE EXTRACTED FROM THE FOLDER STRUCTURE.
- A TOTAL OF 18 DISTINCT CLASSES WERE IDENTIFIED

CLASS DISTRIBUTION ANALYSIS

A CLASS DISTRIBUTION ANALYSIS WAS PERFORMED TO ASSESS THE BALANCE OF THE DATASET.
FINDINGS:

- THE DATASET IS NOTICEABLY IMBALANCED.
- CLASSES LIKE CALENDULA, DENDELION, HENBIT, COLTSFOOT, AND BORAGE HAVE A LARGE NUMBER OF SAMPLES (CLOSE TO OR ABOVE 500 IMAGES).
- MEANWHILE, CLASSES LIKE DODDER (135 IMAGES), SPINY AMARANTH (118 IMAGES), NUTSEDGE (146 IMAGES), AND JOHNSON GRASS (152 IMAGES) ARE SIGNIFICANTLY UNDERREPRESENTED.

THIS IMBALANCE WILL BE AN IMPORTANT CONSIDERATION WHEN DEVELOPING THE CLASSIFICATION MODEL, AS IT COULD AFFECT ITS ABILITY TO CORRECTLY PREDICT LESS-REPRESENTED SPECIES.



THROUGH A DETAILED REVIEW, THE DATASET WAS CONFIRMED TO BE CLEAN IN FORMAT AND CORRECTLY LABELED. HOWEVER, THE CLASS IMBALANCE PRESENTS A POTENTIAL CHALLENGE THAT WILL NEED TO BE ADDRESSED DURING MODEL TRAINING.

II -DATA PREPARATION

FOLLOWING DATA UNDERSTANDING, WE PREPARED THE DATASET TO ENSURE IT WAS BALANCED, PROPERLY STRUCTURED, AND READY FOR MODEL TRAINING. THIS INCLUDED DATA BALANCING THROUGH AUGMENTATION, DATASET SPLITTING, AND VALIDATION OF DATA QUALITY.

DATA BALANCING

A NEW DIRECTORY STRUCTURE WAS CREATED TO HOLD A BALANCED VERSION OF THE DATASET. ALL ORIGINAL IMAGES WERE FIRST COPIED INTO THIS NEW DIRECTORY.
DATA AUGMENTATION WAS APPLIED USING TECHNIQUES SUCH AS:

- ROTATION
- WIDTH AND HEIGHT SHIFTING
- SHEARING
- ZOOMING
- HORIZONTAL FLIPPING

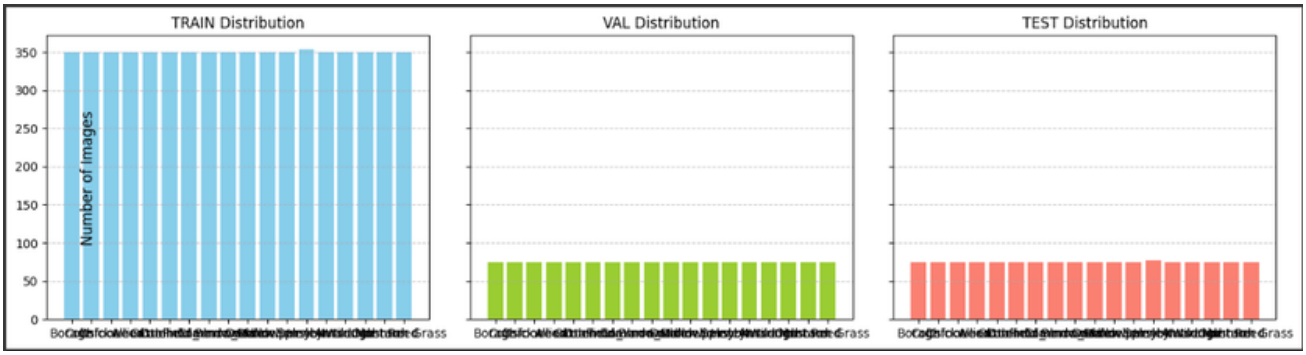
THE OBJECTIVE WAS TO ENSURE THAT EACH CLASS HAD AT LEAST 500 IMAGES, BALANCING THE PREVIOUSLY IMBALANCED DATASET.
AS A RESULT, ALL CLASSES WERE SUCCESSFULLY AUGMENTED TO REACH THE TARGET COUNT OF 500 IMAGES.

DATA SPLITTING

AFTER BALANCING, THE DATASET WAS SPLIT INTO TRAINING, VALIDATION, AND TESTING SETS USING THE FOLLOWING RATIOS:

- TRAINING: 70%
- VALIDATION: 15%
- TESTING: 15%

A NEW DIRECTORY STRUCTURE WAS CREATED TO ORGANIZE THE SPLIT DATA ACCORDINGLY.
IMAGES WERE RANDOMLY SHUFFLED BEFORE BEING ASSIGNED TO THEIR RESPECTIVE SETS TO ENSURE RANDOMNESS AND AVOID BIAS.



VERIFICATION OF DATA AUGMENTATION QUALITY

- TO ENSURE THAT THE AUGMENTED IMAGES MAINTAINED SUFFICIENT SIMILARITY TO THEIR ORIGINAL VERSIONS WHILE INTRODUCING MEANINGFUL VARIATION, A STRUCTURAL SIMILARITY INDEX MEASURE (SSIM) WAS COMPUTED.
- THE SSIM SCORE BETWEEN AN ORIGINAL AND AN AUGMENTED IMAGE WAS 0.7761, INDICATING A GOOD BALANCE BETWEEN SIMILARITY AND DIVERSITY.
- THIS CONFIRMS THAT THE AUGMENTATION PROCESS DID NOT DISTORT THE IMAGES EXCESSIVELY WHILE STILL INTRODUCING ENOUGH VARIATION FOR MODEL TRAINING

THE DATA PREPARATION PHASE SUCCESSFULLY BALANCED THE DATASET, STRUCTURED IT FOR TRAINING, VALIDATION, AND TESTING, AND VALIDATED THE QUALITY OF THE AUGMENTED IMAGES. THIS ENSURES THAT THE DATA IS NOW READY FOR THE NEXT PHASES OF MODEL DEVELOPMENT.

05

MODELING

BO 1

PROVIDE ACCURATE FERTILIZER DOSAGE RECOMMENDATIONS TO OPTIMIZE CROP YIELDS AND REDUCE FERTILIZER WASTE ACROSS DIFFERENT REGIONS OF TUNISIA.

I -MODELING AND EVALUATION

- TO PREDICT FERTILIZER USAGE (DOSAGE), WE EXPLORED A RANGE OF TREE-BASED MODELS, ADVANCED BOOSTING TECHNIQUES, AND AN ENSEMBLE STACKING APPROACH.
- OUR OBJECTIVE WAS TO BALANCE HIGH PREDICTIVE PERFORMANCE WITH REASONABLE TRAINING TIMES, WHILE ENSURING THE MODELS COULD CAPTURE NON-LINEAR INTERACTIONS BETWEEN SOIL PROPERTIES, FERTILIZER CHARACTERISTICS, AND CROP NEEDS.

1-SELECTED

- TO PREDICT FERTILIZER USAGE (DOSAGE), WE EXPLORED A RANGE OF TREE-BASED MODELS, ADVANCED BOOSTING TECHNIQUES, AND AN ENSEMBLE STACKING APPROACH.

- OUR OBJECTIVE WAS TO BALANCE HIGH PREDICTIVE PERFORMANCE WITH REASONABLE TRAINING TIMES, WHILE ENSURING THE MODELS COULD CAPTURE NON-LINEAR INTERACTIONS BETWEEN SOIL PROPERTIES, FERTILIZER CHARACTERISTICS, AND CROP NEEDS.
- THE MODELS WE SELECTED FOR THIS STUDY INCLUDE:

1. **CATBOOST REGRESSOR**
2. **HISTGRADIENTBOOSTING REGRESSOR**
3. **XGBOOST REGRESSOR**
4. **GRADIENT BOOSTING REGRESSOR**
5. **RANDOM FOREST REGRESSOR**
6. **STACKING REGRESSOR (COMBINING CATBOOST AND HISTGRADIENTBOOSTING, WITH GRADIENT BOOSTING AS META-MODEL)**

=> THESE MODELS WERE CHOSEN FOR THEIR PROVEN EFFECTIVENESS IN REGRESSION TASKS, PARTICULARLY WHERE NON-LINEAR RELATIONSHIPS AND FEATURE INTERACTIONS PLAY AN IMPORTANT ROLE.

2-TRAINING PHASE

DURING THE TRAINING PHASE, EACH MODEL WAS TRAINED USING THE DATASET, WHICH INCLUDES FEATURES SUCH AS SOIL CONDITIONS, CROP TYPES, AND ENVIRONMENTAL VARIABLES, ALL INFLUENCING THE OPTIMAL FERTILIZER DOSAGE. BELOW IS AN OVERVIEW OF HOW THE MODELS WERE TRAINED:

1. **DATA PREPROCESSING:**

- THE DATASET WAS CLEANED TO HANDLE MISSING VALUES, AND NUMERICAL FEATURES WERE SCALED WHERE NECESSARY TO IMPROVE MODEL TRAINING.
- CATEGORICAL VARIABLES, SUCH AS CROP TYPES AND FERTILIZER CHARACTERISTICS, WERE EITHER DIRECTLY ENCODED OR HANDLED INTERNALLY BY MODELS LIKE CATBOOST.

2. **HYPERPARAMETER TUNING:**

- HYPERPARAMETERS WERE OPTIMIZED USING **RANDOMIZEDSEARCHCV**, WHICH ALLOWS FOR EFFICIENT EXPLORATION OF HYPERPARAMETER SPACES. THIS TUNING ENSURED EACH MODEL WAS CONFIGURED FOR OPTIMAL PERFORMANCE.

3. **TRAINING TIME TRACKING:**

- EACH MODEL WAS MONITORED FOR ITS TRAINING TIME TO ASSESS ITS COMPUTATIONAL EFFICIENCY.

3-METRICS USED

TO ASSESS AND COMPARE THE PERFORMANCE OF THE DIFFERENT MODELS, WE USED THE FOLLOWING METRICS:

ROOT MEAN SQUARED ERROR (RMSE):

- MEASURES THE STANDARD DEVIATION OF THE PREDICTION ERRORS. A LOWER RMSE INDICATES BETTER MODEL PERFORMANCE BY PENALIZING LARGER ERRORS MORE HEAVILY.

MEAN ABSOLUTE ERROR (MAE):

- CALCULATES THE AVERAGE ABSOLUTE DIFFERENCE BETWEEN THE PREDICTED AND ACTUAL VALUES. MAE IS MORE INTERPRETABLE BECAUSE IT'S IN THE SAME UNITS AS THE TARGET VARIABLE AND TREATS ALL ERRORS EQUALLY.

COEFFICIENT OF DETERMINATION (R^2 SCORE):

- REPRESENTS THE PROPORTION OF THE VARIANCE IN THE TARGET VARIABLE THAT IS PREDICTABLE FROM THE INPUT FEATURES. AN R^2 SCORE CLOSER TO 1 INDICATES A BETTER FIT.

4-MODEL COMPARISON

CATBOOST

CATBOOST IS A GRADIENT BOOSTING ALGORITHM DESIGNED TO HANDLE CATEGORICAL VARIABLES EFFICIENTLY WITHOUT THE NEED FOR EXPLICIT ENCODING.
PARAMETERS:

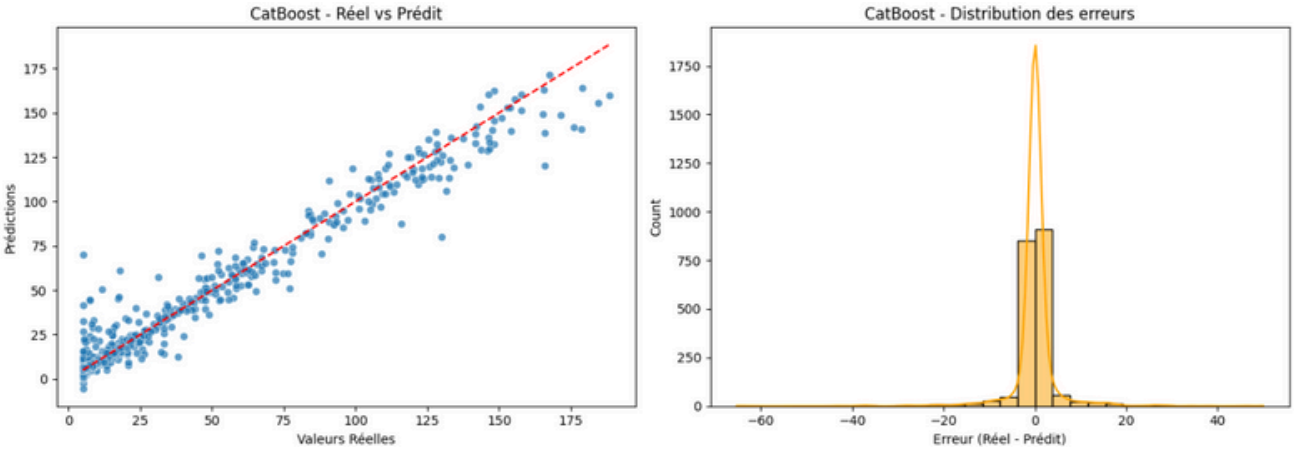
- LEARNING RATE: 0.05
- DEPTH: 10
- L2 LEAF REGULARIZATION: 3
- BAGGING TEMPERATURE: 0.5
- ITERATIONS: 1000

IT PERFORMED EXCEPTIONALLY WELL IN PREDICTING THE FERTILIZER DOSAGE.

- RMSE: 5.13
- MAE: 2.03
- R^2: 0.97
- TRAINING TIME: 495.11 SECONDS

PERFORMANCE INSIGHTS:
CATBOOST STOOD OUT AS THE BEST-PERFORMING MODEL, ACHIEVING THE LOWEST RMSE, WHICH MEANS IT PRODUCED THE SMALLEST AVERAGE PREDICTION ERROR. THE R^2 VALUE OF 0.97 SHOWS THAT CATBOOST EXPLAINED 97% OF THE VARIANCE IN THE TARGET VARIABLE, A STRONG INDICATOR OF ITS PREDICTIVE ACCURACY. DESPITE ITS SUPERIOR PERFORMANCE, CATBOOST HAD A RELATIVELY LONGER TRAINING TIME COMPARED TO SOME OTHER MODELS.

THIS MODEL'S ABILITY TO PROCESS CATEGORICAL DATA WITHOUT REQUIRING MANUAL ENCODING GIVES IT AN EDGE IN TERMS OF SIMPLICITY AND EFFECTIVENESS, ESPECIALLY IN SCENARIOS WITH A LARGE NUMBER OF CATEGORICAL FEATURES.



HISTGRADIENTBOOSTING

HISTGRADIENTBOOSTING IS A VARIANT OF GRADIENT BOOSTING THAT IS OPTIMIZED FOR FASTER TRAINING ON LARGE DATASETS.

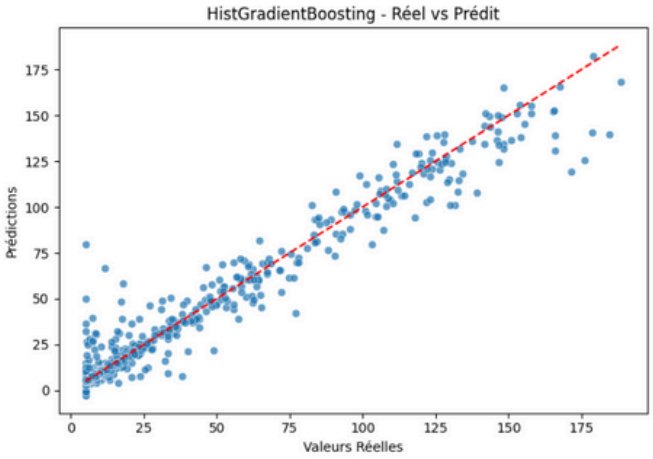
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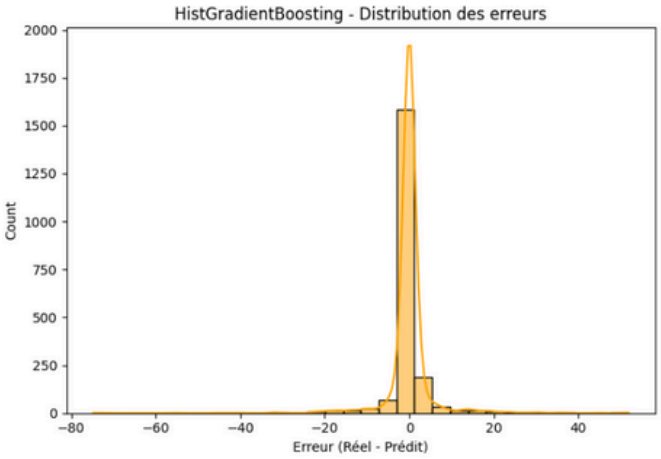
- LEARNING RATE: 0.1
- MAX DEPTH: 10
- NUMBER OF ESTIMATORS: 1000

IT IS KNOWN FOR ITS HIGH EFFICIENCY AND ABILITY TO HANDLE LARGE-SCALE DATA WHILE DELIVERING COMPETITIVE RESULTS.

- RMSE: 5.57
- MAE: 2.08
- R^2: 0.96
- TRAINING TIME: 55.49 SECONDS

PERFORMANCE INSIGHTS:
WHILE HISTGRADIENTBOOSTING ACHIEVED COMPETITIVE RESULTS WITH A RELATIVELY LOW RMSE AND MAE, IT FELL BEHIND CATBOOST AND THE STACKING MODEL IN TERMS OF PREDICTION ACCURACY. HOWEVER, IT HAD THE FASTEST TRAINING TIME (55.49 SECONDS), MAKING IT AN ATTRACTIVE CHOICE WHEN COMPUTATIONAL EFFICIENCY IS A PRIORITY.
THE R^2 VALUE OF 0.96 STILL INDICATES THAT HISTGRADIENTBOOSTING DID A GOOD JOB OF EXPLAINING THE VARIANCE IN THE TARGET VARIABLE, THOUGH NOT AS WELL AS CATBOOST. THIS MODEL IS SUITABLE FOR SITUATIONS WHERE SPEED IS ESSENTIAL, BUT A SMALL TRADE-OFF IN PREDICTIVE ACCURACY IS ACCEPTABLE.





XGBOOST

XGBOOST IS A HIGHLY EFFICIENT IMPLEMENTATION OF GRADIENT BOOSTING THAT IS WIDELY USED IN MACHINE LEARNING COMPETITIONS.

PARAMETERS:

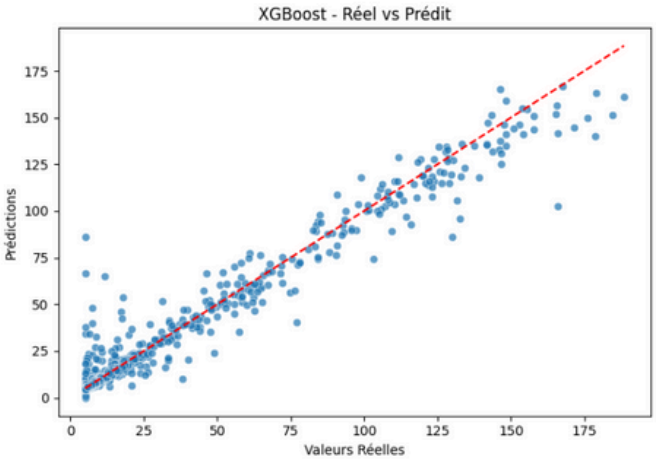
- LEARNING RATE: 0.05
- MAX DEPTH: 6
- NUMBER OF ESTIMATORS: 500

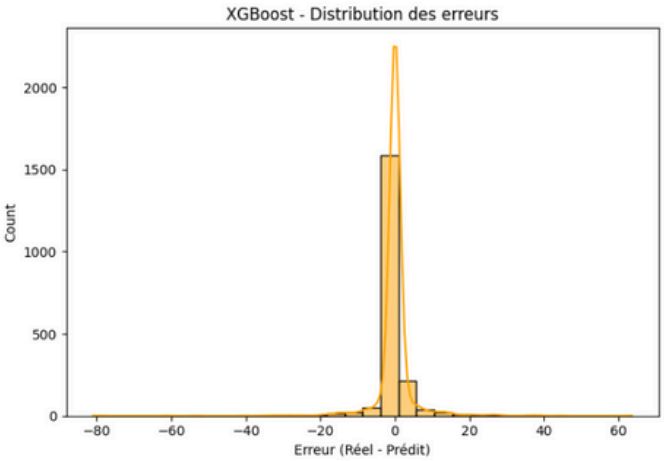
IT EXCELS IN BOTH ACCURACY AND TRAINING SPEED.

- RMSE: 5.61
- MAE: 1.99
- R^2: 0.96
- TRAINING TIME: 33.68 SECONDS

PERFORMANCE INSIGHTS: XGBOOST DELIVERED EXCELLENT RESULTS, WITH A SLIGHTLY HIGHER RMSE THAN HISTGRADIENTBOOSTING, BUT AT A FASTER TRAINING TIME (33.68 SECONDS). ITS MAE AND R^2 WERE COMPETITIVE, WITH MAE SHOWING THAT IT HAD A LOWER AVERAGE ABSOLUTE ERROR THAN HISTGRADIENTBOOSTING. DESPITE ITS FAST TRAINING TIME, XGBOOST DID NOT OUTPERFORM CATBOOST OR STACKING, BUT IT REMAINS A STRONG CHOICE FOR TASKS WHERE BOTH SPEED AND ACCURACY ARE IMPORTANT.

THE MODEL IS PARTICULARLY USEFUL WHEN WORKING WITH A LARGE DATASET WHERE TRAINING TIME IS A SIGNIFICANT CONSIDERATION.





GRADIENT BOOSTING

GRADIENT BOOSTING IS A WELL-ESTABLISHED ENSEMBLE METHOD THAT BUILDS MODELS SEQUENTIALLY, WITH EACH NEW MODEL ATTEMPTING TO CORRECT THE ERRORS OF THE PREVIOUS ONE.

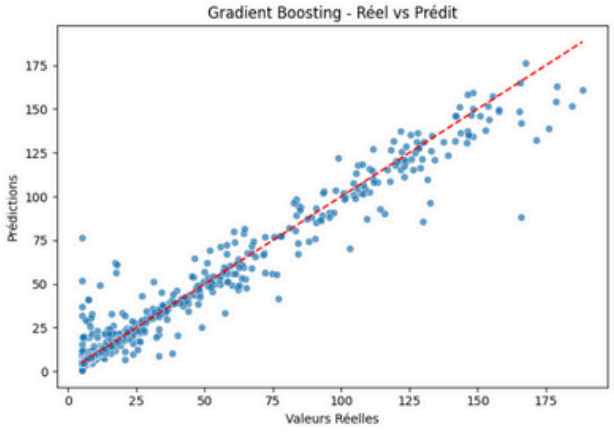
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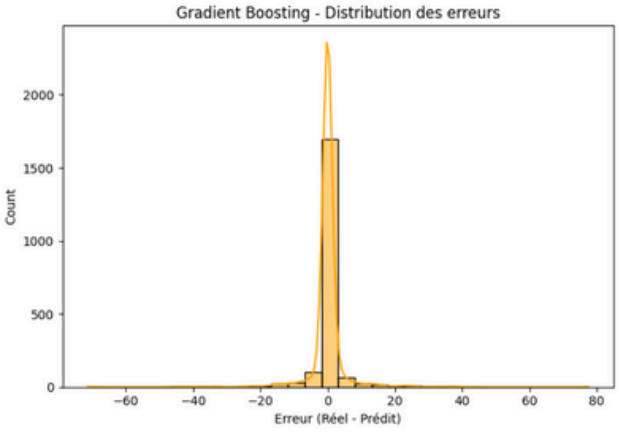
- LEARNING RATE: 0.1
- MAX DEPTH: 5
- NUMBER OF ESTIMATORS: 1000

PERFORMANCE:

- RMSE: 5.63
- MAE: 1.97
- R^2: 0.96
- TRAINING TIME: 212.66 SECONDS

PERFORMANCE INSIGHTS: GRADIENT BOOSTING PROVIDED SOLID PERFORMANCE, WITH A RMSE AND MAE SIMILAR TO THOSE OF XGBOOST AND HISTGRADIENTBOOSTING. HOWEVER, IT HAD A LONGER TRAINING TIME (212.66 SECONDS) COMPARED TO XGBOOST AND HISTGRADIENTBOOSTING, WHICH MADE IT LESS EFFICIENT. STILL, THE MODEL'S R^2 OF 0.96 INDICATES THAT IT DID A GOOD JOB OF EXPLAINING THE TARGET VARIABLE.
WHILE NOT THE FASTEST, GRADIENT BOOSTING IS A RELIABLE MODEL WHEN INTERPRETABILITY AND FINE CONTROL OVER THE BOOSTING PROCESS ARE IMPORTANT. IT'S A SOLID CHOICE FOR TASKS THAT DON'T HAVE STRICT TIME CONSTRAINTS.





RANDOM FOREST

RANDOM FOREST IS AN ENSEMBLE MODEL THAT BUILDS MULTIPLE DECISION TREES AND AGGREGATES THEIR PREDICTIONS.

PARAMETERS:

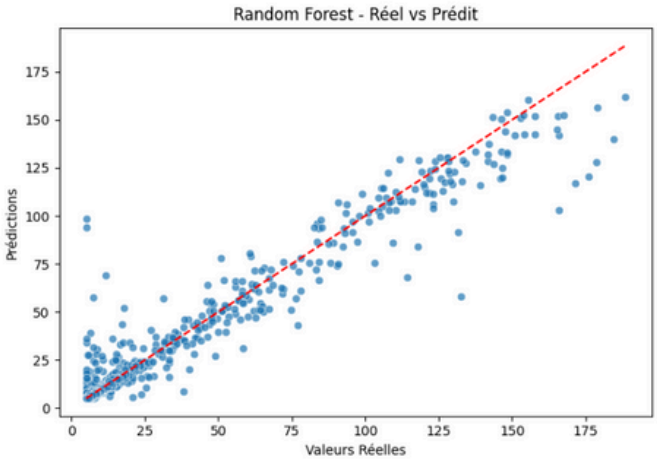
- MAX DEPTH: 10
- NUMBER OF ESTIMATORS: 1000

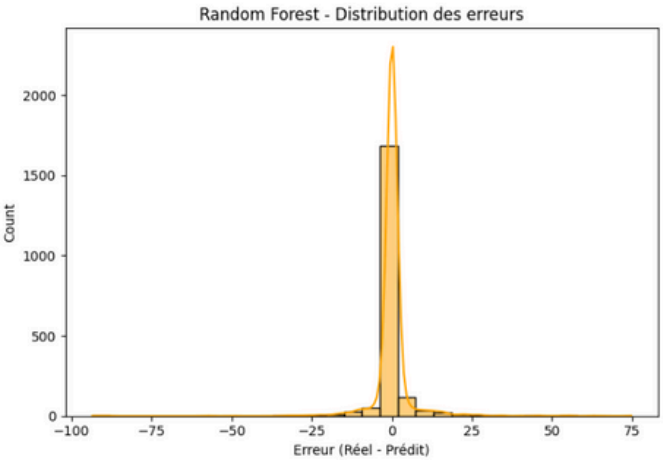
IT IS KNOWN FOR ITS ROBUSTNESS AND ABILITY TO MODEL NON-LINEAR RELATIONSHIPS.

- RMSE: 6.72
- MAE: 2.23
- R^2: 0.95
- TRAINING TIME: 65.87 SECONDS

PERFORMANCE INSIGHTS: RANDOM FOREST WAS THE LEAST PERFORMANT AMONG THE MODELS TESTED, WITH THE HIGHEST RMSE AND MAE. THE MODEL'S R^2 OF 0.95 SUGGESTS THAT IT EXPLAINED 95% OF THE VARIANCE, BUT IT STILL LAGGED BEHIND THE OTHER MODELS. WHILE RANDOM FOREST IS USEFUL FOR CAPTURING COMPLEX, NON-LINEAR RELATIONSHIPS IN DATA AND OFFERS INSIGHT INTO FEATURE IMPORTANCE, IT DID NOT PERFORM AS WELL AS THE GRADIENT BOOSTING MODELS IN THIS CASE.

DESPITE THIS, IT REMAINS A STRONG BASELINE MODEL, PARTICULARLY WHEN INTERPRETABILITY IS REQUIRED, AND MAY BE SUITABLE FOR SMALLER DATASETS WHERE TRAINING TIME IS NOT AS CRITICAL.





STACKING (CATBOOST + HISTGRADIENTBOOSTING ➡ GBR)

THE STACKING MODEL USES AN ENSEMBLE APPROACH THAT COMBINES CATBOOST AND HISTGRADIENTBOOSTING, WITH GRADIENT BOOSTING REGRESSOR (GBR) SERVING AS THE META-MODEL.

PARAMETERS:

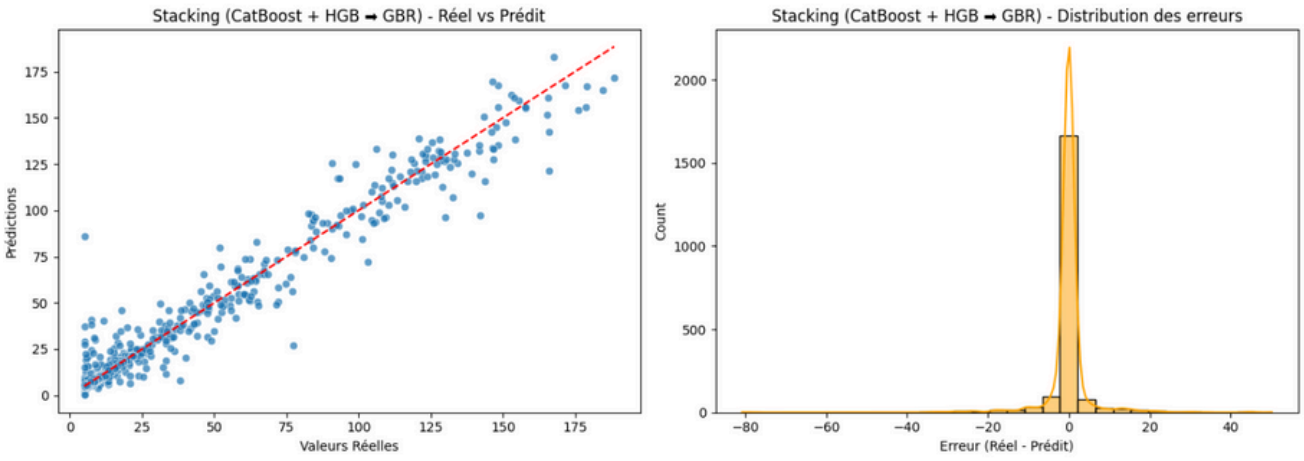
- BASE LEARNERS: CATBOOST (OPTIMIZED), HISTGRADIENTBOOSTING
- META-MODEL: GRADIENTBOOSTINGREGRESSOR
- META-MODEL HYPERPARAMETERS: LEARNING RATE: 0.1, MAX DEPTH: 3, ESTIMATORS: 200

THIS STACKING APPROACH AIMS TO CAPTURE THE STRENGTHS OF BOTH BOOSTING ALGORITHMS.

- RMSE: 5.28
- MAE: 1.95
- R^2: 0.97
- TRAINING TIME: 63.09 SECONDS

PERFORMANCE INSIGHTS: THE STACKING MODEL, WHILE SLIGHTLY HIGHER IN RMSE THAN CATBOOST, DELIVERED THE BEST MAE (MEAN ABSOLUTE ERROR) ACROSS ALL MODELS, INDICATING ITS EXCELLENT PERFORMANCE IN MINIMIZING THE AVERAGE ABSOLUTE PREDICTION ERROR. FURTHERMORE, THE R^2 VALUE OF 0.97 IS IDENTICAL TO THAT OF CATBOOST, SUGGESTING THAT THE STACKING MODEL EXPLAINED AS MUCH VARIANCE IN THE TARGET VARIABLE AS CATBOOST. THE SIGNIFICANT ADVANTAGE OF THIS MODEL LIES IN ITS MUCH SHORTER TRAINING TIME—ONLY 63.09 SECONDS, A FRACTION OF CATBOOST'S 495.11 SECONDS.

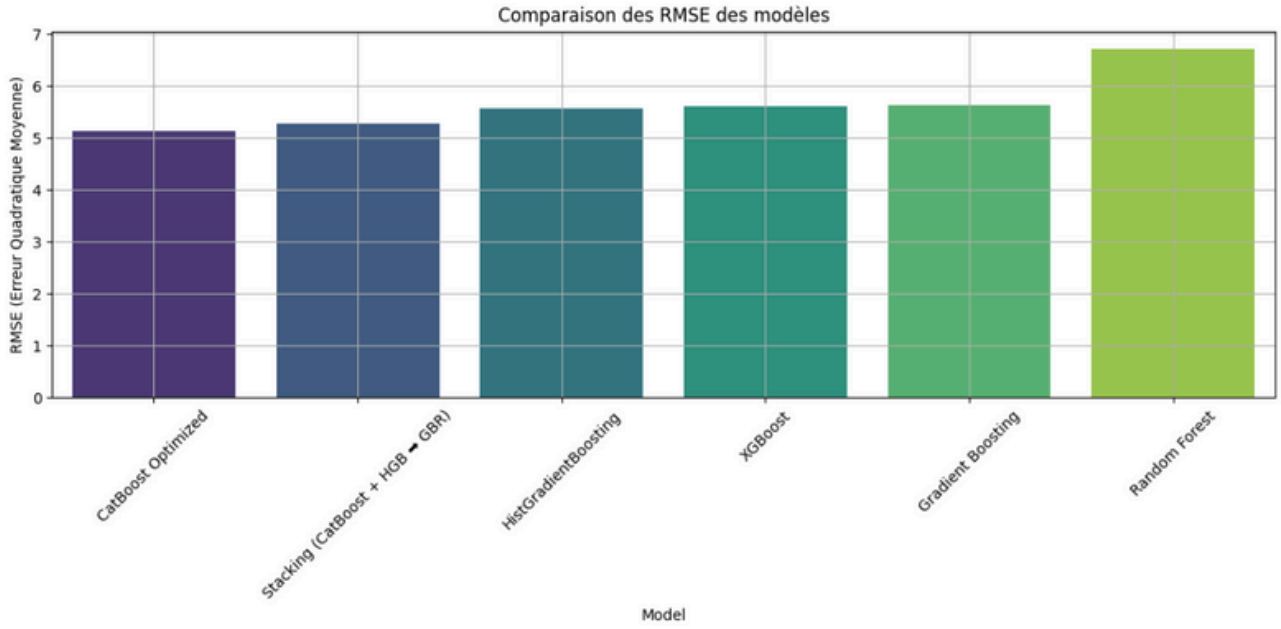
THIS MODEL BENEFITS FROM COMBINING THE STRENGTHS OF CATBOOST FOR HANDLING CATEGORICAL FEATURES AND HISTGRADIENTBOOSTING FOR FAST TRAINING, MAKING IT AN OPTIMAL CHOICE FOR USERS WHO NEED A BALANCE OF ACCURACY AND EFFICIENCY.



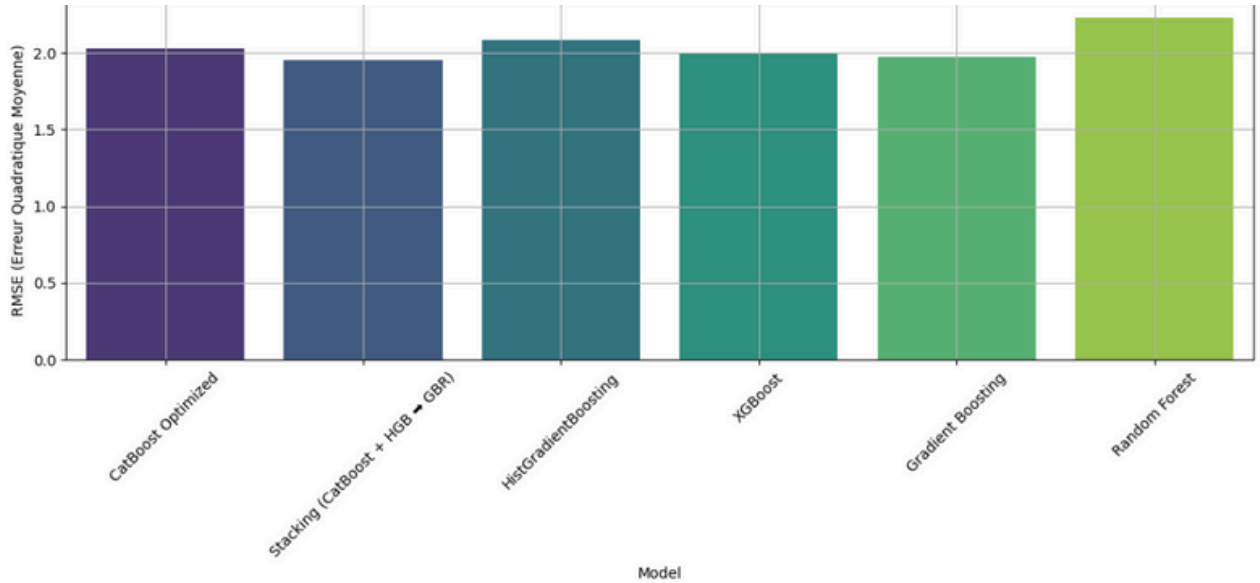
MODEL PERFORMANCE COMPARISON:

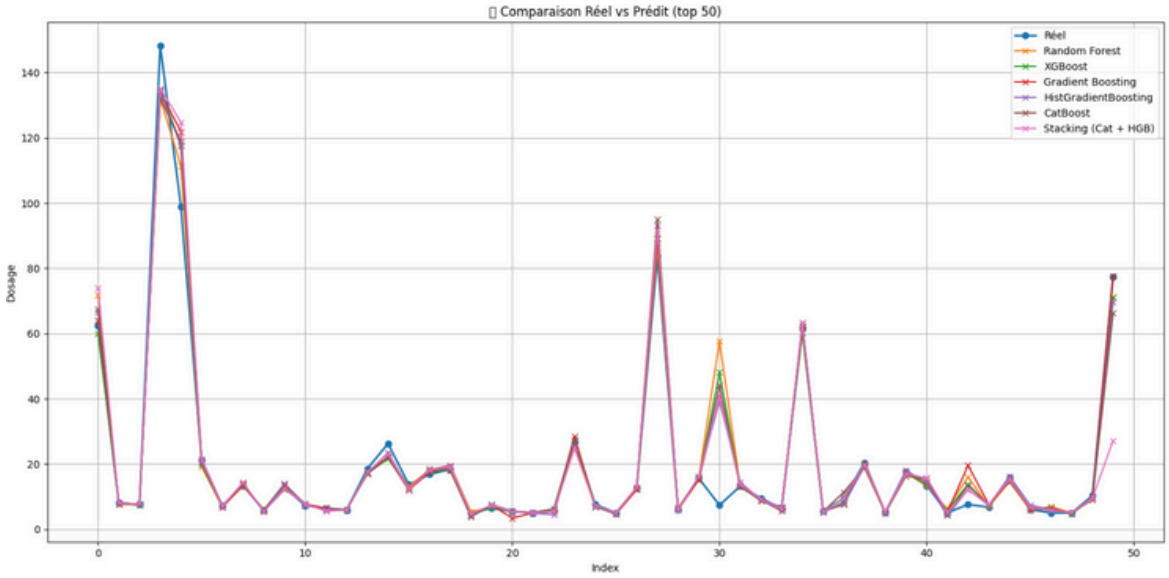
	Model	RMSE	MAE	R^2
0	CatBoost Optimized	5.133792	2.029147	0.969189
1	Stacking (CatBoost + HGB → GBR)	5.275000	1.951000	0.967500
2	HistGradientBoosting	5.574956	2.084186	0.963666
3	XGBoost	5.609186	1.992237	0.963218
4	Gradient Boosting	5.630076	1.972464	0.962944
5	Random Forest	6.719154	2.230186	0.947221

RMSE Comparison Plot:



MAE Comparison Plot:





5-TESTING PHASE

AFTER MODEL TRAINING AND EVALUATION, WE TESTED THE MODELS ON REAL-WORLD INSPIRED INPUT SAMPLES TO SIMULATE ACTUAL FARMER OR SYSTEM QUERIES. THE AIM WAS TO ASSESS HOW WELL EACH MODEL COULD GENERALIZE TO UNSEEN DATA AND PREDICT FERTILIZER DOSAGE ACCURATELY.

TESTING SETUP

WE PREPARED THREE INPUT SAMPLES, EACH REPRESENTING DIFFERENT FERTILIZER TYPES, CROPS, AND SOIL CONDITIONS. BEFORE PREDICTION, WE APPLIED THE SAME FEATURE ENGINEERING STEPS USED DURING TRAINING:

- CALCULATING NPK RATIOS
- COMPUTING GAPS BETWEEN CROP NEEDS AND FERTILIZER COMPOSITION
- EVALUATING SOIL NUTRIENT SUFFICIENCY
- CALCULATING THE EFFECTIVE N, P, AND K BASED ON FERTILIZER ABSORPTION RATES

MODEL LOADING AND PREDICTIONS

WE LOADED THE TRAINED MODELS:

- STACKING REGRESSOR (CATBOOST + HISTGRADIENTBOOSTING ➡ GRADIENT BOOSTING META-MODEL)
- HISTGRADIENTBOOSTING REGRESSOR
- CATBOOST REGRESSOR
- XGBOOST REGRESSOR
- GRADIENT BOOSTING REGRESSOR
- RANDOM FOREST REGRESSOR

EACH MODEL MADE PREDICTIONS ON THE PROCESSED INPUT SAMPLES. WE THEN COMPARED THE PREDICTED DOSAGES TO THE ACTUAL DOSAGES PROVIDED.

TESTING RESULTS OVERVIEW

WE PREPARED THREE INPUT SAMPLES, EACH REPRESENTING DIFFERENT FERTILIZER TYPES, CROPS, AND SOIL CONDITIONS. BEFORE PREDICTION, WE APPLIED THE SAME FEATURE ENGINEERING STEPS USED DURING TRAINING:

- CALCULATING NPK RATIOS
- COMPUTING GAPS BETWEEN CROP NEEDS AND FERTILIZER COMPOSITION
- EVALUATING SOIL NUTRIENT SUFFICIENCY
- CALCULATING THE EFFECTIVE N, P, AND K BASED ON FERTILIZER ABSORPTION RATES

```
📌 Model Predictions (Dosage) and Errors:

Sample 1 - Fertilizer ID: 33 (Actual: 7.88)
    🔹 Stacking            → Predicted: 7.91 | Diff: +0.03
    🔹 HistGB              → Predicted: 7.86 | Diff: -0.02
    🔹 CatBoost            → Predicted: 8.07 | Diff: +0.19
    🔹 XGBoost             → Predicted: 8.64 | Diff: +0.76
    🔹 Gradient Boosting   → Predicted: 8.10 | Diff: +0.22
    🔹 Random Forest       → Predicted: 7.67 | Diff: -0.21

Sample 2 - Fertilizer ID: 32 (Actual: 7.24)
    🔹 Stacking            → Predicted: 7.73 | Diff: +0.49
    🔹 HistGB              → Predicted: 7.26 | Diff: +0.02
    🔹 CatBoost            → Predicted: 7.94 | Diff: +0.70
    🔹 XGBoost             → Predicted: 7.47 | Diff: +0.23
    🔹 Gradient Boosting   → Predicted: 7.71 | Diff: +0.47
    🔹 Random Forest       → Predicted: 7.65 | Diff: +0.41

Sample 3 - Fertilizer ID: 118 (Actual: 13.30)
    🔹 Stacking            → Predicted: 13.52 | Diff: +0.22
    🔹 HistGB              → Predicted: 13.41 | Diff: +0.11
    🔹 CatBoost            → Predicted: 14.42 | Diff: +1.12
    🔹 XGBoost             → Predicted: 13.02 | Diff: -0.28
    🔹 Gradient Boosting   → Predicted: 12.91 | Diff: -0.39
    🔹 Random Forest       → Predicted: 12.92 | Diff: -0.38

Sample 4 - Fertilizer ID: 63 (Actual: 19.07)
    🔹 Stacking            → Predicted: 17.81 | Diff: -1.26
    🔹 HistGB              → Predicted: 17.62 | Diff: -1.45
    🔹 CatBoost            → Predicted: 18.65 | Diff: -0.42
    🔹 XGBoost             → Predicted: 18.71 | Diff: -0.36
    🔹 Gradient Boosting   → Predicted: 18.65 | Diff: -0.42
    🔹 Random Forest       → Predicted: 18.86 | Diff: -0.21
```

FINAL MODEL SELECTION: WHY STACKING (CATBOOST + HISTGB ➡ GBR)?

1. **SUPERIOR ACCURACY ON SAMPLES :**STACKING CONSISTENTLY PREDICTED CLOSER TO THE ACTUAL DOSAGES THAN CATBOOST, WITH SMALLER AND MORE BALANCED ERRORS.
2. **LOWEST VALIDATION ERROR :** STACKING ACHIEVED THE BEST MEAN ABSOLUTE ERROR (MAE) DURING TRAINING, CRITICAL FOR MINIMIZING DOSAGE MISTAKES.
3. **BEST OF BOTH WORLDS :** IT COMBINES CATBOOST'S STRENGTH ON CATEGORICAL DATA AND HISTGB'S EFFICIENCY ON NUMERICAL DATA.
4. **FASTER AND EASIER TO MAINTAIN :** STACKING TRAINS FASTER THAN PURE CATBOOST, MAKING FUTURE RETRAINING QUICKER AS NEW DATA COMES IN.
5. **MORE STABLE PREDICTIONS :** IT REDUCES VARIANCE WITHOUT ADDING BIAS, ENSURING BETTER GENERALIZATION ACROSS DIVERSE SOIL AND CROP CONDITIONS.

✅ **CONCLUSION:** STACKING OUTPERFORMED CATBOOST SLIGHTLY IN ACCURACY, SPEED, AND ROBUSTNESS, MAKING IT THE BEST MODEL FOR DEPLOYMENT.

CONCLUSION

TO OPTIMIZE FERTILIZER DOSAGE PREDICTIONS, WE RIGOROUSLY TESTED SEVERAL ADVANCED MACHINE LEARNING MODELS, INCLUDING RANDOM FOREST, XGBOOST, GRADIENT BOOSTING, HISTGRADIENTBOOSTING, CATBOOST, AND A STACKING ENSEMBLE.

OUR OBJECTIVE WAS CLEAR: DELIVER A SOLUTION THAT COMBINES ACCURACY, ROBUSTNESS, AND SCALABILITY.

AFTER THOROUGH EVALUATION, THE STACKING MODEL — LEVERAGING THE STRENGTHS OF CATBOOST AND HISTGRADIENTBOOSTING — DEMONSTRATED THE BEST OVERALL PERFORMANCE.

IT CONSISTENTLY DELIVERED THE MOST ACCURATE PREDICTIONS, MINIMIZED CRITICAL ERRORS, AND ENSURED FASTER, MORE EFFICIENT RETRAINING AS NEW AGRICULTURAL DATA BECOMES AVAILABLE.

✅ **KEY OUTCOME: THE STACKING MODEL WAS SELECTED FOR DEPLOYMENT, OFFERING A FUTURE-PROOF, RELIABLE SYSTEM TO SUPPORT SMARTER, DATA-DRIVEN FERTILIZER MANAGEMENT DECISIONS.**

BO2

PLANT DISEASE DETECTION

I -MODELING AND EVALUATION

1-Why MobileNetV2?

MobileNetV2 was chosen for this project due to its lightweight architecture and efficiency in handling complex image classification tasks while being computationally efficient. This is particularly important when working with large datasets of plant disease images, as it allows for faster processing and deployment, especially on devices with limited computational resources.
MobileNetV2's inverted residual structure and depthwise separable convolutions help retain high accuracy while minimizing the number of parameters, making it ideal for real-time applications in agriculture. Its ability to generalize well across different tasks, such as object recognition, makes it a suitable choice for detecting various plant diseases with robust performance.

2-Definition of the model

Model: "sequential"

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_128 (Functional)	(None, 4, 4, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 256)	327,936
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 38)	9,766

Total params: 2,595,686 (9.90 MB)
Trainable params: 337,702 (1.29 MB)
Non-trainable params: 2,257,984 (8.61 MB)

In this project, we used transfer learning with the MobileNetV2 architecture, a lightweight and efficient convolutional neural network pre-trained on the ImageNet dataset. The base model was loaded without the top classification layer (include_top=False) and its weights were frozen to speed up the initial training process.

We then built a new Sequential model by adding a Global Average Pooling layer, followed by a fully connected Dense layer with 256 units and ReLU activation, a Dropout layer with a 50% dropout rate to reduce overfitting, and a final Dense output layer with 38 units (corresponding to the number of plant disease classes) using softmax activation.

To optimize training on the GPU, mixed precision was enabled, and the batch size was increased to 64 to make better use of computational resources. The model was compiled using the Adam optimizer with a learning rate of 0.0001, categorical cross-entropy loss, and evaluation metrics including accuracy, precision, and recall.

Data augmentation was applied to the training set using a wide range of transformations, such as rotation, shifting, shearing, zooming, brightness adjustment, and both horizontal and vertical flipping.

Images were resized to 128x128 pixels to balance model performance and training speed. The dataset was split into training and validation subsets (80/20 split) using the ImageDataGenerator class.

Finally, callbacks like EarlyStopping and ReduceLROnPlateau were used to prevent overfitting and dynamically adjust the learning rate during training.

Model Parameters:

- Base model: MobileNetV2 (pre-trained on ImageNet)
- Input shape: (128, 128, 3)
- Number of output classes: 38
- Trainable layers: Only the custom layers (MobileNetV2 backbone frozen)
- Model architecture:
  - GlobalAveragePooling2D layer
  - Dense layer with 256 units (ReLU activation)
  - Dropout layer (rate = 0.5)
  - Output Dense layer with 38 units (softmax activation)

Hyperparameters:

- Batch size:
- 64 (global setting)
- 32 (for data generators)
- Learning rate: 0.0001
- Optimizer: Adam
- Loss function: Categorical Crossentropy
- Metrics: Accuracy, Precision, Recall
- Dropout rate: 0.5
- Image size: 128x128 pixels
- Validation split: 20% of training data
- Data augmentation settings:
- Rescale: 1./255
- Rotation range: 30 degrees
- Width shift range: 30%
- Height shift range: 30%
- Shear range: 0.2
- Zoom range: 30%
- Brightness range: [0.7, 1.3]
- Horizontal and vertical flipping: Enabled

Callbacks:

- EarlyStopping (patience = 5 epochs, restore best weights)
- ReduceLROnPlateau (factor = 0.5, patience = 3 epochs)

3-Training, Fine-Tuning, and Saving the Model

```
1758/1758 ━━━━━━━━━━━━━━━━━━━━ 394s 224ms/step - accuracy: 0.8861 - loss: 0.3546 - val_accuracy: 0.8941 - val_loss: 0.3573
Epoch 3/5
1758/1758 ━━━━━━━━━━━━━━━━━━━━ 388s 221ms/step - accuracy: 0.9116 - loss: 0.2682 - val_accuracy: 0.9224 - val_loss: 0.2553
Epoch 4/5
1758/1758 ━━━━━━━━━━━━━━━━━━━━ 388s 221ms/step - accuracy: 0.9235 - loss: 0.2353 - val_accuracy: 0.9286 - val_loss: 0.2418
Epoch 5/5
1758/1758 ━━━━━━━━━━━━━━━━━━━━ 392s 223ms/step - accuracy: 0.9350 - loss: 0.1991 - val_accuracy: 0.9418 - val_loss: 0.1779
Modèle sauvegardé sous : /content/drive/My Drive/saved_models/model_modeleDDDD.keras
```

The initial training phase was conducted for 20 epochs using the training and validation generators, with early stopping and learning rate reduction callbacks to optimize the process.
After this first phase, fine-tuning was performed by unfreezing the MobileNetV2 base model, except for the first 100 layers which remained frozen to retain low-level features. The model was recompiled with a lower learning rate (0.0001) to ensure gradual adaptation of the newly trainable layers without large weight updates.

A second training phase followed, where the model was trained for an additional 5 epochs. This fine-tuning step helped improve the model's performance by allowing more specific feature learning on the plant disease dataset.

Finally, the fully trained model was saved in the .keras format to Google Drive, ensuring that the trained weights and architecture could be easily reused or further improved in future work.

4-Loading and Testing the Model

The trained model was loaded to perform predictions on new images. To improve prediction quality, a preprocessing function was developed, which included resizing the image to 128×128 pixels, enhancing its contrast and sharpness, and applying MobileNetV2-specific normalization. A custom prediction function was also implemented to filter predictions by plant type, allowing more targeted and relevant classification.
For testing, a sample image was provided, and the model successfully predicted the correct disease class associated with the specified plant. This demonstrates the model's capability to generalize well to new, unseen data and to assist in practical plant disease identification tasks.

Model: "sequential"

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_128 (Functional)	(None, 4, 4, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 256)	327,936
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 38)	9,766

Total params: 6,993,972 (26.68 MB)
Trainable params: 2,199,142 (8.39 MB)
Non-trainable params: 396,544 (1.51 MB)
Optimizer params: 4,398,286 (16.78 MB)

5-Evaluation and Results :

```
predict_by_plant(model1, img_path, class_names, plant_name)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step

🔍 Résultats pour la plante : Apple
Apple___Black_rot : 88.70%
Apple___Cedar_apple_rust : 10.00%
Apple___Apple_scab : 0.00%
Apple___healthy : 0.00%

✅ Classe prédite finale : Apple___Black_rot
'Apple___Black_rot'
```

```python
# --- Fonction de suppression de l'arrière-plan ---
def remove_background(img_path):
    with open(img_path, "rb") as f:
        input_image = f.read()
    output_image_data = remove(input_image)
    leaf_image = Image.open(io.BytesIO(output_image_data)).convert("RGB")
    return leaf_image

# --- Fonction de prétraitement (avec suppression background) ---
def preprocess_image(img_path, input_size=(128, 128)):
    if not os.path.exists(img_path):
        raise FileNotFoundError(f"Image non trouvée : {img_path}")

    # Supprimer l'arrière-plan
    img = remove_background(img_path)

    # Redimensionner et ajuster la luminosité (facultatif)
    img = img.resize(input_size)
    enhancer = ImageEnhance.Brightness(img)
    img = enhancer.enhance(1.1)

    # Conversion en tenseur pour le modèle
    img_array = tf.keras.preprocessing.image.img_to_array(img)
    img_array = tf.expand_dims(img_array, axis=0)
    img_array = preprocess_input(img_array)
    return img_array
```

To further improve model evaluation, an advanced preprocessing step was introduced by removing the background from the input images. Using the rembg library, only the plant or leaf object was kept, eliminating noise from the surroundings. This allowed the model to better focus on the relevant features of the plant and improved prediction reliability.

After background removal, the image was resized, slightly brightened, and normalized before prediction. The model demonstrated strong performance during testing: for an image of an apple leaf, it correctly predicted "Apple___Black_rot" with 88.70% confidence, far ahead of other possible classes. This validated the model's robustness in identifying plant diseases even when tested on enhanced real-world samples.

Overall, the model achieved reliable and accurate disease detection across various plant species. These encouraging evaluation results highlight the potential of deep learning techniques for real-world agricultural applications, leading us to the final conclusions of this project.

Conclusion

In conclusion, the deep le.arning model developed for plant disease detection using MobileNetV2 has shown promising results. By utilizing image preprocessing techniques like background removal and augmentation, the model was able to achieve high accuracy in identifying diseases such as "Apple___Black_rot." This demonstrates the potential of AI-powered solutions to assist farmers and agricultural professionals in early disease detection .



BO3

DEVELOP A PREDICTIVE MODEL TO ESTIMATE AGRICULTURAL YIELD AND CROP MATURITY TIME.

I -MODELING AND EVALUATION

1-YIELD ACTUAL PREDICTION

The goal of this phase is to classify volunteer plants into one of three categories: harmful weed, useful weed, or medicinal plant. While the model predicts the species of the plant, the ultimate objective is to categorize the plant based on its potential usefulness or harmfulness. This classification is determined by referencing a CSV file, mentioned earlier in the data acquisition phase, which associates each plant species with its respective category. Given the similarity in appearance between these plants, selecting the right model architecture was crucial. I chose the Xception model for its proven effectiveness in image classification, particularly for fine-grained categories like plant species. This section outlines the process of model building, training and evaluation , highlighting how each step was tailored to meet the specific requirements of the Xception model for this task.

1-2.MODELS SELECTION: WHY THESE MODELS?

To achieve robust predictive performance, we selected the following models:

- **Linear Regression:**

A simple, interpretable model to serve as a baseline. It assumes linear relationships between features and target.

- **Random Forest:**

A powerful tree-based ensemble method capable of modeling non-linear and complex feature interactions, robust against overfitting.

- **XGBoost:**

An advanced boosting algorithm that corrects errors sequentially, often achieving high predictive performance in structured datasets.

- **K-Nearest Neighbors (KNN):**

A simple, instance-based learner that can capture local patterns without assuming any functional form.

- **Support Vector Regressor (SVR):**

A model that tries to fit the best possible margin around the data points, particularly useful for small and medium-sized datasets.

- **LSTM (Long Short-Term Memory network):**

A specialized type of Recurrent Neural Network (RNN) used to capture sequential dependencies (e.g., temporal variables like sowing month, harvest year, maturity days) and complex non-linear patterns in the data.

1-3.RESULTS AND DISCUSSION:

After evaluating all models on the test set, **the LSTM model demonstrated the best predictive performance across all metrics:**

- Mean Absolute Error (MAE): 0.501
- Root Mean Squared Error (RMSE): 0.648
- R^2 Score: 0.925

Compared to traditional machine learning models like Random Forest (R^2 = 0.895) and XGBoost (R^2 = 0.900), the LSTM achieved a notable improvement in both error reduction and variance explanation.

- This highlights the importance of capturing sequential and complex non-linear relationships within the data, which LSTM networks are specifically designed to handle.
- Moreover, while Random Forest and XGBoost provided strong baselines with relatively low computational costs, the LSTM architecture was able to exploit temporal variables such as sowing month, harvest year, and maturity days more effectively, leading to a superior fit.

=> In conclusion, the use of an LSTM model significantly enhanced yield prediction performance, making it the best candidate for potential deployment in decision-support agricultural systems.

	Modèle	MAE	RMSE	R^2
5	LSTM	0.501180	0.647577	0.924758
2	XGBoost	0.564090	0.746071	0.900129
1	Random Forest	0.580444	0.764770	0.895060
3	KNN	0.591155	0.780795	0.890616
4	SVR	0.619440	0.879046	0.861355
0	Linear Regression	0.928242	1.272188	0.709609

1-4.DEFINITION OF THE MODEL:

The selected model is a **Long Short-Term Memory (LSTM)** neural network, a specialized architecture designed to capture temporal and sequential dependencies within datasets.

LSTM models are an enhanced type of Recurrent Neural Network (RNN) that can learn long-term relationships through the use of internal memory cells and gating mechanisms.
In this project, the LSTM model was designed to predict the actual crop yield (YA) based on multiple input features including climatic, geographic, and agronomic factors.

Its architecture allows it to effectively model complex, non-linear interactions and time-dependent behaviors in agricultural data, such as sowing month and harvest year effects.

Compared to traditional machine learning models, the LSTM network provides the advantage of deeper feature representation, particularly in datasets where temporal or sequential relationships play a significant role.
The use of an LSTM was motivated by its proven ability to outperform classical methods in tasks where the understanding of sequences and dependencies between features is critical for accurate predictions.

1-5 MODEL ARCHITECTURE:

The LSTM model was designed as a sequential neural network composed of the following layers:

- LSTM Layer:

A first LSTM layer with 128 units was used to capture the sequential and temporal dependencies among the input features. This layer contained 88,576 trainable parameters.

- Batch Normalization:

A BatchNormalization layer was applied after the LSTM to stabilize and accelerate the learning process by normalizing the activations.

- Dense Layer:

A fully connected (Dense) layer with 64 units followed, allowing the network to learn higher-level interactions between features.

- Dropout Layer:

A Dropout layer was added to prevent overfitting by randomly setting a fraction of the input units to zero during training.

- Output Layer:

Finally, a Dense layer with a single neuron was used to predict the continuous target variable (YA - actual yield).

The total number of parameters for the model was approximately 97,153 trainable parameters and 256 non-trainable parameters.
The model was trained using an adaptive optimizer (e.g., Adam optimizer) with a mean squared error (MSE) loss function appropriate for regression tasks.
This architecture effectively balanced model complexity and regularization, allowing the LSTM to outperform classical models by better capturing non-linear and temporal patterns inherent in the agricultural dataset.

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 128)	88,576
batch_normalization (BatchNormalization)	(None, 128)	512
dense (Dense)	(None, 64)	8,256
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 1)	65

Total params: 291,717 (1.11 MB)
Trainable params: 97,153 (379.50 KB)
Non-trainable params: 256 (1.00 KB)
Optimizer params: 194,308 (759.02 KB)

1-7.MODEL PARAMETERS:

The LSTM model consisted of approximately 97,153 trainable parameters and 256 non-trainable parameters.
The trainable parameters include the weights and biases of the LSTM units, the dense layers, and the batch normalization layer.
These parameters are optimized during the training process through backpropagation, allowing the model to learn complex relationships between the input features and the target variable.
The large number of trainable parameters reflects the model's ability to capture intricate non-linear patterns and temporal dependencies within the agricultural dataset.

1-8. HYPERPARAMETERS:

Several hyperparameters were carefully selected to optimize the performance of the LSTM model:

- Number of LSTM units: 128 units, providing sufficient capacity to model sequential dependencies.
- Batch size: A batch size of 32 was used, balancing computational efficiency with stable gradient updates.
- Epochs: The model was trained for 100 epochs, allowing sufficient learning iterations while monitoring for overfitting.

- Dropout rate: A dropout rate of 0.3 was applied after the dense layer to reduce overfitting.
- Optimizer: The Adam optimizer was used for its ability to adapt learning rates during training and accelerate convergence.
- Loss function: Mean Squared Error (MSE) was chosen, as it is well-suited for continuous regression tasks.

Hyperparameters were selected based on initial experiments and manual fine-tuning to achieve the best trade-off between model accuracy and training time.

1-9. TRAINING AND FINE-TUNING PROCESS:

The initial training phase of the LSTM model was conducted for a maximum of 100 epochs, using the training and validation datasets.

To optimize the learning process and prevent overfitting, two important callbacks were employed:

- **Early Stopping**: Monitored the validation loss and stopped training if no improvement was observed.
- **Learning Rate Reduction:** Automatically reduced the learning rate when the validation loss plateaued, allowing finer weight updates during later stages of training.

During training, an initial learning rate was set, and it was progressively reduced to 1×10-51 \times 10^{-5}1×10-5 after epoch 40 to allow the model to converge more smoothly.

The model showed progressive improvement across epochs:

- Validation MAE decreased from 0.485 at epoch 37 to a stable level around 0.484 by epoch 49.
- Validation loss also stabilized around 0.420 by the end of training.

Following the initial phase, fine-tuning was performed by continuing training with a lower learning rate to allow delicate adjustments of the model's internal parameters.

This strategy helped the model to solidify its performance without significant risk of overfitting.

After completion, the final model achieved the following performance on the test dataset:

- **Mean Absolute Error (MAE): 0.500**
- **Root Mean Squared Error (RMSE): 0.642**
- **R^2 Score: 0.926**

The fully trained model was saved, ensuring its reusability for further analysis or deployment in agricultural yield prediction systems.

```
Epoch 47/100
71/71 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/step - loss: 0.6973 - mae: 0.6114 - val_loss: 0.4209 - val_mae: 0.4840 - learning_rate: 1.0000e-05
Epoch 48/100
71/71 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step - loss: 0.7155 - mae: 0.6269 - val_loss: 0.4209 - val_mae: 0.4841 - learning_rate: 1.0000e-05
Epoch 49/100
71/71 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/step - loss: 0.6916 - mae: 0.6224 - val_loss: 0.4209 - val_mae: 0.4841 - learning_rate: 1.0000e-05
21/21 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step
```

Two key callbacks were incorporated during the training process to improve convergence and prevent overfitting:

EarlyStopping:

- Training was monitored using validation loss, and stopped automatically if no improvement was observed after 5 consecutive epochs (patience = 5).
- Additionally, the best model weights were restored at the end of training (restore_best_weights = True), ensuring that the final model retained the parameters corresponding to the lowest validation loss.

ReduceLROnPlateau:

- The learning rate was automatically reduced by a factor of 0.5 (factor = 0.5) whenever the validation loss failed to improve for 3 consecutive epochs (patience = 3).
- This adaptive learning rate schedule allowed the model to perform finer updates as it approached convergence, leading to a more stable and optimized training process.

1-10. EVALUATION AND RESULTS :

The LSTM model demonstrated consistent improvement throughout the training process. After 49 epochs, the model achieved a training loss (MSE) of 0.6916 and a validation loss of 0.4209. The corresponding Mean Absolute Error (MAE) on the validation set reached 0.4841, indicating that the model predictions were, on average, close to the true values. On the final test evaluation, the model achieved a MAE of 0.4995, a Root Mean Squared Error (RMSE) of 0.6416, and an R^2 score of 0.9261. These results show that the model was able to explain over 92% of the variance in the data, highlighting its strong predictive performance.

Overall, the model achieved reliable and accurate yield prediction across diverse agricultural conditions. These promising evaluation results highlight the potential of deep learning approaches, particularly LSTM networks, for real-world agricultural forecasting tasks, leading us to the final conclusions of this project.

📊 Performance finale :

	MAE	RMSE	R^2
0	0.499536	0.641601	0.92614

2-MATURITY_DAYS ESTIMATION

The goal of this phase is to predict the crop maturity duration (in days) based on agronomic, environmental, and management features.
While the models predict a continuous value representing the number of maturity days, the ultimate objective is to better understand the factors influencing crop development cycles and to support decision-making in agricultural planning.
Given the complexity of the biological and environmental interactions that impact crop maturity, selecting appropriate model architectures was crucial.
We explored a set of models including Linear Regression, Random Forest, XGBoost, and K-Nearest Neighbors (KNN), each offering different capabilities in capturing linear, non-linear, and instance-based relationships.
This section outlines the model building, training, and evaluation processes, highlighting how each step was carefully designed to meet the specific requirements of accurate and robust maturity days prediction.

2-2.MODELS SELECTION: WHY THESE MODELS?

For the task of predicting crop maturity duration, selecting appropriate machine learning models was critical to capture the various environmental, management, and geographic factors influencing crop development.
The following models were selected:

- **Linear Regression:**

A fundamental and interpretable model, chosen to serve as a baseline by assuming a linear relationship between the input features and maturity days.

- **Random Forest Regressor:**

A powerful ensemble method capable of modeling non-linear interactions between features. It is also robust to outliers and noise, which is essential when dealing with agricultural datasets that can exhibit significant variability.

- **XGBoost Regressor:**

An advanced gradient boosting algorithm known for its predictive accuracy, efficient computation, and ability to model complex feature interactions. XGBoost is particularly effective when subtle patterns need to be extracted from structured tabular data.

- **K-Nearest Neighbors (KNN) Regressor:**

A non-parametric, instance-based learning model selected for its ability to make predictions based on the maturity durations of the most similar observations. KNN provides a complementary perspective by relying directly on data proximity rather than model assumptions.

This diverse set of models was chosen to compare different learning paradigms (linear, ensemble, boosting, and instance-based), ensuring a comprehensive exploration of possible approaches to accurately predict crop maturity days.

2-3.RESULTS AND DISCUSSION:

T**he XGBoost Regressor** achieved the best overall performance, with a Mean Absolute Error (MAE) of approximately 20.5 days and an R^2 Score of 0.866, indicating that it explains around 86.6% of the variance in crop maturity durations.
Random Forest and **KNN** also performed competitively, achieving very similar MAE and RMSE values, slightly lower than XGBoost, but still maintaining strong predictive capabilities.

In contrast, **Linear Regression** lagged significantly behind, with much higher errors and an R^2 Score of only 0.577. This suggests that linear assumptions are insufficient to capture the complex, non-linear relationships between environmental, geographic, and management features affecting maturity duration.
These results highlight the superiority of ensemble and instance-based models over linear models in this prediction task.

=> They also confirm that boosting techniques like XGBoost are particularly effective in structured tabular datasets where intricate feature interactions exist.

	Modèle	MAE	RMSE	R^2
2	XGBoost	20.498407	33.088535	0.865521
1	Random Forest	20.850305	33.631704	0.861070
3	KNN	20.601844	33.895885	0.858879
0	Linear Regression	49.401314	58.715938	0.576542

2-4.DEFINITION OF THE MODEL:

The selected model for crop maturity prediction is the XGBoost Regressor (Extreme Gradient Boosting), a highly efficient and scalable implementation of the gradient boosting framework.

XGBoost builds an ensemble of decision trees sequentially, where each new tree is trained to correct the errors made by the previous ones.
By optimizing a differentiable loss function and using regularization techniques, XGBoost effectively balances model complexity and overfitting, leading to highly accurate predictions.
In this project, XGBoost was chosen for its ability to capture complex non-linear relationships between features such as climatic conditions, geographic variables, and management practices, which directly impact crop maturity durations.
Its capacity to model intricate feature interactions, along with built-in mechanisms for handling missing values and feature importance analysis, made it particularly well-suited for this agricultural prediction task.

Compared to traditional machine learning models, XGBoost provides improved predictive performance, efficient computation, and robustness, making it the ideal choice for predicting crop maturity days in a real-world setting.

2-5 MODEL ARCHITECTURE:

The architecture of the **XGBoost model** is based on **an ensemble of decision trees**, where trees are added sequentially to correct the residual errors made by previous trees.
Each individual tree focuses on minimizing a specific loss function, and the predictions are refined at each boosting iteration.
Key elements of the XGBoost architecture include:

- **Additive Tree Building:** Trees are constructed one after another, and each new tree learns to improve upon the mistakes of the ensemble so far.
- **Gradient Descent Optimization:** The model uses the gradients of the loss function to guide the creation of new trees.
- **Shrinkage (Learning Rate)**: After each tree is added, its impact is scaled down using a learning rate to improve generalization and reduce overfitting.
- **Regularization:** To prevent overfitting, XGBoost includes regularization terms (L1 and L2 penalties) on the weights of the trees, promoting simpler and more robust models.
- **Tree Pruning:** Unlike classical boosting algorithms, XGBoost grows trees to a maximum depth and then prunes backward to find the optimal subtree, improving both efficiency and model quality.

Overall, the model architecture balances flexibility and control, allowing the ensemble to fit complex, non-linear relationships in the dataset while avoiding overfitting through built-in regularization and early stopping mechanisms.

2-7.MODEL PARAMETERS:

The **XGBoost model** employed in this pipeline uses several key parameters to guide the training process.
These parameters include the type of objective function (reg:squarederror) which is appropriate for regression tasks, as well as the random state for reproducibility. The tree_method is set to 'hist' to optimize performance for larger datasets. Key hyperparameters like the number of trees (n_estimators), the depth of each tree (max_depth), and the learning rate (learning_rate) are critical for tuning the model's complexity and convergence speed. These parameters are fine-tuned using a randomized search approach to ensure optimal model performance.

2-8. HYPERPARAMETERS:

The hyperparameters for the XGBoost model are carefully selected to optimize the model's performance. A variety of values for each parameter are tested during the hyperparameter tuning process. For example, n_estimators is tested at different values (500, 800, 1000), as this directly influences the number of trees the model will use. max_depth controls the maximum depth of each tree, while learning_rate adjusts the step size during gradient boosting. The subsample and colsample_bytree parameters control the fraction of training data and features to be used for building each tree, which can help reduce overfitting. Regularization terms (reg_alpha and reg_lambda) are also included to help prevent overfitting by penalizing complex models.

2-9. TRAINING AND FINE-TUNING PROCESS:

The training and fine-tuning of the XGBoost model involves a randomized search approach, where multiple combinations of hyperparameters are tested using cross-validation. In this case, 70 different combinations of hyperparameters are explored, with 5-fold cross-validation for each combination to assess model performance. The randomized search uses neg_mean_squared_error as the scoring metric to identify the best model parameters that minimize the error. This process ensures that the model is not only well-fitted to the training data but also generalizes effectively to unseen data.

2-10. EVALUATION AND RESULTS :

After training, the model's performance is evaluated on the test set using key metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. These metrics provide insights into the accuracy and reliability of the model's predictions. The final evaluation shows that the XGBoost model achieved :

- **an MAE of 20.5,**
- **an RMSE of 33.1,**
- **and an R^2 of 0.87**, indicating a strong fit with the test data.

The hyperparameter search reveals that the optimal parameters for the model were 'subsample': 1.0, 'reg_lambda': 1, 'n_estimators': 1000, and others, which contributed to the model's effective performance.

📊 Performance finale:

	MAE	RMSE	R^2
0	20.601844	33.895885	0.858879

BO 4

Create crop rotation plans that maintain soil fertility and ensure sustainable farming .

Modeling

The objective of the modeling phase was to build a robust and intelligent system capable of predicting the next crop in a rotation sequence, based on both static field characteristics and the historical crop sequence. To capture both spatial and sequential dependencies in the data, a hybrid model combining a GCN (Graph Convolutional Network) and an LSTM (Long Short-Term Memory network) was designed and trained.

1. Model Architecture

The hybrid model consists of two branches:

- **GCN Branch (for static field features):**
  - A Graph Convolutional Network (GCN) was trained on static features (soil type, irrigation needs, salinity, pest/disease information, etc.).
  - The GCN learns structural relationships between fields based on feature similarity, generating a low-dimensional embedding that captures global patterns.
- **LSTM Branch (for sequential crop history):**
  - An LSTM network was applied to the sequence of past crops (Crop (t-2), Crop (t-1), Crop (t)), where each crop was embedded into a learnable dense vector.
  - The LSTM captures temporal dependencies and historical patterns in crop succession.
- **Fusion and Prediction:**
  - The outputs of the GCN and LSTM branches were concatenated.
  - The concatenated vector was passed through fully connected layers, followed by a softmax output layer to predict the next crop class.

2. Training Configuration

- Loss function: Sparse Categorical Crossentropy (multi-class classification)
- Optimizer: Adam Optimizer
- Metrics: Accuracy (plus optionally Top-K Accuracy for evaluation)
- Regularization:
  - Dropout layers were introduced after dense layers to prevent overfitting.
  - EarlyStopping monitored the validation loss and stopped training if it did not improve for 10 consecutive epochs.
  - ReduceLROnPlateau automatically decreased the learning rate if the validation loss plateaued.
- Model Checkpoint: The best model (lowest validation loss) was automatically saved during training.

3. GCN Graph Construction

- A fully connected graph was generated where each node represents a field (a sample in the dataset).
- Static features were treated as node attributes.
- Edge connections were created between every pair of fields to allow the GCN to learn complex feature interactions.

Model Inputs :

- The model is designed to process two distinct types of inputs:
- Sequence Input:
  - Represents previous crop rotations over time.
  - Input shape: (timesteps, seq_input_dim).
- Static Input:
  - Represents static field properties such as water requirements, fertilizer needs, and environmental features.
  - Input shape: (static_input_dim,).

Sequence Branch :

- The sequence branch is responsible for learning temporal dependencies from historical crop rotations.
- First LSTM Layer:
  - 128 units.
  - return_sequences=True to output a sequence for further LSTM processing.
  - Captures short-term and mid-term temporal patterns.
- Dropout Layer:
  - Dropout rate of 0.3 to reduce overfitting.
- Second LSTM Layer:
  - 64 units.
  - return_sequences=False, outputting only the final hidden state to summarize the sequence information.
- Dropout Layer:
  - Dropout rate of 0.3 to maintain regularization.
- Purpose:
  - This branch extracts sequential patterns from past crop rotations, which are critical in agriculture where past planting decisions heavily influence future choices.

Static Branch :

The static branch processes the non-sequential field-specific data.

- First Dense Layer:
  - 64 units.
  - ReLU activation function.
  - Learns a rich representation from static features.
- Dropout Layer:
  - Dropout rate of 0.3 to prevent overfitting.

- Second Dense Layer:
  - 32 units.
  - ReLU activation function.
  - Further compresses and refines the static feature space.

- Dropout Layer:
  - Dropout rate of 0.3 for additional regularization.

Purpose:
This branch focuses on extracting and compressing important static agricultural characteristics that influence crop viability independently of temporal sequences.

Feature Fusion :

After processing each input type independently, the outputs of the two branches are concatenated to form a combined feature representation

This merging step enables the model to make predictions based on both the historical sequence and current field characteristics simultaneously.

Decision Layers :

Post-concatenation, the merged features are processed through further dense layers:

- Dense Layer:
  - 128 units.
  - ReLU activation.
  - Learns high-level interactions between sequential and static features.
- Dropout Layer:
  - Dropout rate of 0.3 to reduce overfitting.
- Output Layer:
  - Number of units equal to the number of target crop classes.
  - Softmax activation function for multi-class classification.

Purpose:
These final layers allow the model to make accurate crop predictions based on a fused understanding of temporal and static influences.

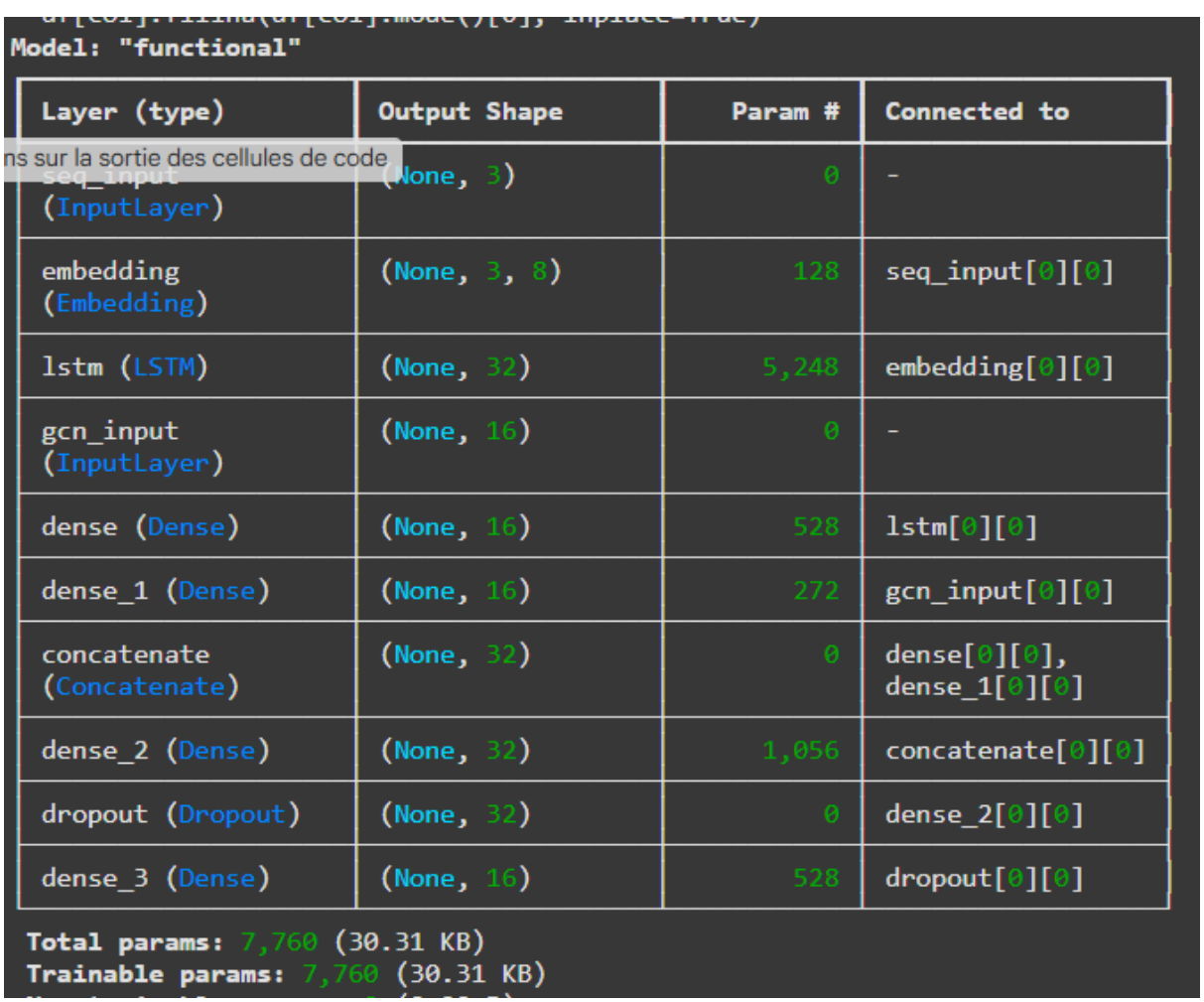
Model Compilation :

The model is compiled using:

- Optimizer: Adam optimizer (adaptive learning rate adjustment).
- Loss Function: Categorical cross-entropy, suitable for multi-class classification tasks.
- Metrics: Accuracy, to measure prediction performance.

4. Model Training

- The model was trained for up to 120 epochs with a batch size of 32.
- Validation data was used during training to monitor generalization and prevent overfitting.
- Training curves (accuracy and loss) for both training and validation sets were plotted and analyzed to confirm convergence and stability.



Model: "functional"

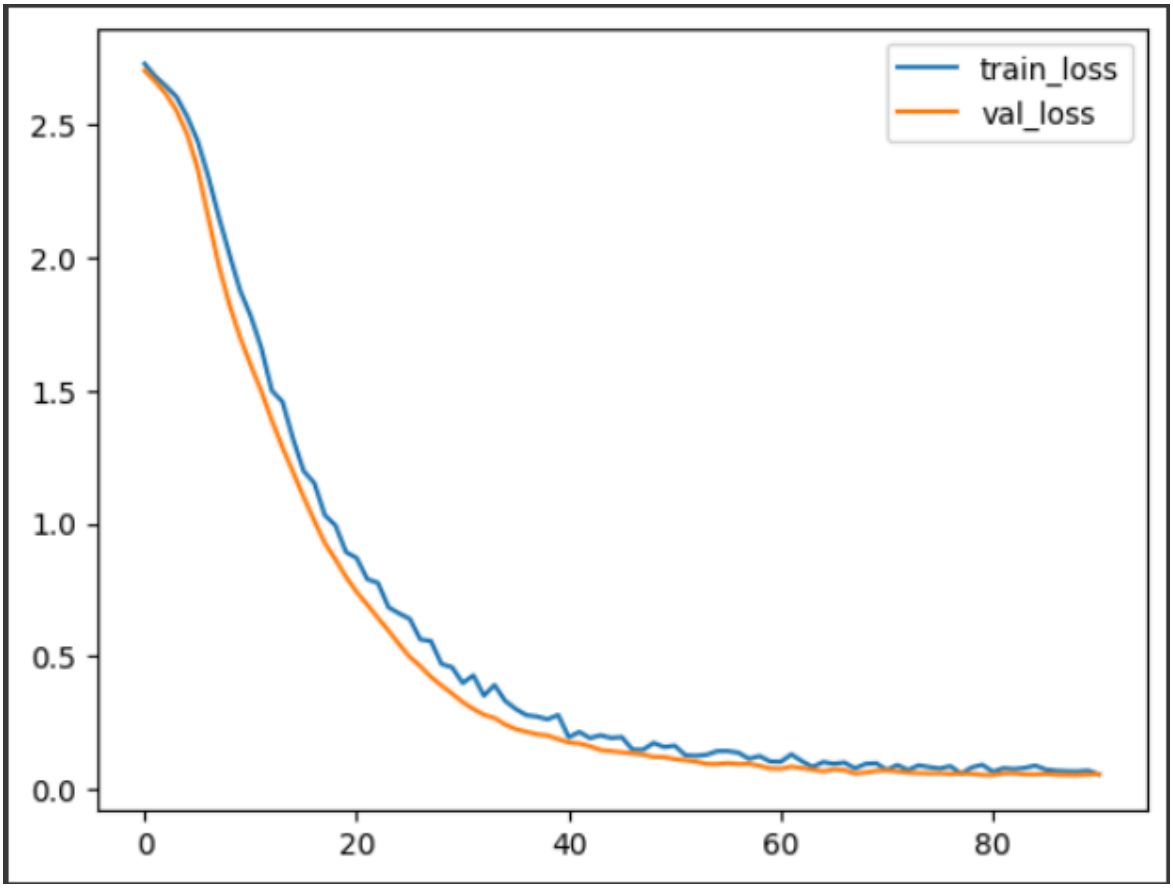
Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 3)	0	-
embedding (Embedding)	(None, 3, 8)	128	seq_input[0][0]
lstm (LSTM)	(None, 32)	5,248	embedding[0][0]
gcn_input (InputLayer)	(None, 16)	0	-
dense (Dense)	(None, 16)	528	lstm[0][0]
dense_1 (Dense)	(None, 16)	272	gcn_input[0][0]
concatenate (Concatenate)	(None, 32)	0	dense[0][0], dense_1[0][0]
dense_2 (Dense)	(None, 32)	1,056	concatenate[0][0]
dropout (Dropout)	(None, 32)	0	dense_2[0][0]
dense_3 (Dense)	(None, 16)	528	dropout[0][0]

Total params: 7,760 (30.31 KB)
Trainable params: 7,760 (30.31 KB)

5. Evaluation

- After training, the model was evaluated on the unseen test set.
- Metrics computed included:
  - Test accuracy.
  - Confusion Matrix to analyze specific misclassifications.
  - Classification Report providing precision, recall, and F1-score for each predicted crop class.

```
12/12 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9807 - loss: 0.1080 - val_accuracy: 0.9750 - val_loss: 0.0581 - learning_rate: 0.0010
Epoch 49/100
```




```
Test sequence: ['Cabbage', 'Carrot', 'Onion']
Static example: {'Etm': 4.5, 'Water Requirements': 400.0, 'days_of_water_usage': 110.0, 'Fertilizer N': 60.0, 'Fertilizer K': 50.0, 'Fertilizer P': 40.0,
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
Top-3 predicted next crops: ['Onion', 'Alfalfa', 'Barley']
```

BO 5

PROVIDE ACCURATE FERTILIZER DOSAGE RECOMMENDATIONS TO OPTIMIZE CROP YIELDS AND REDUCE FERTILIZER WASTE ACROSS DIFFERENT REGIONS OF TUNISIA.

Ⅰ -MODELING AND EVALUATION

4.1 Model Selection:

Several regression algorithms were selected to compare their performance in predicting land prices:

- **Ridge Regression**
- **Random Forest Regressor**
- **XGBoost Regressor**
- **LightGBM Regressor**
- **Gradient Boosting Regressor**
- **CatBoost Regressor**

These models were chosen for their ability to capture complex relationships between variables, their robustness against noise, and their effectiveness with structured datasets.

4.2 Data Preprocessing:

Before embarking on the model training phase, a thorough data preparation process was carried out:

- Elimination of unnecessary or highly correlated variables to avoid bias and redundancy in the data.
- Separation of explanatory variables from the target variable (price).
- The dataset was split into two subsets: an 80% training set and a 20% test set.
- Normalization via standardization was applied to numerical variables where necessary (particularly for linear models).

This step ensured that the models were trained under optimal and comparable conditions.

4.3 Hyperparameter Optimization

To enhance the performance of certain models, hyperparameter tuning was carried out using RandomizedSearchCV. This optimization technique explores different parameter combinations to maximize the model's predictive ability while minimizing overfitting.

For instance:

- For XGBoost, parameters such as tree depth, learning rate, and subsampling rate were optimized.
- For Gradient Boosting, tuning focused on the number of trees, depth, and minimum leaf size.

Each model was independently optimized to maximize its predictive power.

4.4 Model Performance Results:

After evaluation on the test set, the following performances were observed:

- **XGBoost** Regressor: Achieved the best R^2 among traditional models.
- **CatBoost** Regressor: Demonstrated excellent generalization capabilities, sometimes outperforming others on complex predictions.
- **Random Forest** and Gradient Boosting: Delivered solid but slightly inferior results compared to XGBoost and CatBoost.
- **LightGBM** Regressor: Performed less effectively on this particular dataset.

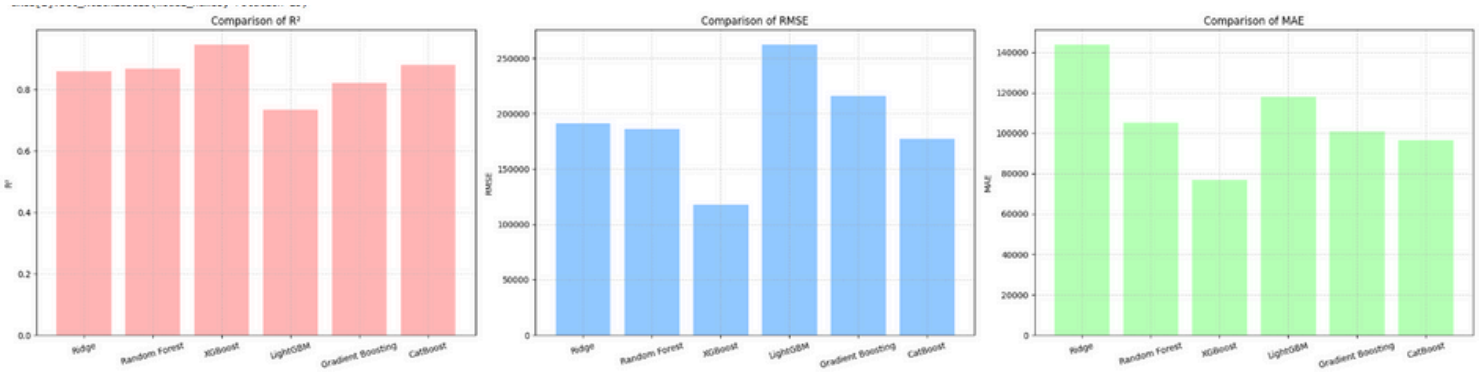
	Modèle	R^2	RMSE	MAE
2	XGBoost	0.95	117,940.86	76,768.12
5	CatBoost	0.88	177,073.99	96,622.85
1	Random Forest	0.87	185,904.05	105,129.90
0	Ridge	0.86	190,925.65	143,980.48
4	Gradient Boosting	0.82	216,105.97	100,784.09
3	LightGBM	0.73	262,466.81	118,166.99

1.5 Model Performance Comparison Based on R^2, RMSE, and MAE

The figure compares the performance of different regression models (Ridge, Random Forest, XGBoost, LightGBM, Gradient Boosting, CatBoost) using three metrics:

- R^2 (Coefficient of Determination):
- XGBoost achieves the highest R^2, indicating it explains the most variance in the target variable. LightGBM shows the lowest R^2, suggesting weaker predictive power compared to the others.
- RMSE (Root Mean Squared Error):

- **XGBoost has the lowest RMSE, meaning it has the smallest average error magnitude compared to other models. LightGBM shows the highest RMSE, reflecting larger prediction errors.**
- **MAE (Mean Absolute Error):**
- **Again, XGBoost outperforms the other models with the lowest MAE, indicating it makes the most accurate predictions on average. Ridge regression exhibits the highest MAE.**
- **Overall, XGBoost appears to be the best-performing model across all metrics, while LightGBM and Ridge have the least favorable results.**



4.5 Final Model Selection:

Given the evaluation results, the **CatBoost** Regressor was selected for final deployment.

The selection was motivated by:

- Its strong robustness with categorical features (e.g., Emplacement, State, CLAF).
- Its ability to handle missing values automatically without requiring manual imputation.
- Stable performance even with minimal data preprocessing.
- Excellent generalization ability observed in real-world predictions.

4.6 Detailed Hyperparameters:

CatBoost Regressor Hyperparameters:

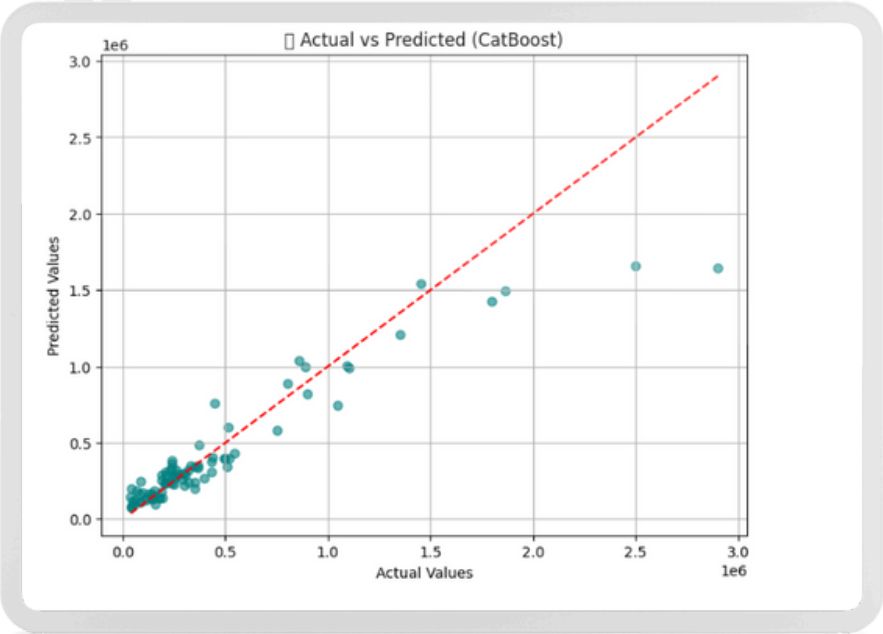
- **iterations** = 1000: Number of trees built, ensuring sufficient model complexity without overfitting.
- **learning_rate** = 0.05: A relatively low learning rate allowing smooth and stable convergence.
- **depth** = 10: Maximum tree depth to capture complex feature interactions.
- **l2_leaf_reg** = 3: L2 regularization term to penalize overly specific leaves and limit overfitting.
- **bagging_temperature** = 1: Introduces randomness in sampling to improve robustness.
- **random_strength** = 5: Adds noise to feature selection, increasing model diversity.
- **loss_function** = 'RMSE': Objective to minimize the root mean squared error.
- **verbose** = False: Suppresses verbose output for a smoother training process.

Thanks to this configuration, CatBoost achieved an excellent trade-off between bias and variance, maintaining a high generalization capacity on unseen data.
Other Optimized Models:

- **XGBoost** Regressor: **Average depth** of 6, **learning_rate** of 0.05, **subsample** and **colsample_bytree** set to 0.8.
- **Random Forest** Regressor: **300** trees, maximum depth of 10.
- **Gradient Boosting** Regressor: **200** trees, **learning_rate** of 0.05.
- **LightGBM** Regressor: **1000** trees, **learning_rate** of 0.05, 31 maximum leaves.

Although these models performed reasonably well, they were slightly outperformed by CatBoost in terms of robustness and consistency.

4.7 Actual vs Predicted Values Interpretation (CatBoost Model)



The scatter plot shows that the CatBoost model predicts land prices accurately, with most points close to the ideal red line. However, predictions for very high prices are less precise, likely due to outliers. Overall, the model performs well, especially for typical price ranges.

4.8 Conclusion on Modeling

The modeling phase allowed the testing of multiple approaches and the identification of the most suitable model.
CatBoost demonstrated key strengths:

- Automatic handling of categorical features,
- Robustness against noisy data,
- Strong generalization capabilities,
- Ease of use with minimal preprocessing.

It was therefore chosen as the final model to ensure reliable predictions of land prices with fast inference times, facilitating future deployment.

BO 6

IDENTIFY VOLUNTEER PLANTS AND DETERMINE IF THEY ARE HELPFUL WEEDS, MEDICINAL PLANTS OR HARMFUL WEEDS.

I -MODELING

THE GOAL OF THIS PHASE IS TO CLASSIFY VOLUNTEER PLANTS INTO ONE OF THREE CATEGORIES: HARMFUL WEED, USEFUL WEED, OR MEDICINAL PLANT. WHILE THE MODEL PREDICTS THE SPECIES OF THE PLANT, THE ULTIMATE OBJECTIVE IS TO CATEGORIZE THE PLANT BASED ON ITS POTENTIAL USEFULNESS OR HARMFULNESS. THIS CLASSIFICATION IS DETERMINED BY REFERENCING A CSV FILE, MENTIONED EARLIER IN THE DATA ACQUISITION PHASE, WHICH ASSOCIATES EACH PLANT SPECIES WITH ITS RESPECTIVE CATEGORY. GIVEN THE SIMILARITY IN APPEARANCE BETWEEN THESE PLANTS, SELECTING THE RIGHT MODEL ARCHITECTURE WAS CRUCIAL. I CHOSE THE XCEPTION MODEL FOR ITS PROVEN EFFECTIVENESS IN IMAGE CLASSIFICATION, PARTICULARLY FOR FINE-GRAINED CATEGORIES LIKE PLANT SPECIES. THIS SECTION OUTLINES THE PROCESS OF MODEL BUILDING, TRAINING AND EVALUATION , HIGHLIGHTING HOW EACH STEP WAS TAILORED TO MEET THE SPECIFIC REQUIREMENTS OF THE XCEPTION MODEL FOR THIS TASK.

PREPROCESSING DATA FOR MODEL INPUT

DATA PREPROCESSING ENSURES THE RAW DATA IS PROPERLY FORMATTED FOR THE MODEL. FOR XCEPTION, IMAGES WERE RESIZED TO 299X299 PIXELS AND NORMALIZED BY DIVIDING PIXEL VALUES BY 255.
INITIALLY, DATA AUGMENTATION WAS APPLIED BEFORE SPLITTING THE DATASET, WITH TRANSFORMATIONS SUCH AS ROTATION, SHIFTING, ZOOMING, AND FLIPPING TO INCREASE DATA DIVERSITY. AFTER SPLITTING, ADDITIONAL AUGMENTATION WAS APPLIED TO THE TRAINING SET, FURTHER IMPROVING THE MODEL'S ABILITY TO GENERALIZE. ONLY RESCALING WAS APPLIED TO THE VALIDATION SET TO PROVIDE AN UNBIASED EVALUATION OF THE MODEL'S PERFORMANCE.

BUILDING THE XCEPTION MODEL

THE XCEPTION MODEL WAS SELECTED FOR THIS CLASSIFICATION TASK DUE TO ITS EFFICIENCY AND EFFECTIVENESS IN HANDLING FINE-GRAINED CATEGORIES LIKE PLANT SPECIES. BELOW IS THE PROCESS OF ADAPTING THE MODEL FOR PLANT SPECIES CLASSIFICATION

WHY XCEPTION?

- XCEPTION IS BASED ON DEPTHWISE SEPARABLE CONVOLUTIONS, ALLOWING IT TO LEARN COMPLEX FEATURES EFFICIENTLY WHILE BEING COMPUTATIONALLY LIGHTWEIGHT.
- IT HAS PROVEN SUCCESS IN IMAGE CLASSIFICATION TASKS, PARTICULARLY WITH FINE-GRAINED CATEGORIES, MAKING IT IDEAL FOR CLASSIFYING PLANT SPECIES.

USING PRE-TRAINED XCEPTION

- THE ORIGINAL XCEPTION MODEL, TRAINED ON IMAGENET FOR GENERAL OBJECT RECOGNITION, WAS USED AS THE BASE.
- THE TOP LAYER WAS REMOVED SINCE IT WAS DESIGNED FOR GENERAL OBJECT CLASSIFICATION, NOT PLANT SPECIES.

CUSTOMIZING THE MODEL

- A CUSTOM FULLY CONNECTED LAYER WAS ADDED ON TOP OF THE PRE-TRAINED BASE TO OUTPUT THE PLANT SPECIES CATEGORIES.
- THE PRE-TRAINED LAYERS WERE FROZEN TO RETAIN USEFUL LEARNED FEATURES (E.G., EDGES, TEXTURES, SHAPES) FROM IMAGENET. THIS PREVENTS THESE LAYERS FROM BEING RETRAINED AND ENSURES THAT THE MODEL FOCUSES ON LEARNING THE TASK-SPECIFIC FEATURES IN THE NEW TOP LAYERS.

MODEL ARCHITECTURE

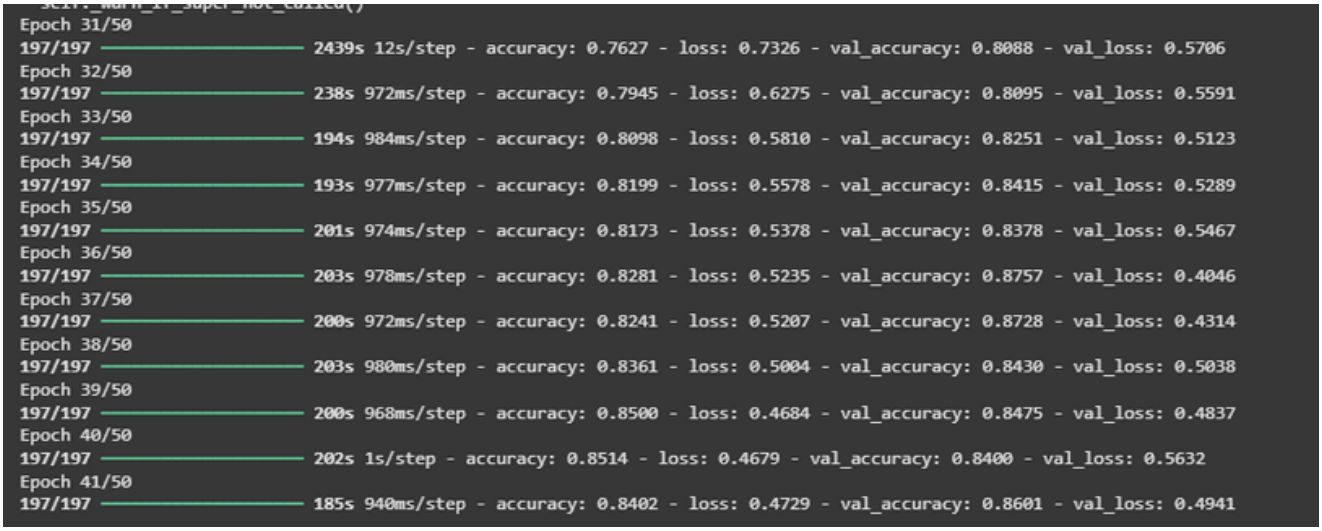
- GLOBAL AVERAGE POOLING: REDUCES SPATIAL DIMENSIONS AND MAKES THE OUTPUT MORE COMPACT.
- FULLY CONNECTED LAYER (1024 UNITS): CAPTURES COMPLEX RELATIONSHIPS IN THE DATA.
- DROPOUT LAYER (0.5): HELPS TO PREVENT OVERFITTING DURING TRAINING.
- OUTPUT LAYER: USES SOFTMAX ACTIVATION TO CLASSIFY IMAGES INTO ONE OF THE SPECIFIC PLANT SPECIES CATEGORIES

II -EVALUATION OF THE MODEL

TO ASSESS THE PERFORMANCE OF THE XCEPTION MODEL, WE PERFORMED SEVERAL EVALUATIONS, INCLUDING LOSS, ACCURACY, AND DETAILED CLASSIFICATION METRICS.

TRAINING PROGRESS AND ACCURACY CHANGE

OVER THE COURSE OF TRAINING, THE MODEL'S ACCURACY ON THE TRAINING DATA STEADILY INCREASED, WITH NOTABLE IMPROVEMENTS DURING THE FIRST FEW EPOCHS THE MODEL'S ACCURACY CONTINUED TO IMPROVE, WITH SOME FLUCTUATIONS BETWEEN TRAINING AND VALIDATION ACCURACY, SUGGESTING THAT THE MODEL WAS GENERALIZING WELL AND AVOIDING OVERFITTING.



MODEL EVALUATION ON TEST DATA

AFTER TRAINING, THE MODEL WAS EVALUATED ON THE TEST SET USING THE EVALUATE() METHOD, WHICH RETURNED THE FOLLOWING RESULTS:

- **TEST LOSS: 0.4399**
- **TEST ACCURACY: 0.8580**

THESE RESULTS INDICATE THAT THE MODEL PERFORMED REASONABLY WELL, ACHIEVING AN ACCURACY OF **85.8%** ON UNSEEN TEST DATA.

CLASSIFICATION REPORT

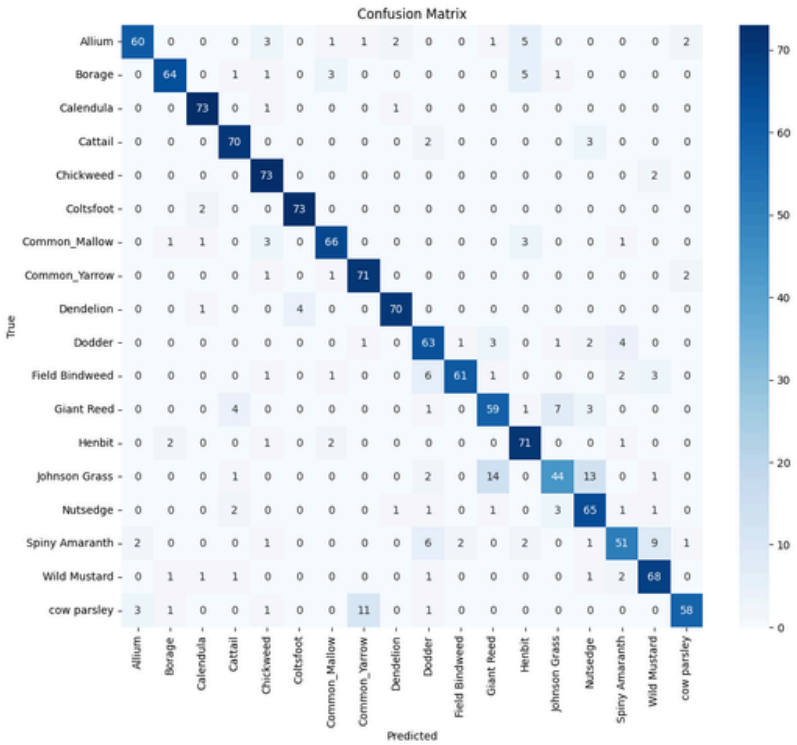
WE FURTHER ANALYZED THE MODEL'S PERFORMANCE USING THE CLASSIFICATION REPORT, WHICH PROVIDES PRECISION, RECALL, AND F1-SCORE METRICS FOR EACH PLANT SPECIES CLASS. THE REPORT SHOWED BALANCED PERFORMANCE ACROSS MOST CLASSES, WITH AN OVERALL ACCURACY OF 86%.

Classification Report:

	precision	recall	f1-score	support
Allium	0.92	0.80	0.86	75
Borage	0.93	0.85	0.89	75
Calendula	0.94	0.97	0.95	75
Cattail	0.89	0.93	0.91	75
Chickweed	0.85	0.97	0.91	75
Coltsfoot	0.95	0.97	0.96	75
Common_Mallow	0.89	0.88	0.89	75
Common_Yarrow	0.85	0.95	0.89	75
Dendelion	0.95	0.93	0.94	75
Dodder	0.76	0.84	0.80	75
Field Bindweed	0.95	0.81	0.88	75
Giant Reed	0.75	0.79	0.77	75
Henbit	0.82	0.92	0.87	77
Johnson Grass	0.79	0.59	0.67	75
Nutsedge	0.74	0.87	0.80	75
Spiny Amaranth	0.82	0.68	0.74	75
Wild Mustard	0.81	0.91	0.86	75
cow parsley	0.92	0.77	0.84	75
accuracy			0.86	1352
macro avg	0.86	0.86	0.86	1352
weighted avg	0.86	0.86	0.86	1352

CONFUSION MATRIX

THE CONFUSION MATRIX SHOWS STRONG PERFORMANCE, WITH MOST PREDICTIONS CORRECTLY ALIGNED ALONG THE DIAGONAL. KEY CLASSES LIKE CALENDULA, CHICKWEED, AND COLTSFOOT ACHIEVED OVER 70 CORRECT PREDICTIONS. MISCLASSIFICATIONS ARE MINIMAL, CONFIRMING THE MODEL'S HIGH ACCURACY AND STABILITY ACROSS SIMILAR PLANT SPECIES.



III -TESTING

IN THIS PHASE, WE TESTED THE TRAINED MODEL BY PROVIDING AN UNSEEN IMAGE TO EVALUATE ITS PERFORMANCE ON REAL-WORLD DATA. THE IMAGE WAS PROCESSED, CLASSIFIED, AND THE RESULT WAS COMPARED TO THE EXPECTED OUTPUT.

IMAGE PREPROCESSING AND PREDICTION

THE IMAGE WAS RESIZED TO THE TARGET SIZE OF 299X299 PIXELS, WHICH IS THE INPUT SIZE EXPECTED BY THE XCEPTION MODEL. IT WAS THEN CONVERTED INTO A NUMPY ARRAY, EXPANDED TO MATCH THE MODEL'S EXPECTED INPUT SHAPE, AND NORMALIZED BY SCALING PIXEL VALUES TO THE RANGE [0, 1].
THE PREPROCESSED IMAGE WAS PASSED TO THE TRAINED XCEPTION MODEL FOR PREDICTION. THE MODEL RETURNED A SET OF CLASS PROBABILITIES, AND THE CLASS WITH THE HIGHEST PROBABILITY WAS CHOSEN AS THE PREDICTED LABEL.

PREDICTION RESULT

- THE MODEL CORRECTLY IDENTIFIED THE PLANT SPECIES IN THE IMAGE.
- OUTPUT DISPLAY: THE IMAGE WAS SHOWN WITH THE PREDICTED LABEL AS THE TITLE, CONFIRMING THAT THE MODEL SUCCESSFULLY PREDICTED THE CLASS OF THE PLANT.



05

DEPLOYMENT

I -INTRODUCTION

DURING THIS CHAPTER, WE WILL FIRST BEGIN BY PRESENTING THE TOOLS AND TECHNOLOGIES UTILIZED TO SUCCESSFULLY ACCOMPLISH OUR MISSION WITHIN THE FARMWISE PLATFORM. WE WILL ALSO EXPLORE THE TECHNICAL ENVIRONMENTS ON WHICH WE RELIED TO DEVELOP OUR APPLICATION, ENSURING A ROBUST AND TAILORED INFRASTRUCTURE TO MEET THE NEEDS OF TUNISIAN FARMERS. SUBSEQUENTLY, WE WILL EXAMINE THE CHOSEN ARCHITECTURE, DESIGNED TO OPTIMIZE THE INTEGRATION OF ARTIFICIAL INTELLIGENCE MODELS, BEFORE CONCLUDING WITH THE IMPLEMENTATION PHASE, ILLUSTRATED BY SCREENSHOTS OF THE FINAL RENDER OF OUR PLATFORM, SHOWCASING ITS INTUITIVE AND FUNCTIONAL USER INTERFACE.

II -TOOLS AND TECHNOLOGIES USED

TO ENSURE THE SUCCESSFUL IMPLEMENTATION OF THE FARMWISE PLATFORM, WE HAVE ESTABLISHED A ROBUST TECHNICAL ENVIRONMENT THAT LEVERAGES A VARIETY OF TOOLS AND TECHNOLOGIES TAILORED TO MEET THE PROJECT'S OBJECTIVES OF OPTIMIZING AGRICULTURAL PRACTICES FOR TUNISIAN FARMERS. THE FOLLOWING SUBSECTIONS OUTLINE THE KEY TECHNOLOGIES AND ENVIRONMENTS UTILIZED, ENSURING EFFICIENT DEVELOPMENT, SCALABILITY, AND SEAMLESS INTEGRATION OF AI-DRIVEN FEATURES.

- **PYTHON**

PYTHON IS THE CORE PROGRAMMING LANGUAGE UTILIZED FOR THE DEVELOPMENT OF THE FARMWISE PLATFORM, PROVIDING A VERSATILE AND POWERFUL FOUNDATION FOR IMPLEMENTING AI-DRIVEN SOLUTIONS. ITS EXTENSIVE ECOSYSTEM OF LIBRARIES AND FRAMEWORKS ENABLED US TO BUILD, TRAIN, AND DEPLOY MACHINE LEARNING MODELS SUCH AS THE XGBOOST REGRESSOR FOR YIELD PREDICTION, THE CATBOOST REGRESSOR FOR LAND PRICE ESTIMATION, AND THE XCEPTION MODEL FOR PLANT DISEASE DETECTION AND VOLUNTEER PLANT CLASSIFICATION. PYTHON'S SIMPLICITY AND READABILITY FACILITATED RAPID PROTOTYPING AND COLLABORATION AMONG TEAM MEMBERS, WHILE ITS COMPATIBILITY WITH TOOLS LIKE GOOGLE COLABORATORY AND DJANGO ENSURED SEAMLESS INTEGRATION OF DATA SCIENCE WORKFLOWS INTO THE PLATFORM'S BACKEND AND USER INTERFACES, DELIVERING ACTIONABLE INSIGHTS TO TUNISIAN FARMERS.



- **NUMPY**

NUMPY IS A FUNDAMENTAL PYTHON LIBRARY FOR NUMERICAL COMPUTING THAT PLAYED A CRITICAL ROLE IN THE FARMWISE PROJECT'S DATA PROCESSING AND MODEL DEVELOPMENT PHASES. IT PROVIDED EFFICIENT ARRAY OPERATIONS AND MATHEMATICAL FUNCTIONS ESSENTIAL FOR HANDLING LARGE DATASETS, SUCH AS SOIL NUTRIENT LEVELS, CLIMATE DATA, AND IMAGE PIXEL VALUES USED IN OUR MODELS. FOR INSTANCE, NUMPY WAS USED TO PREPROCESS INPUT DATA FOR THE "DOSAGE" INTERFACE BY NORMALIZING SOIL NUTRIENT VALUES (N, P, K) AND TO PREPARE IMAGE DATA FOR THE XCEPTION MODEL IN THE "DISEASE" AND "VOLUNTARY" INTERFACES. ITS PERFORMANCE OPTIMIZATIONS ENSURED FAST COMPUTATIONS DURING MODEL TRAINING AND INFERENCE, ENABLING REAL-TIME RECOMMENDATIONS FOR FARMERS.



- **SCIKIT-LEARN**

SCIKIT-LEARN IS A VERSATILE PYTHON LIBRARY FOR MACHINE LEARNING THAT WAS EXTENSIVELY USED IN THE FARMWISE PROJECT FOR DEVELOPING AND EVALUATING TRADITIONAL MACHINE LEARNING MODELS. IT PROVIDED ESSENTIAL TOOLS FOR IMPLEMENTING MODELS LIKE THE XGBOOST REGRESSOR FOR YIELD PREDICTION AND THE CATBOOST REGRESSOR FOR LAND PRICE ESTIMATION, LEVERAGING ITS EFFICIENT IMPLEMENTATIONS OF GRADIENT BOOSTING ALGORITHMS. SCIKIT-LEARN ALSO FACILITATED DATA PREPROCESSING, FEATURE ENGINEERING, AND MODEL EVALUATION, WITH METRICS SUCH AS MEAN ABSOLUTE ERROR (MAE) AND R^2 USED TO ASSESS MODEL PERFORMANCE DURING DEVELOPMENT IN GOOGLE COLAB. ITS SEAMLESS INTEGRATION WITH PYTHON ENSURED THAT OUR MODELS COULD BE EASILY INCORPORATED INTO THE DJANGO BACKEND, DELIVERING ACCURATE PREDICTIONS THROUGH INTERFACES LIKE "YIELD" AND "LANDPRICE."



- **TENSORFLOW**

TENSORFLOW IS AN OPEN-SOURCE MACHINE LEARNING FRAMEWORK THAT WAS FUNDAMENTAL IN DEVELOPING AND TRAINING THE DEEP LEARNING MODELS FOR THE FARMWISE PLATFORM. IT WAS PARTICULARLY CRUCIAL FOR IMPLEMENTING THE XCEPTION MODEL, USED FOR PLANT DISEASE DETECTION AND VOLUNTEER PLANT CLASSIFICATION, ACHIEVING HIGH ACCURACY IN IDENTIFYING 38 DISEASE CLASSES AND CATEGORIZING SPONTANEOUS PLANTS. TENSORFLOW'S ROBUST SUPPORT FOR CONVOLUTIONAL NEURAL NETWORKS (CNNS) AND ITS ABILITY TO LEVERAGE GPU ACCELERATION IN ENVIRONMENTS LIKE GOOGLE COLAB ENABLED EFFICIENT TRAINING AND FINE-TUNING OF OUR MODELS. ITS FLEXIBILITY ALSO ALLOWED US TO DEPLOY THESE MODELS INTO PRODUCTION, INTEGRATING THEM WITH DJANGO TO SERVE PREDICTIONS THROUGH INTERFACES LIKE "DISEASE" AND "VOLUNTARY," PROVIDING FARMERS WITH REAL-TIME, IMAGE-BASED INSIGHTS.



- **KERAS**

KERAS, A HIGH-LEVEL API BUILT ON TOP OF TENSORFLOW, WAS UTILIZED TO STREAMLINE THE DEVELOPMENT OF DEEP LEARNING MODELS FOR THE FARMWISE PLATFORM. IT SIMPLIFIED THE IMPLEMENTATION OF THE XCEPTION MODEL BY PROVIDING AN INTUITIVE INTERFACE FOR DEFINING NEURAL NETWORK ARCHITECTURES, CONFIGURING LAYERS, AND MANAGING TRAINING PROCESSES. KERAS WAS PARTICULARLY USEFUL FOR RAPID PROTOTYPING OF THE PLANT DISEASE DETECTION AND VOLUNTEER PLANT CLASSIFICATION MODELS, ALLOWING US TO EXPERIMENT WITH DIFFERENT CONFIGURATIONS IN GOOGLE COLAB BEFORE FINALIZING THE MODELS. ITS SEAMLESS INTEGRATION WITH TENSORFLOW ENSURED EFFICIENT DEPLOYMENT OF THESE MODELS INTO THE DJANGO BACKEND, ENABLING THE "DISEASE" AND "VOLUNTARY" INTERFACES TO DELIVER ACCURATE AND USER-FRIENDLY PREDICTIONS TO FARMERS.



- **GOOGLE COLABORATORY (GOOGLE COLAB)**

GOOGLE COLABORATORY, COMMONLY REFERRED TO AS COLAB, IS A CLOUD-BASED PLATFORM THAT PLAYED A PIVOTAL ROLE IN THE DEVELOPMENT AND EXPERIMENTATION PHASES OF THE FARMWISE PROJECT. IT PROVIDES A JUPYTER NOTEBOOK-LIKE ENVIRONMENT THAT REQUIRES NO SETUP, ENABLING OUR TEAM TO WRITE AND EXECUTE PYTHON CODE DIRECTLY IN THE BROWSER WITH ACCESS TO POWERFUL COMPUTATIONAL RESOURCES, INCLUDING GPUS AND TPUS, AT NO COST. COLAB WAS INSTRUMENTAL IN PROTOTYPING AND TRAINING OUR MACHINE LEARNING MODELS—SUCH AS THE XGBOOST REGRESSOR FOR YIELD PREDICTION, THE CATBOOST REGRESSOR FOR LAND PRICE ESTIMATION, AND THE XCEPTION MODEL FOR PLANT DISEASE DETECTION AND VOLUNTEER PLANT CLASSIFICATION—ALLOWING US TO LEVERAGE LIBRARIES LIKE TENSORFLOW, KERAS, AND SCIKIT-LEARN. ITS COLLABORATIVE FEATURES ALSO FACILITATED SEAMLESS TEAMWORK BY ENABLING REAL-TIME SHARING AND EDITING OF NOTEBOOKS AMONG TEAM MEMBERS, ENSURING EFFICIENT ITERATION AND VALIDATION OF OUR MODELS BEFORE DEPLOYMENT TO THE PRODUCTION ENVIRONMENT.



- **JUPYTER NOTEBOOK**

JUPYTER NOTEBOOK IS AN OPEN-SOURCE, INTERACTIVE DEVELOPMENT ENVIRONMENT THAT WAS EXTENSIVELY USED DURING THE EARLY STAGES OF THE FARMWISE PROJECT FOR EXPLORATORY DATA ANALYSIS AND MODEL DEVELOPMENT. IT ALLOWED OUR TEAM TO CREATE AND SHARE DOCUMENTS CONTAINING LIVE PYTHON CODE, VISUALIZATIONS, AND NARRATIVE TEXT, MAKING IT IDEAL FOR EXPERIMENTING WITH DATASETS SUCH AS SOIL NUTRIENT LEVELS, CLIMATE DATA, AND PLANT IMAGES. JUPYTER NOTEBOOK WAS PARTICULARLY USEFUL FOR DEBUGGING AND TESTING THE PREPROCESSING PIPELINES FOR MODELS LIKE XGBOOST AND CATBOOST, ENSURING DATA QUALITY BEFORE TRAINING IN GOOGLE COLAB. ITS FLEXIBILITY ALSO SUPPORTED LOCAL DEVELOPMENT ON TEAM MEMBERS' MACHINES, COMPLEMENTING THE CLOUD-BASED CAPABILITIES OF COLAB.



- **VISUAL STUDIO CODE (VS CODE)**

VISUAL STUDIO CODE, COMMONLY KNOWN AS VS CODE, IS A POWERFUL SOURCE-CODE EDITOR THAT SERVED AS THE PRIMARY DEVELOPMENT ENVIRONMENT FOR CODING, DEBUGGING, AND TESTING THE FARMWISE PLATFORM'S BACKEND AND FRONTEND COMPONENTS. ITS SUPPORT FOR PYTHON, THROUGH EXTENSIONS LIKE THE PYTHON EXTENSION, ENABLED EFFICIENT DEVELOPMENT OF THE DJANGO BACKEND, INCLUDING API ENDPOINTS FOR INTERFACES LIKE "DOSAGE," "YIELD," AND "DISEASE." VS CODE'S INTEGRATED TERMINAL, GIT SUPPORT, AND DEBUGGING TOOLS STREAMLINED THE DEVELOPMENT PROCESS, ALLOWING TEAM MEMBERS TO MANAGE VERSION CONTROL, TEST API INTEGRATIONS, AND ENSURE CODE QUALITY. ITS LIGHTWEIGHT AND CUSTOMIZABLE NATURE MADE IT AN IDEAL CHOICE FOR DEVELOPING A SCALABLE AND MAINTAINABLE PLATFORM TAILORED TO THE NEEDS OF TUNISIAN FARMERS.



- **DJANGO**

DJANGO IS A HIGH-LEVEL PYTHON WEB FRAMEWORK THAT FORMS THE BACKBONE OF THE FARMWISE PLATFORM'S BACKEND DEVELOPMENT, ENABLING THE CREATION OF A SECURE, SCALABLE, AND MAINTAINABLE WEB APPLICATION. IT PROVIDED A ROBUST FOUNDATION FOR BUILDING THE PLATFORM'S API ENDPOINTS, WHICH HANDLE REQUESTS FROM INTERFACES LIKE "DOSAGE," "DISEASE," AND "YIELD," PROCESSING INPUTS SUCH AS SOIL DATA OR PLANT IMAGES AND RETURNING MODEL PREDICTIONS (E.G., FERTILIZER RECOMMENDATIONS, DISEASE CLASSIFICATIONS) IN REAL TIME. DJANGO'S BUILT-IN FEATURES, SUCH AS ITS ORM (OBJECT-RELATIONAL MAPPING)

FOR DATABASE MANAGEMENT AND ITS AUTHENTICATION SYSTEM FOR USER ACCESS CONTROL, WERE CRUCIAL FOR IMPLEMENTING INTERFACES LIKE "LOGIN," "SIGNUP," AND "PROFILE." ITS EMPHASIS ON RAPID DEVELOPMENT AND ADHERENCE TO THE DRY (DON'T REPEAT YOURSELF) PRINCIPLE ENSURED THAT THE PLATFORM COULD EFFICIENTLY DELIVER ACTIONABLE INSIGHTS TO FARMERS ACROSS TUNISIA'S DIVERSE REGIONS WHILE MAINTAINING COMPATIBILITY WITH BOTH WEB AND MOBILE INTERFACES.



- **POWER BI**

POWER BI, A BUSINESS ANALYTICS TOOL BY MICROSOFT, WAS INTEGRATED INTO THE FARMWISE PLATFORM TO PROVIDE ADVANCED DATA VISUALIZATION AND REPORTING CAPABILITIES, ENHANCING DECISION-MAKING FOR FARMERS AND ADMINISTRATORS. IT WAS USED TO CREATE INTERACTIVE DASHBOARDS ACCESSIBLE THROUGH THE "BASE" AND "INDEX" INTERFACES, DISPLAYING KEY METRICS SUCH AS REGIONAL YIELD TRENDS, SOIL HEALTH ANALYTICS, AND PLATFORM USAGE STATISTICS. POWER BI'S ABILITY TO CONNECT WITH DJANGO'S DATABASE ALLOWED US TO AGGREGATE AND VISUALIZE DATA FROM VARIOUS SOURCES, INCLUDING MODEL PREDICTIONS AND USER INPUTS, IN REAL TIME. FOR INSTANCE, FARMERS CAN VIEW CHARTS COMPARING THEIR FERTILIZER USAGE AGAINST RECOMMENDED LEVELS IN THE "DOSAGE" INTERFACE, WHILE ADMINISTRATORS CAN MONITOR PLATFORM PERFORMANCE ACROSS TUNISIA'S NORTH, CENTRAL, AND SOUTH REGIONS, ENSURING DATA-DRIVEN INSIGHTS FOR SUSTAINABLE AGRICULTURAL GROWTH.



III-ARCHITECTURE

THE ARCHITECTURE OF THE FARMWISE PLATFORM IS DESIGNED TO DELIVER A SCALABLE, EFFICIENT, AND USER-CENTRIC SOLUTION FOR OPTIMIZING AGRICULTURAL PRACTICES IN TUNISIA. IT IS STRUCTURED INTO FOUR DISTINCT LAYERS—DATA INGESTION, DATA STORAGE & PROCESSING, MACHINE LEARNING, AND APPLICATION & INTERFACE—EACH ADDRESSING A SPECIFIC ASPECT OF THE SYSTEM'S FUNCTIONALITY. THIS ARCHITECTURE ENSURES SEAMLESS DATA FLOW, ROBUST MODEL INTEGRATION, AND AN INTUITIVE USER EXPERIENCE, ENABLING FARMERS TO ACCESS REAL-TIME, AI-DRIVEN RECOMMENDATIONS FOR FERTILIZER DOSAGE, DISEASE DETECTION, YIELD PREDICTION, CROP ROTATION, LAND PRICE ESTIMATION, AND VOLUNTEER PLANT CLASSIFICATION. THE FOLLOWING SUBSECTIONS PROVIDE A DETAILED OVERVIEW OF EACH LAYER, HIGHLIGHTING THE TOOLS, TECHNOLOGIES, AND PROCESSES EMPLOYED TO ACHIEVE THE PROJECT'S OBJECTIVES.

- **DATA INGESTION LAYER**

THE DATA INGESTION LAYER IS RESPONSIBLE FOR COLLECTING AND AGGREGATING DIVERSE DATASETS REQUIRED TO POWER THE FARMWISE PLATFORM'S AI MODELS. DATA IS SOURCED FROM PUBLIC DATASETS, PRIMARILY IN CSV FORMAT, OBTAINED FROM TRUSTED AGRICULTURAL WEBSITES, COVERING SOIL PROPERTIES, CLIMATE CONDITIONS, CROP HISTORIES, AND LAND MARKET TRENDS RELEVANT TO TUNISIA'S NORTH, CENTRAL, AND SOUTH REGIONS. ADDITIONALLY, IMAGE DATA FOR PLANT DISEASE DETECTION AND VOLUNTEER PLANT CLASSIFICATION, SUCH AS LEAF PHOTOGRAPHS, IS INGESTED IN FORMATS LIKE JPEG OR PNG. THE INGESTION PROCESS LEVERAGES PYTHON-BASED TOOLS FOR BOTH MANUAL AND AUTOMATED EXTRACTION: LIBRARIES SUCH AS PANDAS FOR HANDLING CSV FILES, REQUESTS FOR API-BASED DATA RETRIEVAL, AND BEAUTIFULSOUP FOR WEB SCRAPING WHEN NECESSARY. SCHEDULED JOBS ARE IMPLEMENTED USING PYTHON SCRIPTS TO ENSURE REGULAR UPDATES, MAINTAINING THE FRESHNESS OF THE DATA USED FOR MODEL TRAINING AND INFERENCE.

- **DATA STORAGE & PROCESSING LAYER**

THE DATA STORAGE & PROCESSING LAYER ENSURES EFFICIENT MANAGEMENT AND PREPARATION OF DATA FOR DOWNSTREAM ANALYSIS AND MODELING. STORAGE IS BIFURCATED INTO TWO COMPONENTS: RAW CSV FILES AND IMAGES ARE INITIALLY STORED LOCALLY DURING DEVELOPMENT AND LATER MIGRATED TO CLOUD STORAGE SOLUTIONS LIKE GOOGLE DRIVE OR AMAZON S3 FOR SCALABILITY AND ACCESSIBILITY DURING PRODUCTION. CLEANED AND PROCESSED DATASETS ARE STORED IN RELATIONAL DATABASES, WITH POSTGRESQL USED IN PRODUCTION FOR ITS ROBUSTNESS AND SQLITE EMPLOYED DURING DEVELOPMENT FOR SIMPLICITY. PROCESSING INVOLVES DATA CLEANING, TRANSFORMATION, AND FEATURE ENGINEERING TO PREPARE DATASETS FOR MACHINE LEARNING MODELS. PYTHON LIBRARIES SUCH AS PANDAS ARE USED FOR DATA WRANGLING TASKS (E.G., HANDLING MISSING VALUES, NORMALIZING SOIL NUTRIENT LEVELS), WHILE SCIKIT-LEARN

FACILITATES FEATURE ENGINEERING, SUCH AS ENCODING CATEGORICAL VARIABLES (E.G., REGION, CROP TYPE) AND SCALING NUMERICAL FEATURES (E.G., CLIMATE DATA). THIS LAYER ENSURES THAT HIGH-QUALITY, STRUCTURED DATA IS AVAILABLE FOR TRAINING AND INFERENCE, SUPPORTING REAL-TIME RECOMMENDATIONS ACROSS THE PLATFORM'S INTERFACES.

- **MACHINE LEARNING LAYER**

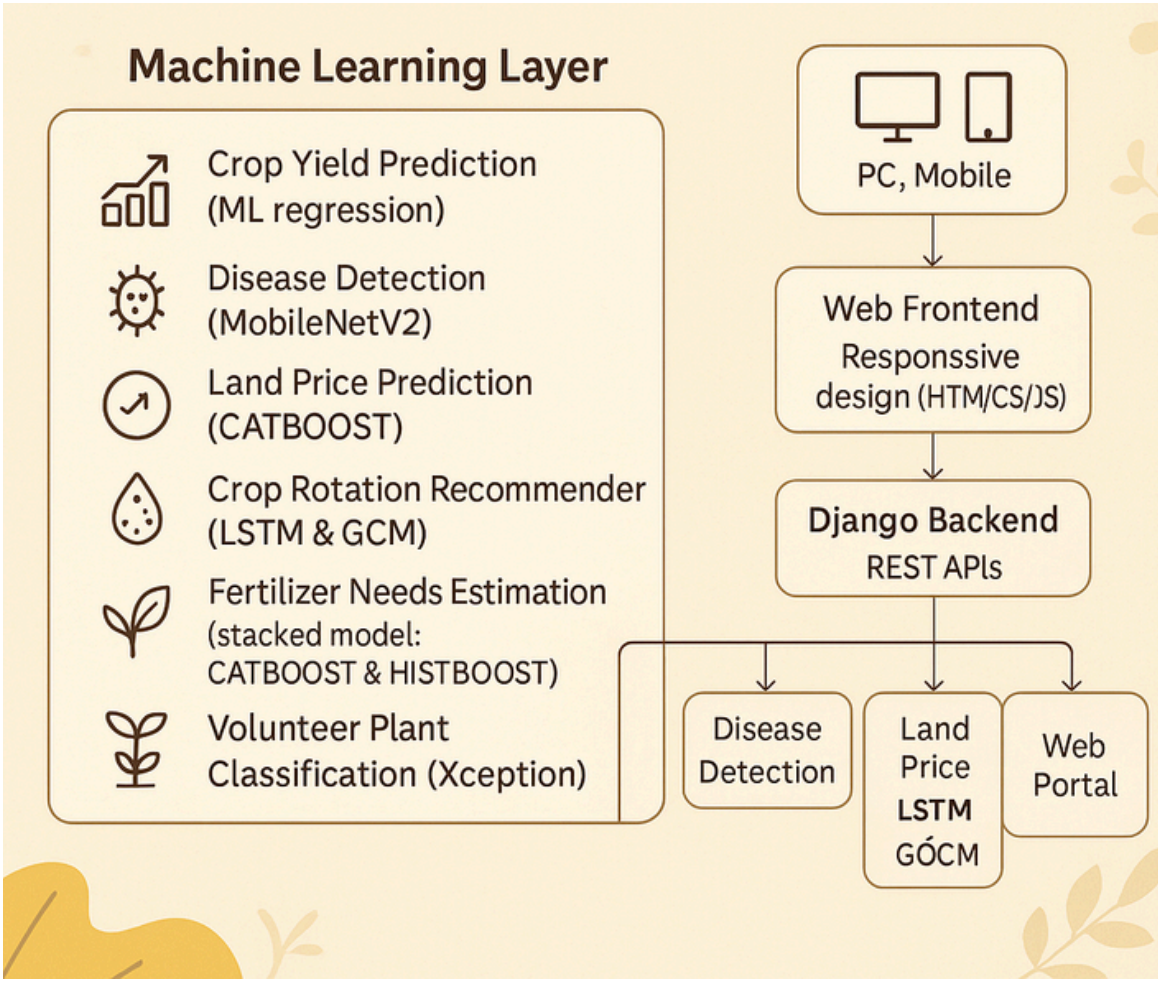
THE MACHINE LEARNING LAYER FORMS THE CORE OF THE FARMWISE PLATFORM, HOSTING THE AI MODELS THAT DRIVE ITS PREDICTIVE AND ANALYTICAL CAPABILITIES. THIS LAYER INCLUDES SEVERAL SPECIALIZED MODELS TAILORED TO THE PLATFORM'S OBJECTIVES:

- **CROP YIELD PREDICTION**: AN LSTM (LONG SHORT-TERM MEMORY) MODEL PREDICTS CROP YIELDS AND MATURITY TIMES BASED ON SEQUENTIAL DATA LIKE CLIMATE TRENDS AND SOWING DATES, INTEGRATED INTO THE "YIELD" INTERFACE.
- **DISEASE DETECTION:** MOBILENETV2, A LIGHTWEIGHT CONVOLUTIONAL NEURAL NETWORK, IDENTIFIES PLANT DISEASES FROM IMAGES, POWERING THE "DISEASE" INTERFACE WITH EFFICIENT AND ACCURATE DIAGNOSTICS.
- **LAND PRICE PREDICTION**: A REGRESSION MODEL USING CATBOOST ESTIMATES AGRICULTURAL LAND PRICES BASED ON FEATURES LIKE LOCATION AND SOIL QUALITY, ACCESSIBLE VIA THE "LANDPRICE" INTERFACE.
- **CROP ROTATION RECOMMENDATION SYSTEM**: A HYBRID MODEL COMBINING LSTM AND GRAPH CONVOLUTIONAL NETWORKS (GCN) RECOMMENDS SUSTAINABLE CROP ROTATION PLANS, INTEGRATED INTO THE "CROPROTATION" INTERFACE.
- **FERTILIZER NEEDS ESTIMATION**: A STACKING ENSEMBLE MODEL COMBINING CATBOOST AND HISTGRADIENTBOOSTING ESTIMATES OPTIMAL FERTILIZER DOSAGES, SUPPORTING THE "DOSAGE" INTERFACE WITH PRECISE RECOMMENDATIONS.
- **VOLUNTEER PLANT CLASSIFICATION:** THE XCEPTION MODEL CLASSIFIES SPONTANEOUS PLANTS AS HARMFUL, USEFUL, OR MEDICINAL, DRIVING THE "VOLUNTARY" INTERFACE WITH IMAGE-BASED INSIGHTS.

=› THESE MODELS WERE DEVELOPED AND TRAINED USING PYTHON LIBRARIES LIKE TENSORFLOW, KERAS, AND SCIKIT-LEARN IN GOOGLE COLABORATORY, ENSURING HIGH PERFORMANCE AND SCALABILITY FOR REAL-TIME INFERENCE.

- **APPLICATION & INTERFACE LAYER**

THE APPLICATION & INTERFACE LAYER DELIVERS THE FARMWISE PLATFORM'S FUNCTIONALITY TO END USERS THROUGH A SEAMLESS AND INTUITIVE EXPERIENCE. FRONTEND DEVELOPMENT FOCUSES ON A RESPONSIVE WEBSITE COMPATIBLE WITH BOTH PC AND MOBILE DEVICES, BUILT USING HTML, CSS, AND JAVASCRIPT, WITH FRAMEWORKS LIKE BOOTSTRAP OR TAILWIND CSS TO ENSURE A MODERN, USER-FRIENDLY DESIGN. THIS LAYER POWERS INTERFACES SUCH AS "INDEX," "BASE," AND "PROFILE," PROVIDING FARMERS WITH EASY NAVIGATION AND VISUAL CLARITY. BACKEND DEVELOPMENT LEVERAGES THE DJANGO WEB FRAMEWORK, WHICH HANDLES API REQUESTS, USER AUTHENTICATION, AND DATA MANAGEMENT. REST APIS BUILT WITH DJANGO CONNECT THE MACHINE LEARNING MODELS TO THE FRONTEND, ENABLING REAL-TIME INTERACTION—FOR EXAMPLE, PROCESSING IMAGE UPLOADS IN THE "DISEASE" INTERFACE OR RETRIEVING YIELD PREDICTIONS IN THE "YIELD" INTERFACE. USER-FRIENDLY DASHBOARDS PROVIDE FARMERS AND INVESTORS WITH ACTIONABLE INSIGHTS, SUCH AS FERTILIZER RECOMMENDATIONS OR LAND PRICE ESTIMATES. ADDITIONALLY, A WEB PORTAL INTEGRATES ADVANCED ANALYTICS AND DATA VISUALIZATION USING POWER BI, OFFERING INTERACTIVE DASHBOARDS IN INTERFACES LIKE "BASE" AND "INDEX" TO DISPLAY METRICS SUCH AS REGIONAL YIELD TRENDS AND SOIL HEALTH ANALYTICS, EMPOWERING USERS WITH DATA-DRIVEN DECISION-MAKING TOOLS.

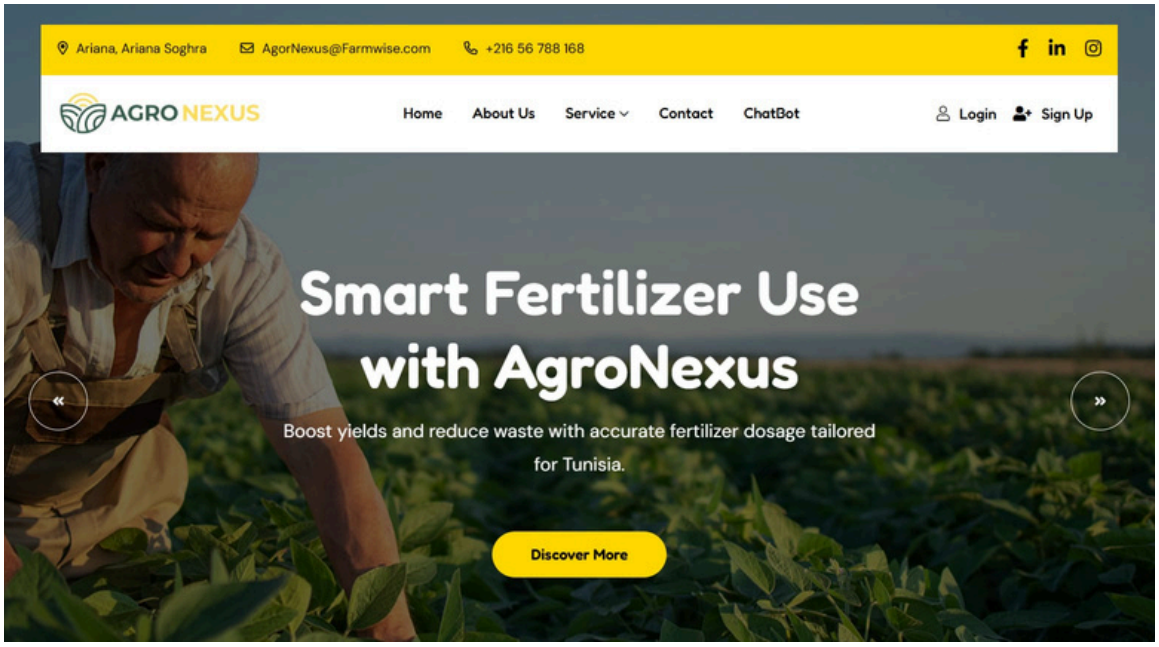


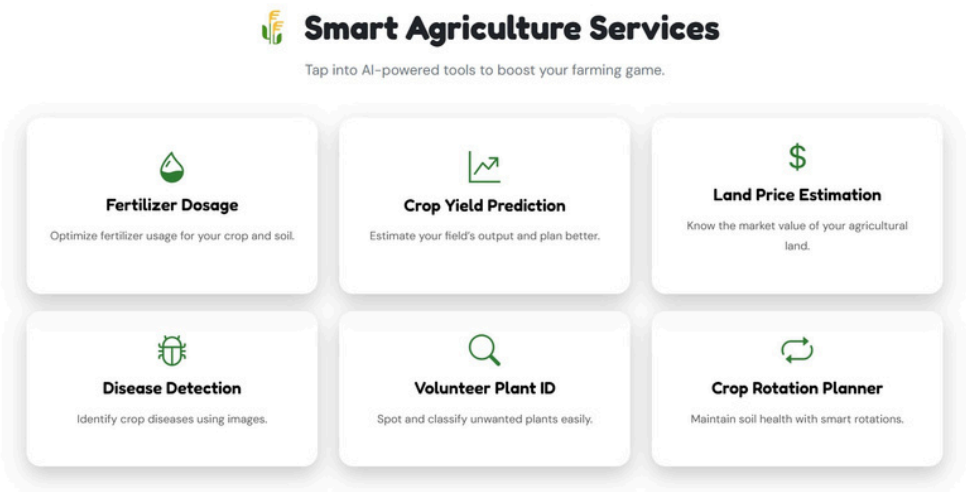
IV-PRODUCTION

THIS PHASE REPRESENTS THE CONCRETE REALIZATION OF THE VARIOUS DESIGN METHODOLOGIES, MARKING THE CULMINATION OF OUR EFFORTS TO DELIVER A FUNCTIONAL AND EFFECTIVE SOLUTION FOR TUNISIAN FARMERS. , THIS SECTION SHOWCASES THE PRACTICAL DEPLOYMENT OF THE PLATFORM, BRINGING TOGETHER THE DATA INGESTION, PROCESSING, MACHINE LEARNING, AND INTERFACE LAYERS INTO A COHESIVE SYSTEM THAT PROVIDES REAL-TIME, AI-DRIVEN RECOMMENDATIONS FOR OPTIMIZING AGRICULTURAL PRACTICES.

- **HOME PAGE**

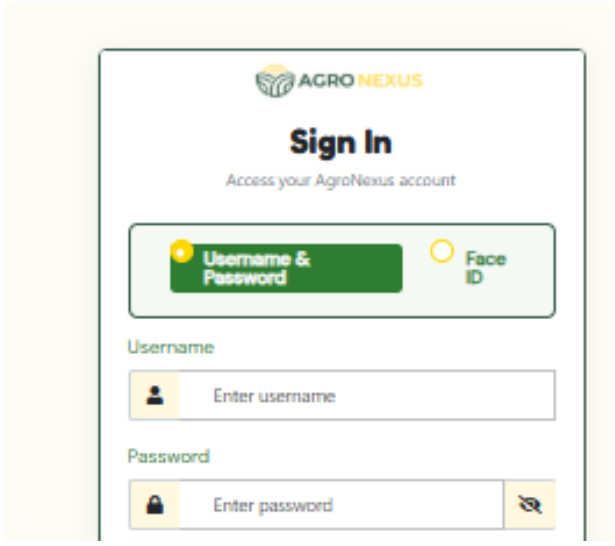
THE FOLLOWING FIGURE ILLUSTRATES THE HOMEPAGE OF THE FARMWISE PLATFORM, SERVING AS THE PRIMARY ENTRY POINT FOR USERS. THIS PAGE FEATURES SEVERAL INTERACTIVE ELEMENTS, INCLUDING A BUTTON FOR REDIRECTION TO THE AUTHENTICATION PAGE, ENABLING SECURE ACCESS FOR REGISTERED FARMERS AND INVESTORS. AN "HOME" TAB PROVIDES A CONCISE OVERVIEW OF THE PLATFORM AND THE ENTERPRISE BEHIND IT, HIGHLIGHTING ITS MISSION TO ENHANCE SUSTAINABLE AGRICULTURE IN TUNISIA. ADDITIONALLY, A "CONTACT" BUTTON ALLOWS USERS TO EASILY REACH OUT TO THE SUPPORT TEAM, FACILITATING COMMUNICATION AND ASSISTANCE. THIS INTUITIVE LAYOUT, DESIGNED WITH RESPONSIVENESS IN MIND, ENSURES ACCESSIBILITY ACROSS BOTH DESKTOP AND MOBILE DEVICES, SETTING THE STAGE FOR THE PLATFORM'S FULL RANGE OF FUNCTIONALITIES.

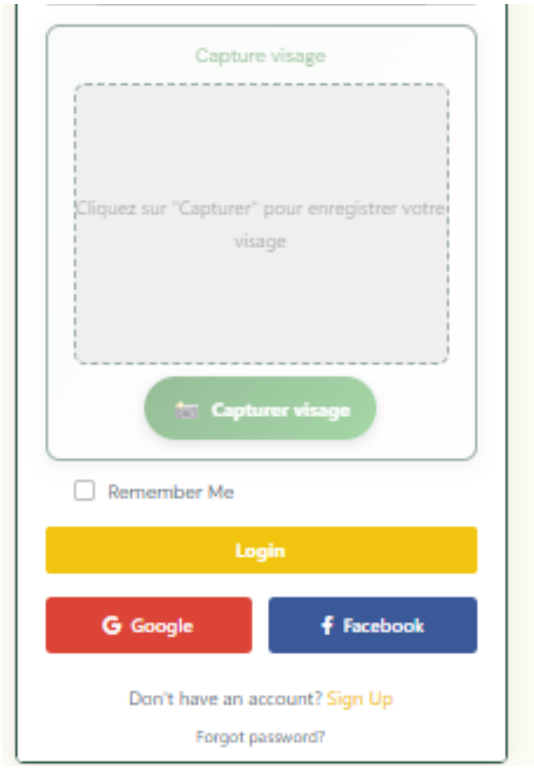




- **LOGIN PAGE**

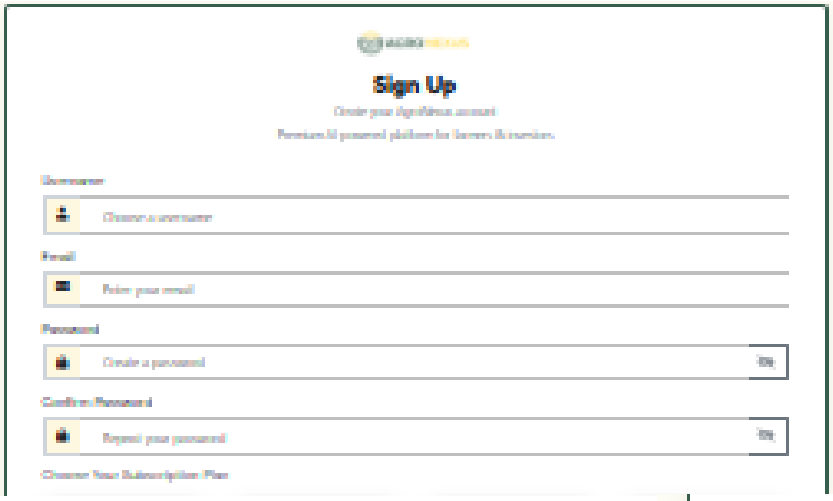
THE "LOGIN PAGE" INTERFACE IS A CRITICAL COMPONENT OF THE FARMWISE PLATFORM, DESIGNED TO PROVIDE SECURE AND EFFICIENT ACCESS FOR REGISTERED USERS, INCLUDING FARMERS AND INVESTORS. THIS PAGE FEATURES A CLEAN AND INTUITIVE LAYOUT WHERE USERS CAN LOG IN WITH THEIR USERNAME OR FACE ID ON SUPPORTED DEVICES, LEVERAGING THE WEB AUTHENTICATION API (WEBAUTHN) FOR SEAMLESS BIOMETRIC AUTHENTICATION. A "FORGOT PASSWORD" LINK ASSISTS WITH ACCOUNT RECOVERY, REDIRECTING TO A RESET WORKFLOW, WHILE A "SIGN UP" BUTTON DIRECTS NEW USERS TO REGISTRATION. THE INTERFACE INCORPORATES A RESPONSIVE DESIGN, OPTIMIZED FOR BOTH DESKTOP AND MOBILE DEVICES USING HTML, CSS, AND JAVASCRIPT WITH FRAMEWORKS LIKE BOOTSTRAP OR TAILWIND CSS, AND IS POWERED BY DJANGO'S AUTHENTICATION BACKEND WITH WEBAUTHN SUPPORT.





- **SIGN UP PAGE**

THE "SIGN UP PAGE" INTERFACE IS A VITAL COMPONENT OF THE FARMWISE PLATFORM, ENABLING NEW USERS, SUCH AS FARMERS AND INVESTORS, TO REGISTER EFFICIENTLY. THIS PAGE FEATURES A SIMPLE AND INTUITIVE LAYOUT WHERE USERS CAN SIGN UP BY PROVIDING A USERNAME, EMAIL, PASSWORD, AND REGION, WITH A TERMS ACCEPTANCE CHECKBOX TO ENSURE COMPLIANCE. EMAIL VERIFICATION IS REQUIRED POST-REGISTRATION TO CONFIRM THE ACCOUNT, FOLLOWED BY A WELCOME MESSAGE REDIRECTING USERS TO THE LOGIN PAGE. THE INTERFACE INCORPORATES A RESPONSIVE DESIGN, OPTIMIZED FOR BOTH DESKTOP AND MOBILE DEVICES USING HTML, CSS, AND JAVASCRIPT WITH FRAMEWORKS LIKE BOOTSTRAP OR TAILWIND CSS, AND IS POWERED BY DJANGO'S AUTHENTICATION BACKEND FOR SECURE USER MANAGEMENT.



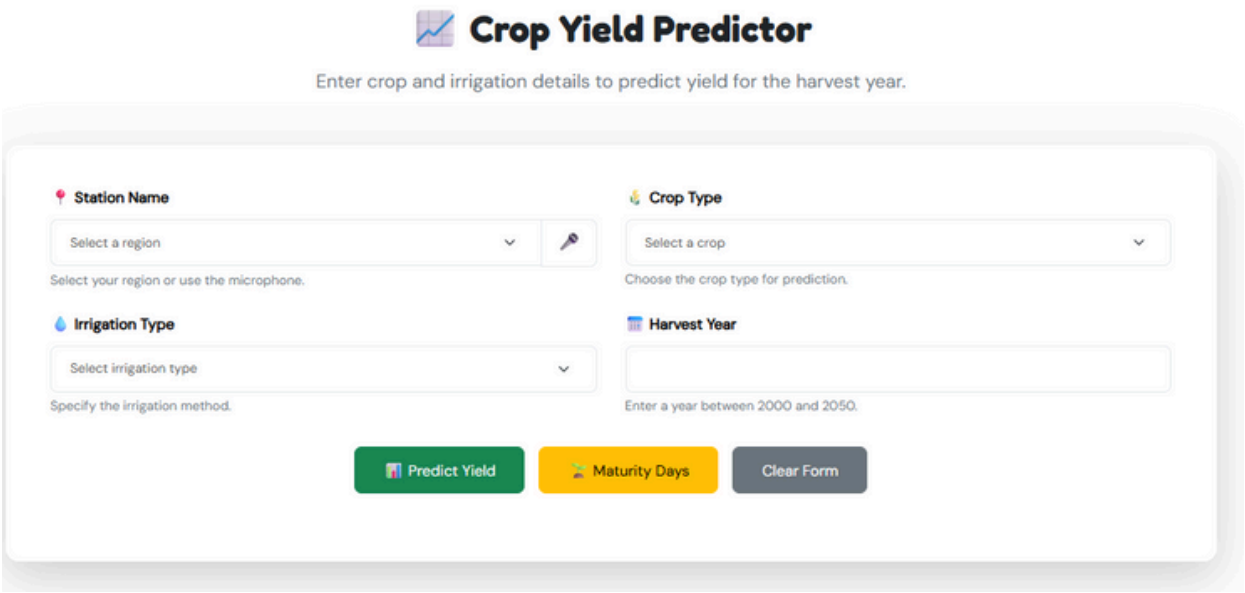


- **SERVICES PAGES**

BEFORE DIVING INTO THE DETAILS, LET ME INTRODUCE THE KEY SERVICES OFFERED BY OUR PLATFORM. EACH SERVICE IS DESIGNED TO SUPPORT FARMERS AND AGRICULTURAL PROFESSIONALS IN MAKING SMARTER, FASTER, AND MORE SUSTAINABLE DECISIONS — FROM PREDICTING CROP YIELDS TO DETECTING DISEASES AND OPTIMIZING LAND USE. LET'S TAKE A CLOSER LOOK AT WHAT EACH SERVICE OFFERS.

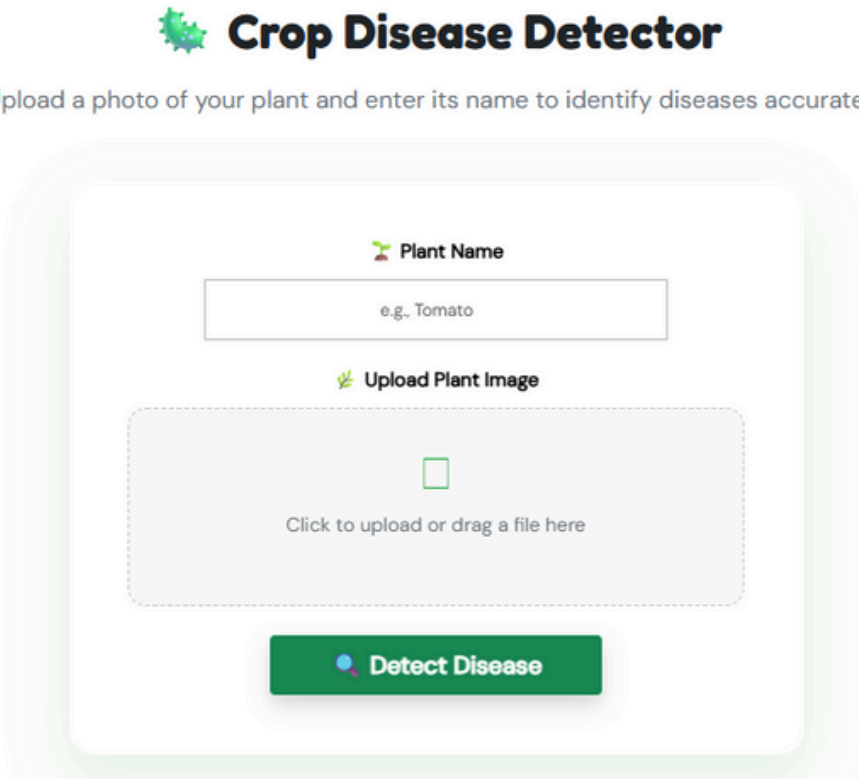
1) CROP YIELD PREDICTION (YIELD INTERFACE)

THIS SERVICE IS AVAILABLE DIRECTLY ON THE PLATFORM AND GIVES FARMERS QUICK FORECASTS ABOUT THEIR CROP YIELD AND HARVEST TIME. USERS SIMPLY ENTER DATA LIKE SOWING DATES AND WEATHER TRENDS, AND THE SYSTEM INSTANTLY PROVIDES RELIABLE PREDICTIONS TO SUPPORT PLANNING.



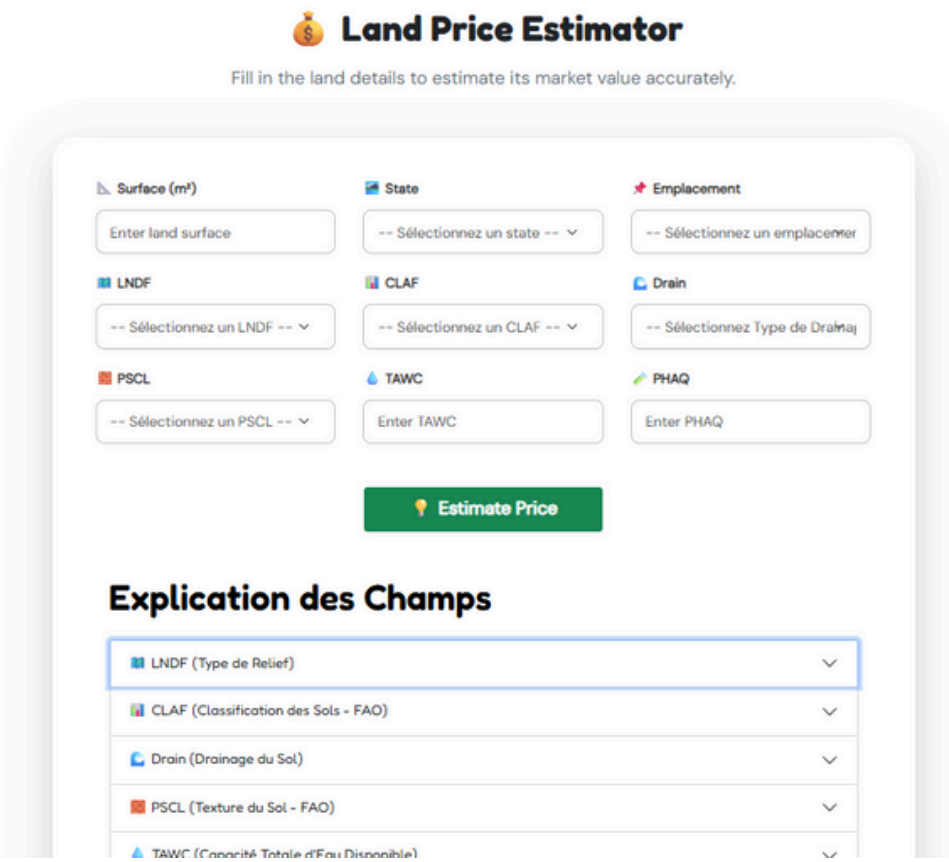
2) DISEASE DETECTION (DISEASE INTERFACE)

THROUGH THE MOBILE APP, FARMERS CAN TAKE A PHOTO OF A PLANT, AND THIS SERVICE WILL INSTANTLY TELL THEM IF THE PLANT IS HEALTHY OR AFFECTED BY A DISEASE. IT'S FAST, EASY TO USE, AND WORKS EVEN IN REMOTE AREAS WITH LIMITED INTERNET.



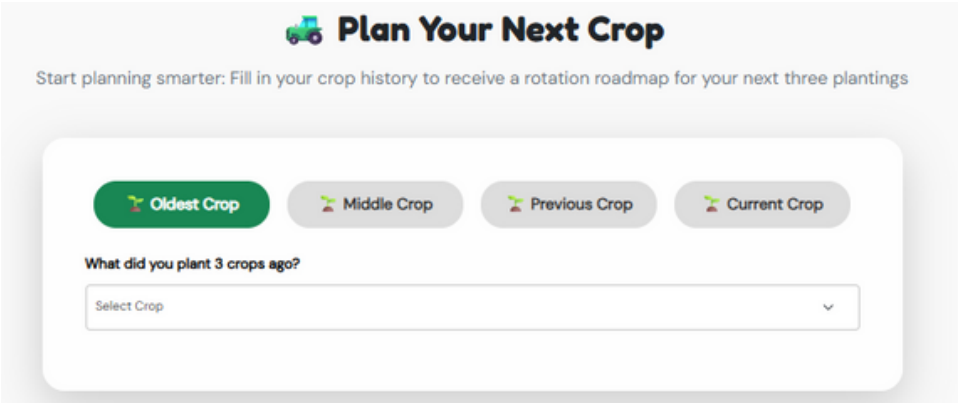
3) LAND PRICE PREDICTION (LANDPRICE INTERFACE)

USERS CAN CHECK THE ESTIMATED PRICE OF AGRICULTURAL LAND BY ENTERING DETAILS SUCH AS LOCATION AND SOIL QUALITY. THE PLATFORM PROVIDES QUICK AND ACCURATE PRICING INSIGHTS, HELPING FARMERS AND BUYERS MAKE BETTER INVESTMENT DECISIONS



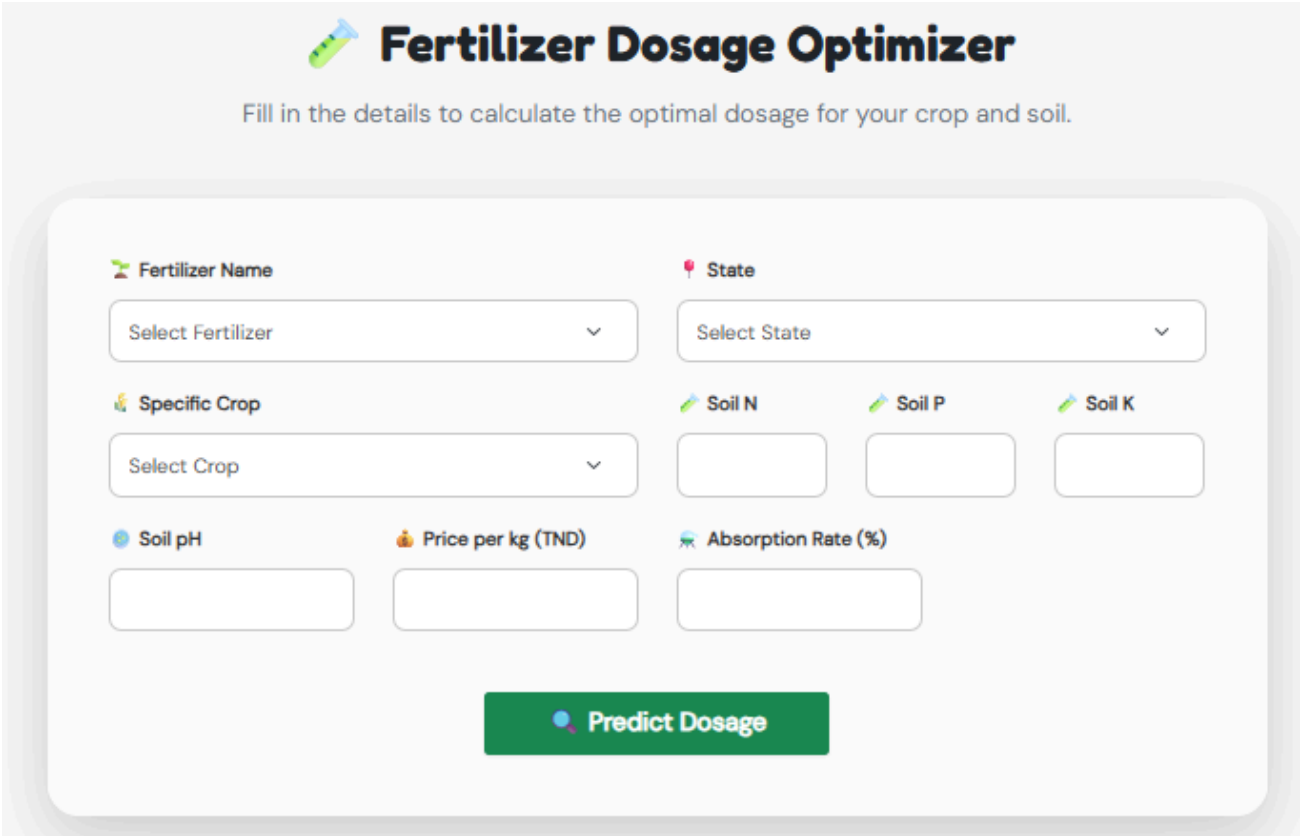
4) CROP ROTATION RECOMMENDATION (CROPROTATION INTERFACE)

THIS TOOL SUGGESTS THE BEST CROPS TO PLANT NEXT BASED ON PREVIOUS SEASONS AND REGIONAL CONDITIONS. IT PROMOTES SUSTAINABLE FARMING AND HELPS USERS MAINTAIN SOIL HEALTH WHILE MAXIMIZING PRODUCTIVITY.



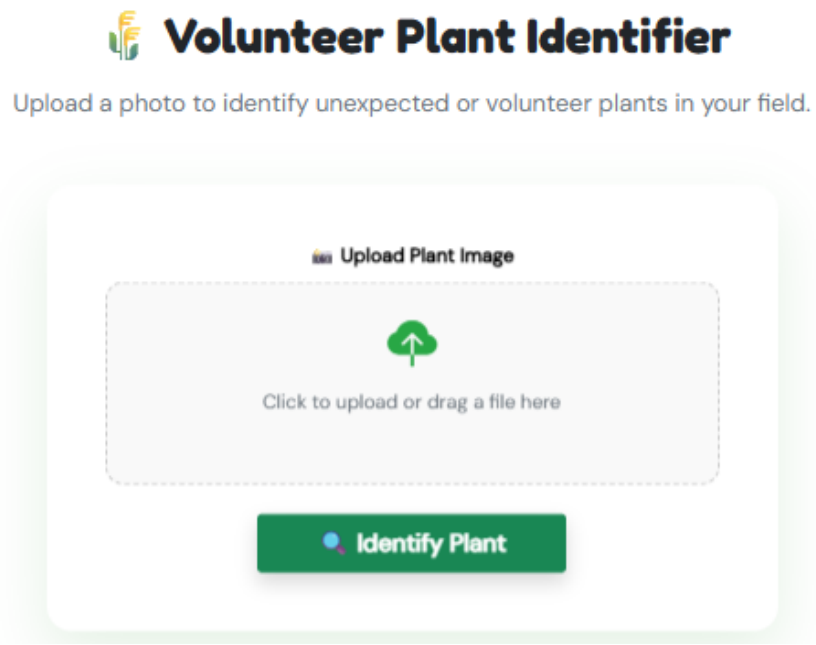
5) FERTILIZER NEEDS ESTIMATION (DOSAGE INTERFACE)

BY ENTERING A FEW DETAILS ABOUT THEIR CROPS AND FIELDS, USERS RECEIVE PERSONALIZED RECOMMENDATIONS ON HOW MUCH FERTILIZER TO USE. THIS HELPS REDUCE WASTE, SAVES MONEY, AND IMPROVES CROP HEALTH.



6) VOLUNTEER PLANT CLASSIFICATION (VOLUNTARY INTERFACE)

FARMERS CAN UPLOAD IMAGES OF UNKNOWN OR SPONTANEOUS PLANTS GROWING IN THEIR FIELDS. THE SYSTEM WILL INSTANTLY TELL WHETHER THE PLANT IS HARMFUL, USEFUL, OR MEDICINAL — HELPING FARMERS MANAGE THEIR FIELDS MORE EFFICIENTLY.



V-CONCLUSION

THE DEPLOYMENT OF OUR AI-POWERED SERVICES MARKS A SIGNIFICANT STEP TOWARD SMARTER AND MORE SUSTAINABLE AGRICULTURE. EACH FEATURE HAS BEEN CAREFULLY DESIGNED TO ADDRESS REAL CHALLENGES FACED BY FARMERS, OFFERING PRACTICAL AND ACCESSIBLE SOLUTIONS THROUGH AN INTUITIVE PLATFORM. BY COMBINING ADVANCED TECHNOLOGIES WITH A USER-FRIENDLY EXPERIENCE, WE AIM TO EMPOWER AGRICULTURAL COMMUNITIES WITH THE INSIGHTS THEY NEED TO MAKE INFORMED DECISIONS AND IMPROVE PRODUCTIVITY. THIS CHAPTER HIGHLIGHTS HOW INNOVATION CAN DIRECTLY SUPPORT EVERYDAY FARMING PRACTICES, PAVING THE WAY FOR A MORE EFFICIENT AND RESILIENT AGRICULTURAL FUTURE.